Geometric Class Field Theory for Curves

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TODO

- Check the references at Bib.
- exclude or minimize the frame-torsor equivliance. and make sure to mention it in ramification settings.
- Go over which results rely on k being algebraically closed (for example, Theorem 58), and verify that this assumption is stated and used consistently throughout the thesis.

Abstract

We calculate the ramification of the symmetric product $\mathcal{F}^{[\deg \mathfrak{m}]}$ of a local system $\mathcal{F}$ on a curve $C \setminus \mathfrak{m}$. Assuming the ramification of $\mathcal{F}$ is bounded by $\mathfrak{m}$, we find that the symmetric product $\mathcal{F}^{[\deg \mathfrak{m}]}$ is at most tamely ramified at the generic point of the exceptional divisor $E_{\mathfrak{m}}$ of the blowup of $C^{(\deg \mathfrak{m})}$ at $\mathfrak{m}$. As a primary application, we utilize this result to prove geometric class field theory for curves. Our approach builds upon the geometric framework for the unramified case originally established by Deligne for the rank-one Langlands correspondence, following the subsequent extensions to the ramified case developed by Takeuchi and Guignard.

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Introduction

In this thesis, we give a direct proof of a key geometric result in Deligne's approach to geometric class field theory. We work primarily over a perfect field k. Let C be a smooth, projective, geometrically connected curve over k, of genus g. Let $\mathfrak{m} = \sum_i n_i P_i$ be an effective divisor on C, called a modulus, and set $U = C \setminus \mathfrak{m}$. Geometric class field theory asserts that every rank-one local system $\mathcal{F}$ on U whose ramification is bounded by $\mathfrak{m}$ arises uniquely as the pullback of a multiplicative rank-one local system on $\mathrm{Pic}_{C,\mathfrak{m}}$ along the Abel–Jacobi morphism:

$$\Phi : U \longrightarrow \mathrm{Pic}_{C,\mathfrak{m}}\,.$$

One of the main geometric ingredients in Deligne's approach to this assertion is the analysis of the symmetric powers $\mathcal{F}^{[d]}$ on $U^{(d)}$, and in particular the proof that, for all sufficiently large d, they descend along the degree d Abel-Jacobi morphism:

$$\Phi_d : U^{(d)} \longrightarrow \mathrm{Pic}^d_{C,\mathfrak{m}}\,.$$

The geometric-fibers of Φ_d (for $d \geq \deg \mathfrak{m} + 2g - 1$) over any point are isomorphic to

$$\begin{cases} \mathbb{A}^{d-\deg \mathfrak{m}-g+1}_{k^{sep}} & \text{if } \mathfrak{m} > 0 \\ \mathbb{P}^{d-g}_{k^{sep}} & \text{if } \mathfrak{m} = 0 \end{cases}$$

Where g is the genus of the curve C. The unramified case ($\mathfrak{m} = 0$) is relatively simple, as the Abel-Jacobi map is proper, surjective with geometrically connected fibers, which follows from the fact that it is a fibration in projective spaces. Thus, by using the homotopy exact sequence for the étale fundamental group, one gets an isomorphism between the étale fundamental group of $U^{(d)} (= C^{(d)})$ and that of $\mathrm{Pic}^d_{C,\mathfrak{m}} (= \mathrm{Pic}^d_C)$.

The ramified case ($\mathfrak{m} > 0$) is more subtle, as the Abel-Jacobi map is not proper anymore, and one needs to analyze the ramification of $\mathcal{F}^{[d]}$ "along the boundary" of $U^{(d)}$ in $C^{(d)}$.

Previous work has generalized Deligne's approach to the ramified case, most notably by Guignard [Gui19] and Takeuchi [Tak19]. Their approaches differ. To descend, Guignard proves that the restriction of $\mathcal{F}^{[d]}$ to any line in the fiber of the degree d Abel-Jacobi map is a constant étale sheaf. He achieves this by demonstrating that the restriction is at most tamely ramified and invoking the triviality of the tame fundamental group of $\mathbb{A}^1_k$. His analysis relies on local geometric class field theory. It is also worth noting that Guignard's method generalizes to relative curves over arbitrary base schemes. Takeuchi, on the other hand, constructs a compactification of $U^{(d)}$ by blowing up $C^{(d)}$ along certain well-chosen centers. This compactification, denoted by $\tilde{C}^{(d)}_{\mathfrak{m}}$, has $U^{(d)}$ as an open subscheme with a codimension 1 closed subscheme H as complement. He then shows that the Abel-Jacobi map extends to a proper morphism from $\tilde{C}^{(d)}_{\mathfrak{m}}$ to $\mathrm{Pic}^d_{C,\mathfrak{m}}$, which is a fibration in

projective spaces. Thus, by the homotopy exact sequence for the étale fundamental group, one gets an isomorphism between the étale fundamental group of $\tilde{C}_{\mathfrak{m}}^{(d)}$ and that of $\mathrm{Pic}_{C,\mathfrak{m}}^{d}$. To conclude the descent, Takeuchi analyzes the ramification of $\mathcal{F}^{[d]}$ along the boundary $H = \tilde{C}_{\mathfrak{m}}^{(d)} \setminus U^{(d)}$ of $U^{(d)}$ in $\tilde{C}_{\mathfrak{m}}^{(d)}$, showing that it is tamely ramified there, which suffices. His method relies on the theory of refined Swan conductors.

For an account of these approaches, see [Gui19] and [Tak19]. For a full approach following Deligne's method in the unramified case, and the tamely ramified case see [Ten15], and [Tót11].

In this thesis, we take a more elementary approach and calculate the ramification of $\mathcal{F}^{[d]}$ at the generic point of H directly, using explicit ramification theory rather than refined Swan conductors. We first reduce the problem to cyclic torsors. We then treat cyclic groups of order prime to the characteristic using Kummer theory and, in characteristic $p > 0$, cyclic groups of order p and p^r, with $r \geq 2$, using Artin–Schreier and Artin–Schreier–Witt theory, respectively. These descriptions provide explicit local equations, which allow us to analyze the relevant valuations after blowing up and prove that the ramification is at most tame. To prove geometric class field theory for curves, we utilize Takeuchi's construction of the compactification $\tilde{C}_{\mathfrak{m}}^{(d)}$ of $U^{(d)}$ via the blowing up of $C^{(d)}$.

We now introduce the notation needed to state geometric class field theory, its standard degree-d reduction, and the ramification theorem proved in this thesis.

Let k be a perfect field, and let C be a projective smooth geometrically connected curve over k, with genus g.

Fix a modulus $\mathfrak{m}$, i.e. an effective Cartier divisor of C and let U be its complement in C. The pairs $(\mathcal{L}, \alpha)$, where $\mathcal{L}$ is an invertible $\mathcal{O}_C$-module and $\alpha : \mathcal{O}_{\mathfrak{m}} \xrightarrow{\sim} \mathcal{L}|_{\mathfrak{m}}$ is a rigidification of $\mathcal{L}$ along $\mathfrak{m}$, are parametrized by a k-group scheme $\mathrm{Pic}_{C,\mathfrak{m}}$, called the rigidified Picard scheme. We write $\mathrm{Pic}_{C,\mathfrak{m}}^{d}$ for its degree-d component. The Abel-Jacobi morphism

$$\Phi : U \to \mathrm{Pic}_{C,\mathfrak{m}}$$

is the morphism which sends a section x of U to the pair $(\mathcal{O}_C(x), 1)$.

Geometric class field theory asserts that rank-one local systems on U, with ramification bounded by $\mathfrak{m}$, arise uniquely as pullbacks of multiplicative rank-one local systems on $\mathrm{Pic}_{C,\mathfrak{m}}$. For a finite commutative coefficient ring Λ, the $\Lambda^\times$-torsor associated with such a multiplicative local system is an isogeny onto $\mathrm{Pic}_{C,\mathfrak{m}}$ with finite constant kernel $\Lambda^\times$. The isogeny formulation will not play a direct role in our setting, which instead focuses on the degree-d descent statement introduced below.

The fundamental result of GCFT can be formulated as:

Theorem 1 (Geometric Class Field Theory)**.** *Let Λ be a finite commutative ring of cardinality invertible in k, and let $\mathcal{F}$ be an étale sheaf of Λ-modules, locally free of rank 1 on U, with ramification bounded by $\mathfrak{m}$. Then, there exists a unique (up to isomorphism) multiplicative[1] étale sheaf of Λ-modules $\mathcal{G}$ on $\mathrm{Pic}_{C,\mathfrak{m}}$, locally free of rank 1, such that the pullback of $\mathcal{G}$ by Φ is isomorphic to $\mathcal{F}$.*

Let d be a positive integer. We denote by $U^{(d)}$ the d-th symmetric power of U over k. For an étale sheaf $\mathcal{F}$ on $U_{\text{ét}}$, we denote by $\mathcal{F}^{[d]}$ the d-th symmetric power of $\mathcal{F}$ on $U^{(d)}$. The degree-d Abel–Jacobi morphism is

$$\Phi_d : U^{(d)} \longrightarrow \mathrm{Pic}_{C,\mathfrak{m}}^{d},$$

[1]Here multiplicative means that the local system is equipped with an isomorphism compatible with the group law on $\mathrm{Pic}_{C,\mathfrak{m}}$, satisfying the usual associativity and unit conditions; see [Gui19].

sending an effective divisor $D = x_1 + \cdots + x_d$ on U to the pair

$$\big(\mathcal{O}_C(D), 1\big).$$

A standard symmetric-power argument[2] reduces Theorem 1 to the following problem: for every sufficiently large d, construct a rank-one local system $\mathcal{G}_d$ on $\mathrm{Pic}^d_{C,\mathfrak{m}}$ whose pullback along Φ_d is isomorphic to $\mathcal{F}^{[d]}$. In our setting, pullback along Φ_d is fully faithful, so any such local system is automatically unique up to isomorphism. Moreover, the canonical compatibility of the symmetric powers $\mathcal{F}^{[d]}$ under the addition of effective divisors descends uniquely to the family $\{\mathcal{G}_d\}$. The resulting multiplicative family determines the multiplicative local system on $\mathrm{Pic}_{C,\mathfrak{m}}$ appearing in Theorem 1. Consequently, it suffices to prove the following reduced statement:

Theorem 2. *Let Λ be a finite commutative ring of cardinality invertible in k, and let $\mathcal{F}$ be an étale sheaf of Λ-modules, locally free of rank 1 on $U_{ét}$, with ramification bounded by $\mathfrak{m}$. Then, for sufficiently large integer d, there exists a unique (up to isomorphism) étale sheaf of Λ-modules $\mathcal{G}_d$ on $\mathrm{Pic}^d_{C,\mathfrak{m}}$, locally free of rank 1, such that the pullback of $\mathcal{G}_d$ by Φ_d is isomorphic to $\mathcal{F}^{[d]}$.*

The main technical input in the proof of Theorem 2 is the following local ramification theorem:

Theorem 3. *Assume that k is algebraically closed. Let Λ be a finite commutative ring of cardinality invertible in k, and let $\mathcal{F}$ be an étale sheaf of Λ-modules, locally free of rank 1 on $U_{ét}$, with ramification bounded by $\mathfrak{m} = \sum n_i P_i$. Let $\mathfrak{n} = n_i P_i \subset \mathfrak{m}$ be a single-point submodulus of $\mathfrak{m}$. Then $\mathcal{F}^{[\deg \mathfrak{n}]}$ is tamely ramified at the generic point $\eta_{\mathfrak{n}}$ of the exceptional divisor $E_{\mathfrak{n}}$ of the blowup $X_{\mathfrak{n}}$ of $C^{(\deg \mathfrak{n})}$ at the closed point $\mathfrak{n}$.*

As a final methodological note, for convenience we carry out the ramification calculations in the language of G-torsors, with $G = \Lambda^\times$, rather than rank-one local systems. By Proposition 9, the two formulations are equivalent, and the results translate directly between them.

The thesis is organized as follows:

Chapter 1 - General Preliminaries establishes the equivalence between rank-one local systems and $G = \Lambda^\times$-torsors and develops the torsor operations, finite étale covers, and symmetric-power constructions used throughout the thesis. It also collects the basic geometric facts about symmetric powers and blowups needed in the subsequent ramification calculations.

Chapter 2 - Ramification After Blowup Is Tame proves Theorem 3. After recalling the necessary ramification theory for discrete valuations and finite abelian torsors, it reduces the problem to cyclic torsors on $\mathbb{P}^1$ and treats the Kummer, Artin–Schreier, and Artin–Schreier–Witt cases through explicit calculations at the exceptional divisor.

Chapter 3 - Geometric Class Field Theory deduces Theorem 2 from the ramification theorem proved in Chapter 2. It recalls the generalized Picard scheme and the Abel–Jacobi morphism, introduces Takeuchi's compactification, and combines the single-point calculations to establish tameness along the full boundary H. It then extends $\mathcal{F}^{[d]}$ across the boundary and descends the resulting local system to $\mathrm{Pic}^d_{C,\mathfrak{m}}$.

Appendix A *Cyclic Torsor Theories and Witt Vectors* collects the group-scheme constructions, classification results, normal forms, and proofs supporting the three cyclic ramification theories used in Chapter 2.

[2] See the last page of [Gui19], Section 8.3 of [Ten15], or the proof of Theorem 1.2 in [Tak19] for details on this reduction.

Appendix B *A Failure of an Artin–Schreier Induction Strategy* examines a tempting attempt to deduce the cyclic p-power calculation inductively from the degree-p Artin–Schreier case, and explains how wild base change obstructs that argument.

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Use of Artificial Intelligence. In preparing this thesis, I used AI-assisted tools including Codex with GPT-5.6 Sol, Claude Code with Claude Opus 4.8 and Claude Fable, and Google Gemini with Gemini 3.1 Pro. These tools assisted with phrasing, editing, restructuring, drafting portions of the exposition, and formulating mathematical statements and arguments. They also assisted in developing and checking parts of proofs, particularly the Artin–Schreier–Witt calculation for cyclic groups of order p^r, as well as several particular cases in Kummer and Artin–Schreier theory that were independently verified by hand. More broadly, these tools supported the organization and completion of the thesis as a whole. Every AI-generated suggestion or draft incorporated into the thesis was fully reviewed and mathematically checked by the author, revised where necessary, and adapted to the notation, presentation, and style of the thesis. No AI-generated material was accepted without human verification, and the author assumes full responsibility for all mathematical and expository content.

Chapter 1

General Preliminaries

This chapter collects the constructions and conventions used in both main parts of the thesis. Rank-one étale local systems are the natural objects in the statement of geometric class field theory. For the geometric and ramification-theoretic arguments, however, it is more convenient to work with torsors: they are represented by finite étale covers, admit concrete contracted products and quotients, and make ramification accessible through extensions of local fields.

If Λ is a finite commutative ring and $G = \Lambda^\times$, the sheaf of frames of a rank-one Λ-local system is a G-torsor, and conversely a G-torsor determines an associated rank-one local system. We record this equivalence once below. Thereafter, the main results are stated in the language of local systems, while the geometric and ramification calculations are carried out with the corresponding torsors.

1.1 Torsors and Rank-One Local Systems

In what follows, we largely adhere to the treatment of torsors and group objects found in published notes of Alex Youcis [You20]. Let $\mathscr{C} = (\mathcal{C}, J)$ be a site and let $\mathcal{E} = Sh(\mathscr{C})$ be the associated topos. Let $\mathcal{G}$ be a group object in $\mathcal{E}$. We denote by $\mathcal{G}\mathcal{E}$ the category of objects in $\mathcal{E}$ endowed with a left $\mathcal{G}$-action.

Definition 4. A **$\mathcal{G}$-torsor** in $\mathcal{E}$ is an object $\mathcal{P}$ of $\mathcal{G}\mathcal{E}$ satisfying the following conditions:

1. The structural morphism $\mathcal{P} \to 1$ is an epimorphism in $\mathcal{E}$ (i.e., $\mathcal{P}$ is locally non-empty).
2. The map $\mathcal{G} \times \mathcal{P} \to \mathcal{P} \times \mathcal{P}$ defined by $(g, p) \mapsto (g \cdot p, p)$ is an isomorphism in $\mathcal{E}$ (i.e., $\mathcal{G}$ acts simply transitively on $\mathcal{P}$).

Since $\mathcal{E}$ is the topos of sheaves on a site $\mathscr{C}$, the definition can be reformulated in terms of covers. A $\mathcal{G}$-sheaf $\mathcal{P}$ is a $\mathcal{G}$-torsor if:

1. For every object $X \in \mathcal{C}$, there exists a covering $\{U_i \to X\} \in J$ such that $\mathcal{P}(U_i) \neq \emptyset$ for all i.
2. For any $X \in \mathcal{C}$ where $\mathcal{P}(X)$ is non-empty, the action of $\mathcal{G}(X)$ on $\mathcal{P}(X)$ is simply transitively.

A fundamental property of torsors is their local triviality: a $\mathcal{G}$-sheaf $\mathcal{P}$ is a $\mathcal{G}$-torsor if and only if it is locally isomorphic to the trivial torsor. Specifically, for every $X \in \mathcal{C}$, there must exist a cover

$\{U_i \to X\}$ such that the restriction $\mathcal{P}|_{U_i}$ is isomorphic, as a $\mathcal{G}|_{U_i}$-sheaf, to $\mathcal{G}|_{U_i}$ acting on itself by left multiplication.

A **morphism of $\mathcal{G}$-torsors** $f : \mathcal{P}_1 \to \mathcal{P}_2$ is a morphism of sheaves that is equivariant with respect to the $\mathcal{G}$-action.

Proposition 5. *Every morphism of $\mathcal{G}$-torsors is an isomorphism. Consequently, the category of $\mathcal{G}$-torsors in $\mathcal{E}$ forms a groupoid.*

Proof. Locally on a cover trivializing both torsors, such a morphism is a $\mathcal{G}$-equivariant endomorphism of the trivial torsor $\mathcal{G}$, i.e. right translation by a section of $\mathcal{G}$, hence an isomorphism; and being an isomorphism may be checked on a cover. □

Definition 6. We denote the groupoid of $\mathcal{G}$-torsors in $\mathcal{E}$ by $\mathbf{Tors}(\mathcal{E}, \mathcal{G})$. The set of isomorphism classes of $\mathcal{G}$-torsors is denoted by $\mathrm{Tors}(\mathcal{E}, \mathcal{G})$.

Torsors exhibit functoriality with respect to the group object:

Definition 7. Let $\varphi : \mathcal{G}_1 \to \mathcal{G}_2$ be a morphism of group sheaves on $\mathscr{C}$, and let $\mathcal{P}$ be a $\mathcal{G}_1$-torsor. We define the **contracted product** $\mathcal{G}_2 \times^{\mathcal{G}_1} \mathcal{P}$ as the quotient sheaf $\mathcal{G}_1 \backslash (\mathcal{G}_2 \times \mathcal{P})$, where $\mathcal{G}_1$ acts on the product by:

$$g_1 \cdot (g_2, p) = (g_2 \varphi(g_1)^{-1}, g_1 \cdot p)$$

The contracted product inherits a natural left $\mathcal{G}_2$-action given on local sections by $h \cdot [g_2, p] = [hg_2, p]$, which endows it with the structure of a $\mathcal{G}_2$-torsor.

This construction yields a functor:

$$\varphi_* : \mathbf{Tors}(\mathcal{E}, \mathcal{G}_1) \to \mathbf{Tors}(\mathcal{E}, \mathcal{G}_2), \quad \mathcal{P} \mapsto \mathcal{G}_2 \times^{\mathcal{G}_1} \mathcal{P}$$

On the level of isomorphism classes, φ_* induces a map of pointed sets $\mathrm{Tors}(\mathcal{E}, \mathcal{G}_1) \to \mathrm{Tors}(\mathcal{E}, \mathcal{G}_2)$, sending the class of the trivial $\mathcal{G}_1$-torsor to the class of the trivial $\mathcal{G}_2$-torsor.

When $\mathcal{G}$ is a **sheaf of abelian groups** (an *abelian sheaf*), the pointed set $\mathrm{Tors}(\mathcal{E}, \mathcal{G})$ inherits the structure of an abelian group:

Definition 8. Assume $\mathcal{G}$ is abelian sheaf. Let $\mathcal{P}_1$ and $\mathcal{P}_2$ be objects of $\mathbf{Tors}(\mathcal{E}, \mathcal{G})$. We define the **product** $\mathcal{P}_1 \otimes^{\mathcal{G}} \mathcal{P}_2$ to be the quotient sheaf $(\mathcal{P}_1 \times \mathcal{P}_2)/\mathcal{G}$, where $\mathcal{G}$ acts on T-points by having:

$$g \cdot (f_1, f_2) := (gf_1, g^{-1} f_2)$$

The group object $\mathcal{G}$ then acts on the resulting quotient via its action on the presheaf quotient, which is given on classes by:

$$g \cdot [(f_1, f_2)] = [(gf_1, f_2)] = [(f_1, gf_2)]$$

where the square brackets denote the class in the quotient set.[1]

Let $[\mathcal{P}_1]$ and $[\mathcal{P}_2]$ be the isomorphism classes of $\mathcal{P}_1$ and $\mathcal{P}_2$ in $\mathrm{Tors}(\mathcal{E}, \mathcal{G})$. We define the sum $[\mathcal{P}_1] + [\mathcal{P}_2]$ to be the class $[\mathcal{P}_1 \otimes^{\mathcal{G}} \mathcal{P}_2]$. This structure turns $\mathrm{Tors}(\mathcal{E}, \mathcal{G})$ into an abelian group, where the identity is the class of the trivial torsor and the inverse is obtained by the opposite action.

[1] Equivantly, the sum is obtained as the contracted product of the $\mathcal{G} \times \mathcal{G}$-torsor $\mathcal{P}_1 \times \mathcal{P}_2$ along the multiplication map $m : \mathcal{G} \times \mathcal{G} \to \mathcal{G}$.

We now record the equivalence used in the thesis. Let Λ be a commutative ring object of $\mathcal{E}$ and let $G = \Lambda^\times$ be its group of units. We write $\mathbf{Pic}(\mathcal{E}, \Lambda)$ for the groupoid of locally free Λ-modules of rank one. For $\mathcal{L}$ in this groupoid, its **frame torsor** is

$$\mathrm{Fr}(\mathcal{L}) := \underline{\mathrm{Isom}}_\Lambda(\Lambda, \mathcal{L}).$$

The group G acts on frames by precomposition with multiplication on Λ. Conversely, if $\mathcal{P}$ is a G-torsor, its **associated Λ-module** is the contracted product

$$\mathcal{P} \times^G \Lambda := (\mathcal{P} \times \Lambda)/G, \qquad g \cdot (p, \lambda) = (gp, g^{-1}\lambda),$$

where G acts on Λ by multiplication.

Proposition 9. *The functors*

$$\mathcal{P} \longmapsto \mathcal{P} \times^G \Lambda, \qquad \mathcal{L} \longmapsto \mathrm{Fr}(\mathcal{L})$$

are mutually quasi-inverse equivalences

$$\mathbf{Tors}(\mathcal{E}, G) \xrightarrow{\sim} \mathbf{Pic}(\mathcal{E}, \Lambda).$$

Proof. There are natural morphisms

$$\mathcal{P} \longrightarrow \mathrm{Fr}(\mathcal{P} \times^G \Lambda), \qquad p \longmapsto \big(\lambda \longmapsto [p, \lambda]\big),$$

and

$$\mathrm{Fr}(\mathcal{L}) \times^G \Lambda \longrightarrow \mathcal{L}, \qquad [\varphi, \lambda] \longmapsto \varphi(\lambda).$$

After passing to a cover on which $\mathcal{P}$ is the trivial torsor and $\mathcal{L}$ is free of rank one, both morphisms become the identity. Hence they are isomorphisms. □

When $\mathcal{E} = \mathrm{Sh}(X_{\text{ét}})$ and Λ is a finite constant ring, the objects of $\mathbf{Pic}(\mathcal{E}, \Lambda)$ are precisely the rank-one Λ-local systems on X. Henceforth, we identify such a local system with its $G = \Lambda^\times$-torsor of frames and pass between the two descriptions without further comment.[2]

Lastly, for any object $\mathcal{X} \in \mathcal{E}$, there is a canonical identification between the slice category $(\mathcal{GE})/\mathcal{X}$ and the category of group objects $\mathcal{G}(\mathcal{E}/\mathcal{X})$, where $\mathcal{X}$ is viewed as having the trivial $\mathcal{G}$-action.

We denote by $\mathbf{Tors}(\mathcal{X}, \mathcal{G})$ the category of G-torsors over $\mathcal{X}$ in $\mathcal{GE}_{/\mathcal{X}}$.

1.2 Étale Torsors, Covers, and Quotients

Fix a scheme X, and let G be a smooth affine X-group scheme. Denote by $X_{\text{Ét}}$ and $X_{\text{ét}}$ the big and small étale sites on X, respectively. The functor of points of G, which we denote by $G(\cdot) = \mathrm{Hom}_X(-, G)$, defines a group sheaf on $X_{\text{ét}}$.

Definition 10. A *principal G-bundle* (or *principal homogeneous space*) for G is a scheme $f : Y \to X$ equipped with a left G-action $\rho : G \times_X Y \to Y$ satisfying:

[2] The equivalence is compatible with pullback and with tensor products, hence with the symmetric-power constructions used below. These compatibilities may be checked after a local trivialization. It also preserves ramification bounds: a rank-one local system and its frame torsor determine, on each connected component, the same character $\pi_1^{\text{et}}(X, \bar{x}) \to \Lambda^\times$, and therefore have the same restrictions to inertia and to the upper ramification groups.

1. The morphism $G \times_X Y \to Y \times_X Y$ sending $(g, y) \mapsto (gy, y)$ is an isomorphism of X-schemes.
2. There exists an étale covering $\{U_i \to X\}$ such that $Y_{U_i} \cong G_{U_i}$ as G-schemes, where G acts on itself by left translation.

Remark. Since $G \to X$ is smooth and Y is étale-locally isomorphic to G, the morphism $f : Y \to X$ is smooth and affine. Note that Y is generally not étale over X unless G is an étale group scheme (e.g., a finite constant group). Similarly, if G is finite then $Y \to X$ is finite.

Similarly to the case of G-torsors of a topos, we define a morphism of principal G-bundles to be a morphism of X-schemes commuting with the G-action. We now have:

Theorem 11. *The morphism sending $Y \mapsto \mathrm{Hom}_X(-, Y)$ is an equivalence of categories from the category of principal G-bundles to the category of G-torsors on $X_{\acute{E}t}$ Similarly, the morphism sending Y to $\mathrm{Hom}_X(-, Y)$ to the category of G-torsors on $X_{\acute{e}t}$ is an equivalence.*

Corollary 12. *There is a natural equivalence $\mathbf{Tors}(X_{\acute{e}t}, G) \cong \mathbf{Tors}(X_{\acute{E}t}, G)$ inducing a bijection of pointed sets $\mathrm{Tors}(X_{\acute{e}t}, G) \xrightarrow{\cong} \mathrm{Tors}(X_{\acute{E}t}, G)$ which is an isomorphism of abelian groups if G is abelian. And every G-torsor, in each of the above sites can be realized as a scheme, which is a prinicipal G-bundle.*

Torsors and first cohomology

We use the following standard fact without proof [Mil80, III, §4]. Let G be an abelian sheaf of groups on $X_{\text{ét}}$. Then there is a natural group isomorphism

$$\mathrm{Tors}(X_{\text{ét}}, G) \xrightarrow{\cong} H^1_{\text{ét}}(X, G),$$

and the trivial torsor maps to 0. So an element of $H^1_{\text{ét}}(X, G)$ is the same as an isomorphism class of G-torsors. The Kummer, Artin–Schreier, and Artin–Schreier–Witt classifications (Section A.1, Theorem 86) use this fact whenever they write $[T_a]$ for the class of a torsor in H^1.

Constant Finite Group Torsors

Let G be a finite group, within this section, we denote the associated constant group scheme over X by $\underline{G}$ to maintain a clear distinction between the group as a set and the group as a scheme. In subsequent sections, we shall follow the standard practice of identifying G with its constant group scheme and omit the underline for brevity.

$\underline{G}$ is given by $\coprod_{g \in G} X$ with the action shuffeling the X's according to multiplication.

By following the definitions, one sees that if X be a connected scheme and $f : Y \to X$ is a finite Galois cover with Galois group G. Then, $f : Y \to X$ is a principal $\underline{G}$-bundle. (Recall that a *finite Galois cover* is a finite étale surjection $Y \to X$ with Y connected and such that $G = \mathrm{Aut}(Y/X)$ acts transitively on the geometric points of Y lying over any geometric point of X).

On the otherhand if $f : Y \to X$ is a principal $\underline{G}$-bundle with Y connected, then Y is a finite Galois cover with automorphism group G.

However, not all $\underline{G}$-torsors are connected. If $H \subset G$ is a proper subgroup then any connected finite étale cover $f : Y \to X$ with Galois group H gives rise to a non-connected $\underline{G}$-torsor by looking at the induced $\underline{G}$-torsor $\varphi_*(Y)$ under the inclusion $\varphi : \underline{H} \to \underline{G}$. On the otherhand, if we fix a

geometric point $\overline{x} \to X$, then to give an homomorphism $\rho \in \mathrm{Hom}_{\mathrm{cont}}(\pi_1^{\mathrm{et}}(X,x), G)$ is equivilant to give a connected pointed Galois cover $(Y,\overline{y}) \to (X,\overline{x})$ with Galois group $H = \rho(\pi_1^{\mathrm{et}}(X,x)) \subset G$. Thus, pushing forward to $\underline{G}$ we get a prinicipal $\underline{G}$-bundle. The choice of a different geometric point $\overline{x'} \to X$ differ the homomorphism by an inner automorphism, thus we have:

Theorem 13. *Let X be a connected scheme and $\overline{x}$ a geometric point of X. Suppose in addition that G is a finite abstract group. Define a map*

$$\mathrm{Hom}_{\mathrm{cont}}(\pi_1^{\mathrm{et}}(X,\overline{x}), G)/\mathrm{Inn}(G) \to \mathrm{Tors}(X_{\acute{E}t}, \underline{G}) \tag{1.1}$$

by sending a homomorphism $\rho : \pi_1^{\mathrm{et}}(X,\overline{x}) \to G$ to the principal $\underline{G}$-bundle $\varphi_*(Y)$ where Y is the principal $\underline{\rho(\pi_1^{\mathrm{et}}(X,\overline{x}))}$-bundle obtained above and φ is the inclusion $\rho(\pi_1^{\mathrm{et}}(X,\overline{x})) \hookrightarrow G$. Then, the map is a bijection of pointed sets where the trivial homomorphisms (which is the only element of its $\mathrm{Inn}(G)$-orbit) is the distinguished element of the left hand side.

If G is abelian then $\mathrm{Inn}(G)$ is trivial, and we obtain:

Corollary 14. *Let G be a finite abelian group, X a connected scheme, and $\overline{x}$ a geometric point of X. Then, the map from Theorem 13 induces an isomorphism of abelian groups*

$$\mathrm{Hom}_{\mathrm{cont.}}(\pi_1^{\acute{e}t}(X,\overline{x}), G) \xrightarrow{\cong} \mathrm{Tors}(X_{\acute{E}t}, \underline{G}) \tag{1.2}$$

For $G = \Lambda^\times$, the character in Corollary 14 is also the monodromy character of the associated rank-one Λ-local system. In particular, the torsor and the local system have the same restrictions to inertia and to the upper ramification groups. Thus, one has ramification bounded by a given modulus, or is tamely ramified along a given divisor, if and only if the other does.

Quotients and Decomposition of Torsors

Let $\mathcal{H}$ be a sheaf of groups acting on a sheaf $\mathcal{P}$ on a site $\mathscr{C}$. The **quotient sheaf** $\mathcal{P}/\mathcal{H}$ is the coequalizer in $\mathrm{Sh}(\mathscr{C})$

$$\mathcal{H} \times \mathcal{P} \rightrightarrows \mathcal{P} \longrightarrow \mathcal{P}/\mathcal{H},$$

where the parallel arrows are the action and the projection. Equivalently, it is the sheaf associated with the presheaf of orbits $U \mapsto \mathcal{H}(U)\backslash\mathcal{P}(U)$. We use the notation $\mathcal{P}/\mathcal{H}$ even though our torsor actions are written on the left. In particular, the quotient always exists as a sheaf. It is characterized by the following universal property: for every sheaf $\mathcal{F}$, regarded as an $\mathcal{H}$-sheaf with the trivial action, composition with $\mathcal{P} \to \mathcal{P}/\mathcal{H}$ identifies morphisms $\mathcal{P}/\mathcal{H} \to \mathcal{F}$ with the $\mathcal{H}$-equivariant morphisms $\mathcal{P} \to \mathcal{F}$, equivalently with morphisms f satisfying $f(hp) = f(p)$ on local sections. A quotient sheaf need not in general be representable by a scheme; representability in the finite-torsor situation used here is part of the next proposition.

Let G be a finite abelian group, let $H \subseteq G$ be a subgroup, and let $\pi : G \to G/H$ be the quotient homomorphism.

Proposition 15. *Let $\mathcal{P} \to X$ be a G-torsor in $X_{\acute{e}t}$. The quotient sheaf $\mathcal{P}/H$ is represented by a finite étale X-scheme, and there is a canonical isomorphism of G/H-torsors*

$$\mathcal{P}/H \xrightarrow{\sim} \pi_*\mathcal{P} = (G/H) \times^G \mathcal{P}.$$

Consequently, $\mathcal{P} \to X$ admits a canonical factorization

$$\mathcal{P} \xrightarrow{\phi} \mathcal{P}/H \xrightarrow{\psi} X,$$

where ϕ is an H-torsor and ψ is a G/H-torsor. This factorization is unique up to a unique isomorphism commuting with the projections.

Proof. The canonical morphism

$$\mathcal{P} \longrightarrow \pi_*\mathcal{P}, \qquad p \longmapsto [1, p],$$

is H-invariant because $\pi(H) = 1$. The universal property of the quotient sheaf therefore gives a canonical morphism

$$\mathcal{P}/H \longrightarrow \pi_*\mathcal{P}.$$

Choose an étale cover $U \to X$ that trivializes $\mathcal{P}$. Over U this morphism identifies with the identity of $(G/H) \times U$, since

$$(G \times U)/H \cong (G/H) \times U \qquad \text{and} \qquad \pi_*\mathcal{P}|_U \cong (G/H) \times U.$$

It is therefore an isomorphism of sheaves. The contracted product $\pi_*\mathcal{P}$ is a G/H-torsor and, by Theorem 11, is represented by a finite étale X-scheme. Hence the same is true of $\mathcal{P}/H$.

Under the same local trivialization, the morphism $\mathcal{P} \to \mathcal{P}/H$ is the product of the quotient map with the identity,

$$G \times U \longrightarrow (G/H) \times U,$$

and is thus an H-torsor. Finally, suppose $\mathcal{P} \to \mathcal{Q} \to X$ is another such factorization. Since an H-torsor is locally split, every H-invariant morphism out of $\mathcal{P}$ descends uniquely through $\mathcal{Q}$. Thus $\mathcal{Q}$ satisfies the same universal property as $\mathcal{P}/H$, which gives the unique isomorphism compatible with the projections. □

1.3 Symmetric Powers of Schemes and Torsors

This section reviews the construction of quotients for schemes and torsors under finite group actions, specifically focusing on symmetric powers. To ensure these quotients exist as schemes, we utilize the framework of admissible actions from [SGA1]. Our treatment here closely follows the exposition in [Gui19] The definitions and results presented below are adapted from their work, the difference being that we ask Γ to act on the left, to make the later construction of symmetric powers more natural. This foundation provides the necessary criteria for admissibility and base change required to define the symmetric powers of a scheme X and a G-torsor $\mathcal{P}$ over X.

Let S be a scheme.

Definition 16 (([SGA1], V.1.7).)**.**

- Let T be an object of a category $\mathcal{C}$ endowed with a right action of a group Γ. We say that **the quotient T/Γ exists** in $\mathcal{C}$ if the covariant functor

$$\begin{aligned} \mathcal{C} &\to \text{Sets} \\ U &\mapsto \text{Hom}_{\mathcal{C}}(T, U)^{\Gamma} \end{aligned}$$

  is representable by an object of $\mathcal{C}$.

- Let T be an S-scheme. An action of a finite group Γ on T is **admissible** if there exists an affine Γ-invariant morphism $f : T \to T'$ such that the canonical morphism $\mathcal{O}_{T'} \to f_*\mathcal{O}_T$ induces an isomorphism from $\mathcal{O}_{T'}$ to $(f_*\mathcal{O}_T)^\Gamma$.

Proposition 17. *The following holds:*

1. ***([SGA1] V.1.3).*** *Let T be an S-scheme endowed with an admissible right action of a finite group Γ. If $f : T \to T'$ is an affine Γ-invariant morphism such that the canonical morphism $\mathcal{O}_{T'} \to f_*\mathcal{O}_T$ induces an isomorphism from $\mathcal{O}_{T'}$ to $(f_*\mathcal{O}_T)^\Gamma$, then the quotient T/Γ exists and is isomorphic to T'.*

2. ***([SGA1], V.1.8).*** *Let T be an S-scheme endowed with a right action of a finite group Γ. Then, the action of Γ on T is admissible if and only if T is covered by Γ-invariant affine open subsets.*

3. ***([SGA1], V.1.9).*** *Let T be an S-scheme endowed with an admissible right action of a finite group Γ, and let S' be a flat S-scheme. Then, the action of Γ on the S'-scheme $T \times_S S'$ is admissible, and the canonical morphism*

$$(T \times_S S')/\Gamma \to (T/\Gamma) \times_S S'$$

is an isomorphism.

Now let X be an S-scheme and let $d \geq 0$ be an integer. The group S_d of permutations of $[\![1, d]\!]$ acts on the right on the S-scheme $X^{\times_S d} = X \times_S \cdots \times_S X$ by the formula

$$(x_i)_{i\in[\![1,d]\!]} \cdot \sigma = (x_{\sigma(i)})_{i\in[\![1,d]\!]}.$$

Proposition 18 ([Gui19] Proposition 2.27)**.** *If X is a scheme, Zariski locally quasi-projective over S, then the right action of the symmetric group S_d on the d-fold fiber product $X^{\times_S d}$ is admissible. Consequently, the quotient $\mathrm{Sym}^d_S(X) = X^{\times_S d}/S_d$ exists as a scheme over S.*

Definition 19. Under the hypotheses of the proposition above, we define the **relative symmetric product** of X over S of degree d as the quotient

$$\mathrm{Sym}^d_S(X) \coloneqq X^{\times_S d}/S_d.$$

When the base scheme S is clear from the context, we shall denote this quotient by $X^{(d)}$.

Guignard shows that when $X = \mathrm{Spec}(B)$ and $S = \mathrm{Spec}(A)$ then $\mathrm{Sym}^d_S(X)$ is representable by an affine S-scheme (See [Gui19, Remark 2.28]).

Proposition 20 ([Gui19] Proposition 2.28)**.** *If X is flat and Zariski-locally quasi-projective over S, then $Sym^d_S(X)$ is flat over S. Moreover, for any S-scheme S', the canonical morphism*

$$Sym^d_{S'}(X \times_S S') \to Sym^d_S(X) \times_S S'$$

is an isomorphism.

Geometric Properties of Symmetric Powers

We record the geometric properties of symmetric powers that will be used in both main parts of the thesis.

Theorem 21. *Let X and Y be integral schemes over a field k. If X is geometrically integral, then $X \times_k Y$ is integral. If both X and Y are geometrically integral, then $X \times_k Y$ is geometrically integral.*

Proof. By [Stacks, Tag 038K], a geometrically integral scheme is both geometrically irreducible and geometrically reduced. Since X is geometrically irreducible, the projection

$$X \times_k Y \longrightarrow Y$$

induces a bijection on irreducible components by [Stacks, Tag 038F]. Because Y is integral, it follows that $X \times_k Y$ is irreducible. Moreover, X is geometrically reduced and Y is reduced, so $X \times_k Y$ is reduced by [Stacks, Tag 035Z]. Hence it is integral.

If both X and Y are geometrically integral, the same argument applies after every field extension k'/k. Thus

$$(X \times_k Y)_{k'} \cong X_{k'} \times_{k'} Y_{k'}$$

is integral for every k'/k, and therefore $X \times_k Y$ is geometrically integral. □

Proposition 22. *Let $f : X \to S$ be a smooth, Zariski-locally quasi-projective morphism of relative dimension one. Then the relative symmetric power $X_S^{(d)} = \mathrm{Sym}_S^d(X)$ is smooth over S of relative dimension d.*

Proof. Since $f : X \to S$ is smooth, the d-fold product X_S^d is smooth over S. By Proposition 20, the quotient $X_S^{(d)}$ is flat over S. Smoothness is a fiberwise property for flat morphisms of finite presentation (cf. [Stacks, Tag 01V8]); thus, it suffices to verify the smoothness of the fibers. For every $s \in S$, the fiber of the symmetric power is given by:

$$(\mathrm{Sym}_S^d(X))_s \cong \mathrm{Sym}_{\kappa(s)}^d(X_s)$$

Consequently, we may reduce to the case where $S = \mathrm{Spec}(k)$ for a field k. Since smoothness is preserved under base change to the algebraic closure, we may further assume $k = \bar{k}$. Let $z = \sum_{i=1}^r d_i \cdot x_i \in X^{(d)}$ be a point represented by an effective cycle of degree d, where the points $x_i \in X(k)$ are distinct and $\sum d_i = d$. Choose pairwise disjoint Zariski-open subsets $U_i \subset X$ such that $x_i \in U_i$. Let $W \subset X^{(d)}$ be the open subset consisting of cycles whose support is contained in $\bigcup U_i$ and which meet each U_i with degree exactly d_i. There is a canonical isomorphism:

$$W \cong \prod_{i=1}^r U_i^{(d_i)}$$

Thus, it suffices to show that each $U_i^{(d_i)}$ is smooth of dimension d_i at the point $d_i \cdot x_i$.

We may therefore restrict our attention to the "worst" case: the diagonal point $z = d \cdot x$. Let $R = \mathcal{O}_{X,x}$ be the local ring of the curve at x. Since X is a smooth curve over k, R is a regular local ring of dimension 1, i.e., a discrete valuation ring. Its completion is $\hat{R} \cong k[[t]]$. The completed local ring of the product X^d at the point $(x, \ldots, x)$ is:

$$\hat{\mathcal{O}}_{X^d,(x,\ldots,x)} \cong \hat{R} \hat{\otimes}_k \ldots \hat{\otimes}_k \hat{R} \cong k[[t_1, \ldots, t_d]]$$

where t_i is the uniformizer for the i-th copy of X. The symmetric group S_d acts on this power series ring by permuting the variables t_i. By the *Fundamental Theorem of Symmetric Polynomials*, the ring of invariants is:

$$\big(k[[t_1,\dots,t_d]]\big)^{S^d} = k[[e_1,\dots,e_d]]$$

where e_j is the j-th elementary symmetric power series in the variables t_i.

The ring $k[[e_1,\dots,e_d]]$ is a regular local ring of dimension d. Regularity of a Noetherian local ring is equivalent to regularity of its completion by [Stacks, Tag 07NY]. Hence $\mathcal{O}_{X^{(d)},z}$ is a regular local ring of dimension d. Since k is algebraically closed and $X^{(d)}$ is locally of finite type over k, regularity at z is equivalent to smoothness at z. This completes the proof. □

Corollary 23. *Let C be a smooth, projective, geometrically connected curve over a field k. For every $d \geq 0$, the scheme $C^{(d)}$ is smooth and geometrically integral. Moreover, every finite product of symmetric powers of C is geometrically integral.*

Proof. The smoothness follows from Proposition 22. The curve C is geometrically integral by [Stacks, Tag 0366]. After any field extension, C^d is integral and its image under the quotient map $C^d \to C^{(d)}$ is irreducible. Since $C^{(d)}$ is smooth, it is geometrically reduced, and hence geometrically integral. The assertion about finite products now follows inductively from Theorem 21. □

Symmetric Powers of Torsors

Let G be a finite abelian group, let $\mathcal{P}$ be a G-torsor over an S-scheme X in $S_{\text{Ét}}$.

Proposition 24 ([[SGA1]], IX.5.8)**.** *Assume that $\mathcal{P}$ and X are endowed with right actions from a finite group Γ such that the morphism $\mathcal{P} \to X$ is Γ-equivariant, and that the following properties hold:*

(a) The right Γ-action on $\mathcal{P}$ commutes with the left G-action.

(b) The right Γ-action on X is admissible, and the quotient morphism $X \to X/\Gamma$ is finite.

(c) For any geometric point $\bar{x}$ of X, the action of the stabilizer $\Gamma_{\bar{x}}$ of $\bar{x}$ in Γ on the fiber $\mathcal{P}_{\bar{x}}$ of $\mathcal{P}$ at $\bar{x}$ is trivial.

Then the action of Γ on $\mathcal{P}$ is admissible, and $\mathcal{P}/\Gamma$ is a G-torsor over X/Γ in $S_{Ét}$.

By Theorem 11, $\mathcal{P}$ is representable by a finite étale X-scheme. For each $i \in [\![1,d]\!]$ let $p_i : X^{\times_S d} \to X$ be the projection on i-th factor, and let us consider the G-torsor

$$p_1^{-1}\mathcal{P} \otimes^G \cdots \otimes^G p_d^{-1}\mathcal{P} = K_G \backslash \mathcal{P}^{\times_S d}$$

over $X^{\times_S d}$, where $K_G \subseteq G^d$ is the kernel of the multiplication morphism $G^d \to G$ (the dependence on d is suppressed in the notation). Equivalently, this is the contraction of the G^d-torsor $p_1^{-1}\mathcal{P} \times \cdots \times p_d^{-1}\mathcal{P}$ along the multiplication map $m : G^d \to G$. Under the identification $G^d/K_G \simeq G$ induced by m, the residual G-action is given by

$$g \cdot [(u_1,\dots,u_d)] = [(g_1 u_1,\dots,g_d u_d)],$$

where $(g_1,\dots,g_d)$ is any lift of g, so that $g_1 \cdots g_d = g$. This is independent of the chosen lift because any two lifts differ by an element of K_G. The object $K_G \backslash \mathcal{P}^{\times_S d}$ of $S_{\text{Ét}}$ is too representable by an

S-scheme which is finite étale over $X^{\times_S d}$. The group S_d acts on the right on $K_G\backslash\mathcal{P}^{\times_S d}$ by the formula

$$(p_i)_{i\in[\![1,d]\!]}\cdot\sigma=(p_{\sigma(i)})_{i\in[\![1,d]\!]}.$$

This action of S_d commutes with the left action of G on $K_G\backslash\mathcal{P}^{\times_S d}$.

Proposition 25 ([Gui19] Proposition 2.32.)**.** *If X is Zariski-locally quasi-projective on S, then the right action of S_d on $K_G\backslash\mathcal{P}^{\times_S d}$ is admissible, so that the quotient $\mathcal{P}^{[d]}$ of $K_G\backslash\mathcal{P}^{\times_S d}$ by S_d exists as a scheme over S. Moreover, the canonical morphism $\mathcal{P}^{[d]}\to\mathrm{Sym}^d_S(X)$ is a G-torsor, and the morphism*

$$p_1^{-1}\mathcal{P}\otimes^G\cdots\otimes^G p_d^{-1}\mathcal{P}\to r^{-1}\mathcal{P}^{[d]}$$

where $r:X^{\times_S d}\to\mathrm{Sym}^d_S(X)$ is the canonical projection, is an isomorphism of G-torsors over $X^{\times_S d}$.

Definition 26 (Symmetric powers of rank-one local systems)**.** Let Λ be a finite commutative ring, let $G=\Lambda^\times$, and let $\mathcal{F}$ be a rank-one Λ-local system on X. If $\mathcal{P}=\mathrm{Fr}(\mathcal{F})$, we define

$$\mathcal{F}^{[d]}:=\mathcal{P}^{[d]}\times^G\Lambda$$

on $X^{(d)}$. Thus there is a canonical identification

$$\mathrm{Fr}\big(\mathcal{F}^{[d]}\big)\cong\mathcal{P}^{[d]}.$$

By Proposition 25, its pullback along $r:X^{\times_S d}\to X^{(d)}$ satisfies

$$r^{-1}\mathcal{F}^{[d]}\cong p_1^{-1}\mathcal{F}\otimes_\Lambda\cdots\otimes_\Lambda p_d^{-1}\mathcal{F}.$$

Thus this definition agrees with the usual descent construction of the symmetric power of a rank-one local system.

Proposition 27 (Semidirect-product description)**.** *Let $\Xi_G:=K_G\rtimes S_d$, where S_d acts on K_G by permuting coordinates. Then Ξ_G acts on $\mathcal{P}^{\times_S d}$ by*

$$((g_1,\dots,g_d),\sigma)\cdot(p_1,\dots,p_d)=(g_1p_{\sigma^{-1}(1)},\dots,g_dp_{\sigma^{-1}(d)}),$$

and there is a canonical isomorphism

$$\mathcal{P}^{[d]}\cong\Xi_G\backslash\mathcal{P}^{\times_S d}.$$

Moreover, the formation of $\mathcal{P}^{[d]}$ commutes with flat base change on S. If, in addition, X is flat over S, then it commutes with arbitrary base change; see [Gui19, Proposition 2.33].

Proof. Because G is abelian, the condition $\prod_{i=1}^d g_i=1$ defining K_G is invariant under permutation of the coordinates. Thus the S_d-action normalizes K_G and defines the semidirect product Ξ_G. By Proposition 25,

$$\mathcal{P}^{[d]}=(K_G\backslash\mathcal{P}^{\times_S d})/S_d.$$

The universal properties of the two quotients identify this iterated quotient canonically with $\Xi_G\backslash\mathcal{P}^{\times_S d}$. Formation of the torsor quotient $K_G\backslash\mathcal{P}^{\times_S d}$ commutes with base change, and the final admissible S_d-quotient commutes with flat base change by [SGA1, p. V.1.9]. If X is flat over S, compatibility with arbitrary base change is [Gui19, Proposition 2.33]. □

Additivity of Symmetric Powers

For $d \geq 0$, write

$$X_S^d := \underbrace{X \times_S \cdots \times_S X}_{d \text{ factors}}, \qquad X_S^0 := S.$$

The construction of symmetric powers is compatible with a natural addition law. For any integers $d_1, d_2 \geq 0$, concatenating the factors gives a canonical isomorphism

$$c_{d_1,d_2} : X_S^{d_1} \times_S X_S^{d_2} \xrightarrow{\sim} X_S^{d_1+d_2}.$$

It is equivariant for the inclusion $S_{d_1} \times S_{d_2} \hookrightarrow S_{d_1+d_2}$ and therefore induces the canonical **addition morphism**

$$+_{d_1,d_2} : X^{(d_1)} \times_S X^{(d_2)} \longrightarrow X^{(d_1+d_2)}.$$

Proposition 28. *Let $\mathcal{P}$ be a G-torsor over X. There is a canonical isomorphism of G-torsors over $X^{(d_1)} \times_S X^{(d_2)}$:*

$$+_{d_1,d_2}^{-1}\left(\mathcal{P}^{[d_1+d_2]}\right) \cong r_1^{-1}\mathcal{P}^{[d_1]} \otimes^G r_2^{-1}\mathcal{P}^{[d_2]}.$$

Proof. Let $\pi = r_{d_1} \times r_{d_2}$. By the commutativity of the following diagram

$$\begin{array}{ccc} X_S^{d_1} \times_S X_S^{d_2} & \xrightarrow[\sim]{c_{d_1,d_2}} & X_S^{d_1+d_2} \\ \downarrow{\scriptstyle \pi} & & \downarrow{\scriptstyle r_{d_1+d_2}} \\ X^{(d_1)} \times_S X^{(d_2)} & \xrightarrow{+_{d_1,d_2}} & X^{(d_1+d_2)} \end{array}$$

and applying Proposition 25, we obtain canonical isomorphisms:

$$\pi^{-1}\left(+_{d_1,d_2}^{-1}\mathcal{P}^{[d_1+d_2]}\right) \cong r_{d_1+d_2}^{-1}\mathcal{P}^{[d_1+d_2]} \cong \bigotimes_{i=1}^{d_1+d_2,\,G} p_i^{-1}\mathcal{P}$$

and

$$\pi^{-1}(r_1^{-1}\mathcal{P}^{[d_1]} \otimes^G r_2^{-1}\mathcal{P}^{[d_2]}) \cong \left(\bigotimes_{i=1}^{d_1,\,G} p_i^{-1}\mathcal{P}\right) \otimes^G \left(\bigotimes_{j=d_1+1}^{d_1+d_2,\,G} p_j^{-1}\mathcal{P}\right).$$

Since both pullbacks identify with

$$\bigotimes_{k=1}^{d_1+d_2,\,G} p_k^{-1}\mathcal{P}$$

in an $S_{d_1} \times S_{d_2}$-equivariant manner, the isomorphism descends to the quotient $X^{(d_1)} \times_S X^{(d_2)}$ by Proposition 20. □

1.4 Basic Geometric Lemmas

We conclude with two elementary facts about codimension-one points and blowups. They are placed here because they enter the geometric arguments in both later chapters.

Proposition 29. *Let $f : X \to Y$ be a finite flat morphism between integral schemes of finite type over a field k. If $Z \subset X$ is a prime divisor with generic point η_Z, then $f(Z) \subset Y$ is a prime divisor, and its generic point is $f(\eta_Z)$.*

Proof. f is finite hence proper hence closed so $f(Z)$ is closed subset of Y, it is irreducible as the image of an irreducible. Since $Z = \overline{\{\eta_Z\}}$ we get:

$$\{f(\eta_Z)\} \subset f(Z) = f(\overline{\{\eta_Z\}}) \subseteq \overline{f(\{\eta_Z\})} = \overline{\{f(\eta_Z)\}}$$

And since $f(Z)$ is closed we get $f(Z) = \overline{\{f(\eta_Z)\}}$.

For flat map of integral schemes we have for every $x \in X$, $y = f(x)$ the dimension formula:

$$\dim(\mathcal{O}_{X,x}) = \dim(\mathcal{O}_{Y,y}) + \dim(\mathcal{O}_{X_y,x})$$

And since $\dim(\mathcal{O}_{X_y,x}) = 0$ we get $\dim(\mathcal{O}_{X,x}) = \dim(\mathcal{O}_{Y,y})$ concluding that $f(Z)$ is a prime divisor as well. □

Blowups

Theorem 30 ([Stacks, Tag 0805])**.** *Let $X_1 \to X_2$ be a flat morphism of schemes, let $Z_2 \subset X_2$ be a closed subscheme, and let Z_1 be its inverse image in X_1. If X_i' is the blowup of X_i along Z_i, then the diagram*

$$\begin{array}{ccc} X_1' & \longrightarrow & X_2' \\ \downarrow & & \downarrow \\ X_1 & \longrightarrow & X_2 \end{array}$$

is cartesian.

Proposition 31. *Let X be a smooth k-scheme of pure dimension $d \geq 1$, let $z \in X(k)$, and let*

$$\pi : \widetilde{X} = \mathrm{Bl}_z(X) \longrightarrow X$$

be the blowup at z. Then the exceptional divisor $E = \pi^{-1}(z)$ is isomorphic to $\mathbb{P}_k^{d-1}$. In particular, E is irreducible and has codimension 1 in $\widetilde{X}$. Consequently, it has a unique generic point η_E, and $\overline{\{\eta_E\}} = E$.

Proof. Because X is smooth of dimension d at the k-rational point z, after replacing X by a sufficiently small open neighborhood of z there is an étale morphism

$$f : X \longrightarrow \mathbb{A}_k^d$$

that sends z to the origin 0. We may shrink this neighborhood further so that z is the scheme-theoretic inverse image of 0. Since an étale morphism is flat, Theorem 30 reduces the assertion to the blowup of $\mathbb{A}_k^d$ at the origin.

Write $x_1, \ldots, x_d$ for the affine coordinates on $\mathbb{A}_k^d$ and $[u_1 : \cdots : u_d]$ for the homogeneous coordinates on $\mathbb{P}_k^{d-1}$. The blowup of the origin is the closed subscheme

$$\mathrm{Bl}_0(\mathbb{A}_k^d) = \left\{((x_1, \ldots, x_d), [u_1 : \cdots : u_d]) \;\middle|\; x_i u_j = x_j u_i \text{ for all } i, j\right\} \subset \mathbb{A}_k^d \times_k \mathbb{P}_k^{d-1}.$$

Over the origin all the x_i vanish, so the displayed equations impose no condition on $[u_1 : \cdots : u_d]$. Hence the exceptional fiber is

$$\{0\} \times_k \mathbb{P}^{d-1}_k \cong \mathbb{P}^{d-1}_k.$$

It is therefore irreducible of dimension $d-1$. The blowup has dimension d (for example, each standard chart $u_i \neq 0$ has d affine coordinates), so the exceptional divisor has codimension 1. □

Theorem 32 ([Stacks, Tag 01OF], Lemma 31.33.9)**.** *If X is integral and $Z \subsetneq X$ is a proper closed subscheme, then the blowup $\mathrm{Bl}_Z(X)$ is integral.*

Chapter 2

Ramification After Blowup Is Tame

Throughout this chapter, let C be a smooth, projective, geometrically connected curve over k. Fix a modulus

$$\mathfrak{m} = \sum_i n_i P_i$$

on C, and set $U = C \setminus \mathfrak{m}$.

For each single-point submodulus $\mathfrak{n} = n_i P_i$ of $\mathfrak{m}$, set $d_\mathfrak{n} = \deg(\mathfrak{n})$. The effective divisor $\mathfrak{n}$ determines a k-rational closed point

$$Z_\mathfrak{n} \hookrightarrow C^{(d_\mathfrak{n})} = \mathrm{Sym}_k^{d_\mathfrak{n}}(C).$$

Let

$$\pi_\mathfrak{n} \colon X_\mathfrak{n} := \mathrm{Bl}_{Z_\mathfrak{n}}\big(C^{(d_\mathfrak{n})}\big) \longrightarrow C^{(d_\mathfrak{n})}$$

be the blowup at this point. We denote its exceptional divisor by

$$E_\mathfrak{n} := Z_\mathfrak{n} \times_{C^{(d_\mathfrak{n})}} X_\mathfrak{n}$$

By Propositions 22 and 31, $E_\mathfrak{n} \cong \mathbb{P}_k^{d_\mathfrak{n}-1}$; in particular, it is irreducible and has codimension 1 in $X_\mathfrak{n}$. We denote its unique generic point by $\eta_\mathfrak{n}$, so that $\overline{\{\eta_\mathfrak{n}\}} = E_\mathfrak{n}$. Thus we have the Cartesian diagram

$$\begin{array}{ccc} E_\mathfrak{n} = \overline{\{\eta_\mathfrak{n}\}} & \hookrightarrow & X_\mathfrak{n} \\ \downarrow & & \downarrow{\scriptstyle \pi_\mathfrak{n}} \\ Z_\mathfrak{n} & \hookrightarrow & C^{(d_\mathfrak{n})}. \end{array} \tag{2.1}$$

The purpose of this chapter is to prove Theorem 3. Let $\mathcal{F}$ be the rank-one Λ-local system

appearing there and put $\mathcal{P} = \mathrm{Fr}(\mathcal{F})$. The theorem becomes the following torsor statement.[1]

Theorem 33. *Assume that k is algebraically closed. Let G be a finite abelian group, and let $\mathcal{P} \to U$ be a G-torsor whose ramification is bounded by $\mathfrak{m}$. For every single-point submodulus $\mathfrak{n} = n_i P_i$ of $\mathfrak{m}$, the symmetric-power torsor $\mathcal{P}^{[d_{\mathfrak{n}}]}$ is tamely ramified at $\eta_{\mathfrak{n}}$.*

After recalling the necessary local ramification theory, we make two reductions. First, the general finite abelian case is reduced to cyclic groups. Second, formal–local invariance reduces the geometric calculation on C to a calculation on $\mathbb{G}_m \subset \mathbb{P}^1$. These reductions motivate the Kummer, Artin–Schreier, and Artin–Schreier–Witt theories used in the cyclic calculations that follow.

2.1 Ramification of Discrete Valuations

This section reviews the ramification of discrete valuations, beginning with the general theory of *discrete valuation rings* and their integral closures in finite separable extensions. We then specialize to complete rings in Galois extensions to describe the ramification filtration via lower and upper numbering, following the treatments in [Stacks, Tag 0EXQ] and [Ser79].

Definition 34 ([Stacks, Tag 09E4])**.** We say that $A \to B$ or $A \subset B$ is an extension of discrete valuation rings if A and B are discrete valuation rings and $A \to B$ is injective and local. In particular, if π_A and π_B are uniformizers of A and B, then $\pi_A = u\pi_B^e$ for some $e \geq 1$ and unit u of B. The integer e does not depend on the choice of the uniformizers as it is also the unique integer ≥ 1 such that

$$\mathfrak{m}_A B = \mathfrak{m}_B^e$$

The integer e is called the *ramification index* of B over A. We say that B is *weakly unramified* over A if $e = 1$. If the extension of residue fields $\kappa_A = A/\mathfrak{m}_A \subset \kappa_B = B/\mathfrak{m}_B$ is finite, then we set $f = [\kappa_B : \kappa_A]$ and we call it the *residual degree* or residue degree of the extension $A \subset B$.

Note that we do not require the extension of fraction fields to be finite.

Now let A be a discrete valuation ring with fraction field K, let L/K be a finite separable field extension and let $B \subset L$ be the integral closure of A in L. Then the ring extension $A \subset B$ is finite, hence B is Noetherian. The dimension of B is 1, hence B is a Dedekind domain. Let $\mathfrak{m}_1, \ldots, \mathfrak{m}_n$ be the maximal ideals of B (i.e., the primes lying over $\mathfrak{m}_A$). We obtain extensions of discrete valuation rings

$$A \subset B_{\mathfrak{m}_i}$$

and hence ramification indices e_i and residue degrees f_i. We have

$$[L : K] = \sum\nolimits_{i=1,\ldots,n} e_i f_i$$

Note that if A is henselian (e.g. A is complete) then $n = 1$.

Definition 35 ([Stacks, Tag 09E9])**.** Let A be a discrete valuation ring with fraction field K. Let L/K be a finite separable extension. With B and $\mathfrak{m}_i$, $i = 1, \ldots, n$ as above, we say the extension L/K is

[1] Here one uses $\mathrm{Fr}(\mathcal{F}^{[d_{\mathfrak{n}}]}) \cong \mathcal{P}^{[d_{\mathfrak{n}}]}$ and the fact that a local system and its frame torsor have the same inertia character, as recorded in a footnote in Chapter 1.

1. unramified with respect to A if $e_i = 1$ and the extension $\kappa(\mathfrak{m}_i)/\kappa_A$ is separable for all i,
2. tamely ramified with respect to A if, for every i, the residue field extension $\kappa(\mathfrak{m}_i)/\kappa_A$ is separable and the ramification index e_i is prime to the residue characteristic. Thus, when $\mathrm{char}(\kappa_A) = 0$, the condition on e_i is vacuous (and the residue field extensions are automatically separable), whereas when $\mathrm{char}(\kappa_A) = p > 0$, each e_i must be prime to p, and
3. totally ramified with respect to A if $n = 1$ and the residue field extension $\kappa(\mathfrak{m}_1)/\kappa_A$ is trivial.

If the discrete valuation ring A is clear from context, then we sometimes say L/K is unramified, totally ramified, or tamely ramified for short.

We will use two standard stability properties of tame ramification. First, ramification indices and residue-field extensions are unchanged by passage to the completion; in particular, tameness can be checked after completion [Ser79, Chapter II, §2]. Second, finite tame extensions of a complete discretely valued field are closed under taking Galois closures, finite composita, and subextensions [Ser79, Chapter IV, §2].

For the remainder of this section, assume that A is a discrete valuation ring of equal characteristic $p > 0$, complete with respect to its maximal-ideal-adic topology (cf. [Stacks, Tag 00IE] and [Stacks, Tag 0324]), with uniformizer π and perfect residue field κ. Thus, $\mathrm{char}(A) = \mathrm{char}(\kappa) = p$. The Cohen structure theorem ([Stacks, Tag 032A]; see also [Stacks, Tag 0C0S]) provides a coefficient field $k \subset A$ mapping isomorphically onto κ. Consequently, $A \cong k[[\pi]] \cong k[[t]]$, and its fraction field is $K \cong k((\pi))$. Let L/K be a finite Galois extension with Galois group G. The integral closure B of A in L is itself a complete discrete valuation ring. Since the residue field κ is perfect, the extension of residue fields is separable, and there exists a uniformizer $\Pi \in B$ such that $B = A[\Pi]$.

The *lower numbering ramification filtration* of G, denoted by $(G_i)_{i \geq -1}$, is defined as:

$$G_i = \{\sigma \in G \mid v_B(\sigma(x) - x) \geq i + 1 \text{ for all } x \in B\}$$

where v_B is the valuation on L associated with B. In this indexing, $G_{-1} = G$ and G_0 corresponds to the *inertia group* of the extension L/K. Note that L/K is unramified if and only if $G_0 = \{1\}$, and it is tamely ramified if and only if $G_1 = \{1\}$.

A standard result in the theory of local fields (complete local fields with a discrete valuation and perfect residue field) ensures that the condition in the definition of G_i need only be checked for the uniformizer Π of B. Specifically, if we define the function

$$i_K^L(\sigma) = v_B(\sigma(\Pi) - \Pi)$$

for $\sigma \in G$, then $G_i = \{\sigma \in G \mid i_K^L(\sigma) \geq i + 1\}$. The subgroups G_i are normal in G and become trivial for sufficiently large i.

Consider a tower of fields $K \subset E \subset L$, and let $H = \mathrm{Gal}(L/E) \subset G$. The lower numbering is compatible with subgroups:

$$G_i \cap H = H_i \quad \text{for all } i \geq -1$$

which follows from the identity $i_E^L = i_K^L|_H$. However, the lower numbering does not behave as simply with respect to quotients. If H is normal in G, the image of G_i in G/H is generally not $(G/H)_i$. Instead, we have $G_iH/H = (G/H)_j$, where the index j is given by the formula:

$$j = \frac{1}{e_{L/E}} \sum_{\tau \in H} \min(i_K^L(\tau), i + 1) - 1$$

In the literature, one reindexes the ramification groups by defining the Herbrand function $\phi_{L/K} : [-1, \infty) \to [-1, \infty)$:

$$\phi_{L/K}(i) = \frac{1}{e_{L/K}} \sum_{\sigma \in G} \min(i_K^L(\sigma), i+1) - 1 = \int_0^i \frac{1}{[G_0 : G_t]} dt$$

This function is continuous, strictly increasing, and piecewise linear, making it a bijection. It satisfies the composition law $\phi_{L/K} = \phi_{E/K} \circ \phi_{L/E}$ for a tower of extensions $K \subset E \subset L$. Using this function, we define the *upper numbering ramification groups* as:

$$G^i = G_{\phi_{L/K}^{-1}(i)}$$

The primary advantage of this reindexing is its compatibility with quotients: for any normal subgroup $H \subset G$, we have:

$$G^i H/H = (G/H)^i$$

for all $i \geq -1$.

We record, for orientation, the structure of the inertia group and of the graded pieces of the filtration; these facts are not used in the sequel, but they explain the dichotomy — prime-to-p versus p-power — that shapes the whole chapter.

Proposition 36 ([Ser79], IV Corollary 3)**.** *The quotients G_i/G_{i+1}, $i \geq 1$, are abelian groups, and are direct products of cyclic groups of order p. The group G_1 is a p-group.*

Proposition 37 ([Ser79], IV Corollary 4)**.** *The inertia group G_0 has the following property: It is the semi-direct product of a cyclic group of order prime to p with a normal subgroup whose order is a power of p.*

2.2 Ramification of G-Torsors

Generally speaking, assume we are given a locally Noetherian scheme X, a dense open $U \subset X$, a finite étale morphism $f : Y \to U$ and a prime divisor $Z \subset X$ such that $Z \cap U = \emptyset$ and the local ring $\mathcal{O}_{X,\xi}$ of X at the generic point ξ of Z is a discrete valuation ring. Setting $K_\xi = \mathrm{Frac}(\mathcal{O}_{X,\xi})$ we obtain a cartesian square

$$\begin{array}{ccc} \mathrm{Spec}(K_\xi) & \longrightarrow & U \\ \downarrow & & \downarrow \\ \mathrm{Spec}(\mathcal{O}_{X,\xi}) & \longrightarrow & X \end{array}$$

of schemes. In particular, $Y \times_U \mathrm{Spec}(K_\xi)$ is the spectrum of a finite étale K_ξ-algebra L_ξ. Write its canonical decomposition into fields as

$$L_\xi \cong \prod_{j=1}^{r} L_{\xi,j},$$

where each $L_{\xi,j}/K_\xi$ is a finite separable field extension.

Definition 38. We say:

1. Y is *unramified* over X at (Z, ξ) resp. Y is *tamely ramified* over X at (Z, ξ) if every field extension $L_{\xi,j}/K_\xi$ is unramified, resp. tamely ramified, with respect to $\mathcal{O}_{X,\xi}$ (Definition 35)

2. Y is unramified over X in codimension 1, resp. Y is tamely ramified over X in codimension 1, if the preceding condition holds for every (Z, ξ) as above.

Now back to our original assumptions, C is a projective smooth geometrically connected curve over a field k and $U = C \setminus \mathfrak{m}$. Every (Z, ξ) is now a closed point P and $\mathcal{O}_{C,P}$ is a discrete valuation ring. Assume $f : Y \to U$ is actually a G-torsor $\mathcal{P} \to U$ in $U_{\text{Ét}}$ for G a finite abelian group.

By passing to the completion $\widehat{\mathcal{O}}_{C,P}$, we obtain the complete local field $\widehat{K}_P$. Restricting the torsor $\mathcal{P}$ to the punctured formal neighborhood $\mathrm{Spec}(\widehat{K}_P)$ allows us to study the ramification in a strictly local context. This restriction is a finite étale scheme over the field $\widehat{K}_P$ and therefore, by [Stacks, Tag 02GL], decomposes into a finite disjoint union of spectra of finite separable field extensions:

$$\mathcal{P}|_{\mathrm{Spec}(\widehat{K}_P)} \cong \bigsqcup_{i=1}^{N} \mathrm{Spec}(F_i).$$

These extensions have the following properties:

1. The extensions $F_i/\widehat{K}_P$ are mutually isomorphic and Galois. Since $\widehat{K}_P$ is complete, every F_i is complete for the unique extension of the discrete valuation on $\widehat{K}_P$ [Ser79, Chapter II, §2, Proposition 3].
2. If F denotes any one of the fields F_i, then the Galois group $H = \mathrm{Gal}(F/\widehat{K}_P)$ identifies with the stabilizer of the corresponding connected component of the pullback $\mathcal{P}|_{\mathrm{Spec}(\widehat{K}_P)}$, which is a subgroup of G.
3. The number of components is $N = [G : H]$.

Definition 39. We say that the extension $F/\widehat{K}_P$ *has ramification bounded by* r if the ramification group H^r is trivial. Here H^r denotes the upper-numbering ramification group of $H = \mathrm{Gal}(F/\widehat{K}_P)$ associated with the extension $F/\widehat{K}_P$; it is not a filtration inherited from the ambient group G. This condition is independent of the chosen connected component of $\mathcal{P}|_{\mathrm{Spec}(\widehat{K}_P)}$. We say $\mathcal{P}$ *has ramification bounded by* r *at* P if the corresponding local field extensions $F/\widehat{K}_P$ satisfy this condition.

Definition 40. Let $\mathfrak{m} = \sum n_i P_i$ be a modulus on C. We say the G-torsor $\mathcal{P} \to U$ has **ramification bounded by** $\mathfrak{m}$ if, for each point P_i, the local restriction $\mathcal{P}|_{\mathrm{Spec}(\widehat{K}_{P_i})}$ has ramification bounded by the coefficient n_i.

Fix an algebraic closure $\bar{k}$ of k and write $\bar{s} = \mathrm{Spec}(\bar{k}) \to \mathrm{Spec}(k)$ for the corresponding geometric point.

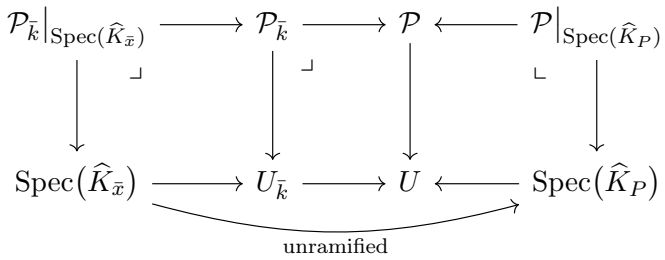
Definition 41 ([Gui19, Definition 5.2])**.** Let $\mathfrak{m} = \sum n_i P_i$ be a modulus on C. The G-torsor $\mathcal{P} \to U$ has **ramification bounded by** $\mathfrak{m}$ if, for every geometric point $\bar{x}$ of $\mathfrak{m}$ with image $\bar{s}$ in $\mathrm{Spec}(k)$, the restriction of $\mathcal{P}_{\bar{k}}$ to the punctured formal neighborhood

$$\mathrm{Spec}\big(\widehat{\mathcal{O}}_{C_{\bar{k}},\bar{x}}\big) \times_{C_{\bar{k}}} U_{\bar{k}}$$

has ramification bounded by the multiplicity of $\mathfrak{m}_{\bar{k}}$ at $\bar{x}$ in the sense of Definition 39.

To compare the two definitions, fix a point $P = P_i$ in the support of $\mathfrak{m}$, let $\bar{x}$ be a geometric point of $C_{\bar{k}}$ lying over P, and abbreviate $\widehat{K}_{\bar{x}} := \mathrm{Frac}\big(\widehat{\mathcal{O}}_{C_{\bar{k}},\bar{x}}\big)$. Since $\widehat{\mathcal{O}}_{C_{\bar{k}},\bar{x}}$ is a complete discrete valuation ring whose closed point lies over $\bar{x} \notin U_{\bar{k}}$ while its generic point lies over the generic point of $C_{\bar{k}}$, the punctured formal neighborhood of Definition 41 is $\mathrm{Spec}(\widehat{\mathcal{O}}_{C_{\bar{k}},\bar{x}}) \times_{C_{\bar{k}}} U_{\bar{k}} \cong \mathrm{Spec}(\widehat{K}_{\bar{x}})$. The local

restriction of Definition 40 (right-hand column) and that of Definition 41 (left-hand column) then fit into a single commutative diagram



in which every square is cartesian. Since cartesian squares compose, the outer rectangle exhibits

$$\mathcal{P}_{\bar{k}}|_{\mathrm{Spec}(\widehat{K}_{\bar{x}})} \cong \mathcal{P}|_{\mathrm{Spec}(\widehat{K}_P)} \times_{\mathrm{Spec}(\widehat{K}_P)} \mathrm{Spec}(\widehat{K}_{\bar{x}}), \tag{2.2}$$

i.e. the geometric restriction is the base change of the closed-point restriction along the bottom bent arrow, which is an unramified extension of complete discretely valued fields. This is the content of the following proposition.

Proposition 42. *The closed-point formulation in Definition 40 and the geometric formulation in Definition 41 are equivalent.*

Proof. Let $P = P_i$, let $\bar{x}$ lie over P, and choose a uniformizer t at P. Because k is perfect, the base change $C_{\bar{k}} \to C$ is ind-étale, so the pullback of t is again a uniformizer at $\bar{x}$. Since $C_{\bar{k}}$ is a smooth curve over the algebraically closed field $\bar{k}$, it follows that $\widehat{\mathcal{O}}_{C_{\bar{k}},\bar{x}} \cong \bar{k}[[t]]$. Thus

$$\widehat{K}_P \cong \kappa(P)((t)) \longrightarrow \widehat{K}_{\bar{x}} \cong \bar{k}((t))$$

is unramified, as claimed in the diagram.

By (2.2), the geometric restriction is obtained from the closed-point restriction by this unramified base change. Ramification bounds are unchanged under unramified extensions of complete discretely valued fields, so the two restrictions have the same ramification bound.

Finally, as $\kappa(P)/k$ is separable the fiber of P_i in $C_{\bar{k}}$ is reduced, so $\mathfrak{m}_{\bar{k}}$ has multiplicity n_i at every $\bar{x}$ above P_i. Thus the geometric condition at $\bar{x}$ is precisely the closed-point condition at P_i. □

Furthermore, the notions of tame ramification and unramifiedness derived here coincide with the codimension-1 criteria in Definition 38.

There is another equivalent formulation through the language of characters. The group G, being a finite abelian group over k, is étale. Consequently, the correspondence between G-torsors and the fundamental group allows us to describe G-torsors via characters. Specifically, there is an isomorphism (Corollary 14):

$$\mathrm{Hom}_{\mathrm{cont.}}(\pi_1^{\text{ét}}(U, \overline{x}), G) \xrightarrow{\cong} \mathrm{Tors}(U_{\text{ét}}, G)$$

In our local setting, where we consider the restriction to the complete local field $\widehat{K}_P$, the fundamental group identifies with the absolute Galois group $G_{\widehat{K}_P} := \mathrm{Gal}(\widehat{K}_P^{\mathrm{sep}}/\widehat{K}_P)$. Thus, the restricted G-torsor $\mathcal{P}|_{\mathrm{Spec}(\widehat{K}_P)}$ corresponds to a unique continuous homomorphism:

$$\rho : G_{\widehat{K}_P} \to G$$

Under this correspondence, the torsor $\mathcal{P}$ has ramification bounded by r at P if and only if the homomorphism ρ trivializes the r-th ramification group in the upper numbering:

$$\rho(G^r_{\widehat{K}_P}) = \{1\}$$

Basic Properties of Ramification of G-Torsors

We now prove some elementary lemmas regarding the ramification of G-torsors.

Lemma 43. *Let G be a finite abelian group and X be a locally Noetherian scheme over a field k. Let $U \subset X$ be a dense open subset and let (Z, ξ) be a prime divisor in the complement $X \setminus U$. If $\mathcal{P}_1$ and $\mathcal{P}_2$ are two G-torsors on $U_{\acute{e}t}$ that are tamely ramified at (Z, ξ), then their product $\mathcal{P}_1 \otimes^G \mathcal{P}_2$ is tamely ramified at (Z, ξ).*

Proof. Let $A = \mathcal{O}_{X,\xi}$ and $K = \mathrm{Frac}(A)$. Tameness may be checked after completing A, so we may assume that K is complete. For $i = 1, 2$, let $L_{i,j}/K$ be the field factors of

$$\mathcal{P}_i \times_U \mathrm{Spec}(K).$$

These extensions are tame by assumption. Let M/K be a finite Galois extension containing the Galois closures of all the $L_{i,j}/K$. Finite tame extensions of a complete discretely valued field are closed under Galois closure and finite compositum, so M/K is tame.

Both $\mathcal{P}_1$ and $\mathcal{P}_2$ become trivial over M. Hence their contracted product also becomes trivial over M. Consequently, every field factor of

$$(\mathcal{P}_1 \otimes^G \mathcal{P}_2) \times_U \mathrm{Spec}(K)$$

is isomorphic to an intermediate extension of M/K: indeed, after tensoring such a factor with M, one obtains a product having M as a factor, and hence a K-embedding of that field factor into M. Since subextensions of a finite tame extension are tame, every field factor is tame. The result follows from Definition 38. □

Lemma 44. *Let $A \to B$ be a local extension of discrete valuation rings. Assume that B/A is weakly unramified and that the residue-field extension κ_B/κ_A is separable. Write $K = \mathrm{Frac}(A)$ and $L = \mathrm{Frac}(B)$. Let $\mathcal{P}$ be a finite G-torsor over $\mathrm{Spec}(K)$, and put*

$$\mathcal{P}_L := \mathcal{P} \times_{\mathrm{Spec}(K)} \mathrm{Spec}(L).$$

Then $\mathcal{P}$ is unramified with respect to A if and only if $\mathcal{P}_L$ is unramified with respect to B. Likewise, $\mathcal{P}$ is tamely ramified with respect to A if and only if $\mathcal{P}_L$ is tamely ramified with respect to B.

Proof. Write the finite étale K-algebra underlying $\mathcal{P}$ as

$$\prod_i E_i,$$

where every E_i/K is finite separable. For every i, write

$$E_i \otimes_K L \cong \prod_j F_{i,j}$$

as a product of finite separable extensions of L. The standard base-change properties of extensions of discrete valuation rings give, for every i and j,

$$e(F_{i,j}/L) = e(E_i/K),$$

and show that $\kappa_{F_{i,j}}/\kappa_L$ is separable if and only if κ_{E_i}/κ_K is separable.

The two assertions now follow immediately from Definition 35: unramifiedness means residue separability and ramification index 1, while tameness means residue separability and ramification index prime to the residue characteristic. □

Corollary (Étale form of Lemma 44)**.** *Let $f : X \to Y$ be a morphism of locally Noetherian schemes. Let $U_X \subset X$ and $U_Y \subset Y$ be dense open subschemes such that $f^{-1}(U_Y) \subset U_X$. Let (Z_X, ξ_X) and (Z_Y, ξ_Y) be prime divisors contained in $X \setminus U_X$ and $Y \setminus U_Y$, respectively, and suppose that $f(\xi_X) = \xi_Y$. Assume that $\mathcal{O}_{X,\xi_X}$ and $\mathcal{O}_{Y,\xi_Y}$ are discrete valuation rings and that f is étale at ξ_X. If $\mathcal{P}$ is a finite G-torsor over U_Y, then $f^{-1}\mathcal{P}$ is unramified (resp. tamely ramified) at ξ_X if and only if $\mathcal{P}$ is unramified (resp. tamely ramified) at ξ_Y.*

Proof. Set $A = \mathcal{O}_{Y,\xi_Y}$ and $B = \mathcal{O}_{X,\xi_X}$. Since f is étale at ξ_X, the induced local homomorphism $A \to B$ is étale. In particular, $\mathfrak{m}_A B = \mathfrak{m}_B$, so B/A is weakly unramified, and the residue-field extension $\kappa(\xi_X)/\kappa(\xi_Y)$ is finite separable. The restriction of $f^{-1}\mathcal{P}$ to $\operatorname{Spec}(\operatorname{Frac}(B))$ is the base change of the restriction of $\mathcal{P}$ to $\operatorname{Spec}(\operatorname{Frac}(A))$. The two equivalences now follow from Lemma 44. □

Lemma 45. *Let G_1, G_2 be finite abelian groups. Let X be a locally Noetherian scheme over a field k, let $U \subset X$ be a dense open subscheme, and let (Z, ξ) be a prime divisor in $X \setminus U$. For $i = 1, 2$, let $\mathcal{P}_i$ be a G_i-torsor on $U_{\text{ét}}$. If both $\mathcal{P}_1$ and $\mathcal{P}_2$ are tamely ramified at (Z, ξ), then the $(G_1 \times G_2)$-torsor $\mathcal{P}_1 \times_U \mathcal{P}_2$ is tamely ramified at (Z, ξ).*

Proof. Let $A = \mathcal{O}_{X,\xi}$ and $K = \operatorname{Frac}(A)$. Tameness may be checked after completing A, so we may assume that K is complete. As in the proof of Lemma 43, there exists a finite tame Galois extension M/K that splits both $\mathcal{P}_1$ and $\mathcal{P}_2$. Their fiber product is therefore split over M. Consequently, every field factor of

$$(\mathcal{P}_1 \times_U \mathcal{P}_2) \times_U \operatorname{Spec}(K)$$

embeds into M. Since subextensions of a finite tame extension are tame, every such field factor is tame. The result follows from Definition 38. □

Lemma 46. *Let A be a discrete valuation ring with fraction field K, let $\iota : H \hookrightarrow G$ be an injective homomorphism of finite abelian groups, and let $\mathcal{Q}$ be an H-torsor over $\operatorname{Spec}(K)$. Let*

$$\iota_* \mathcal{Q} := (\mathcal{Q} \times G)/H$$

be the induced G-torsor, where $h \cdot (q, g) = (qh, \iota(h)^{-1} g)$. Then, as a K-scheme, $\iota_* \mathcal{Q}$ is noncanonically isomorphic to a disjoint union of $[G : H]$ copies of $\mathcal{Q}$. Consequently:

1. *$\iota_* \mathcal{Q}$ is tamely ramified with respect to A if and only if $\mathcal{Q}$ is;*
2. *$\iota_* \mathcal{Q}$ is unramified with respect to A if and only if $\mathcal{Q}$ is;*
3. *if A is complete and its residue field is perfect, then, for every r, $\iota_* \mathcal{Q}$ has ramification bounded by r if and only if $\mathcal{Q}$ does.*

Proof. As a K-scheme, $\iota_*\mathcal{Q}$ is noncanonically isomorphic to a disjoint union of $[G : H]$ copies of $\mathcal{Q}$. Thus it has the same field factors as $\mathcal{Q}$, each repeated $[G : H]$ times, and the assertions follow immediately from the definitions. □

2.3 Reduction to Cyclic Torsors on $\mathbb{P}^1$

Before introducing explicit equations, we explain the strategy of the proof and isolate two reductions that simplify the situation. The theorem concerns a torsor under an arbitrary finite abelian group on an arbitrary smooth curve, whereas the calculation we ultimately wish to perform is local and cyclic. We show that both the structure group and the geometry may be simplified without changing the ramification problem relevant to the exceptional divisor.

The first reduction is algebraic: using the decomposition of a finite abelian group into cyclic factors, together with the compatibility of the symmetric-power construction and ramification bounds with products and quotients, we reduce the theorem to cyclic groups of order prime to p and cyclic groups of order p^r. This explains why the three cyclic theories recalled in the next section suffice. The second reduction is geometric and formal–local: the ramification of the induced torsor at the generic point of the exceptional divisor depends only on the restriction of the original torsor to the punctured formal neighbourhood of the point supporting the modulus. It therefore tells us exactly what the cyclic theories must accomplish: realize the resulting local cyclic torsor by an explicit equation on $\mathbb{G}_m \subset \mathbb{P}^1_k$.

Together, these reductions transform the original problem into three parallel explicit calculations, corresponding to Kummer, Artin–Schreier, and Artin–Schreier–Witt theory. The following section recalls precisely the conclusions from these theories that provide the required realizations.

2.3.1 Reduction from Finite Abelian Groups to Cyclic Groups

Fix a single-point submodulus $\mathfrak{n}$ and write $d = \deg(\mathfrak{n})$. The first reduction concerns the structure group and is independent of the geometry of the curve.

Proposition 47 (Reduction to cyclic groups)**.** *Suppose that the conclusion of Theorem 33 is known whenever the structure group is cyclic of order prime to p, and whenever the structure group is a cyclic p-group. Then it holds for every finite abelian group G.*

Proof. By the structure theorem for finite abelian groups, choose a decomposition

$$G \cong \prod_{i=1}^{s} \mathbb{Z}/e_i\mathbb{Z} \times \prod_{j=1}^{t} \mathbb{Z}/p^{r_j}\mathbb{Z}, \qquad (e_i, p) = 1.$$

Let $\rho : \pi_1^{\text{ét}}(U, \bar{x}) \to G$ be the character corresponding to $\mathcal{P}$. Composing ρ with the projections to the displayed cyclic factors gives torsors $\mathcal{P}_i$ under $\mathbb{Z}/e_i\mathbb{Z}$ and torsors $\mathcal{Q}_j$ under $\mathbb{Z}/p^{r_j}\mathbb{Z}$. Since the character of each factor is a quotient of ρ, the ramification bound on $\mathcal{P}$ passes to every $\mathcal{P}_i$ and $\mathcal{Q}_j$.

The original torsor is recovered as their fiber product over U:

$$\mathcal{P} \cong \mathcal{P}_1 \times_U \cdots \times_U \mathcal{P}_s \times_U \mathcal{Q}_1 \times_U \cdots \times_U \mathcal{Q}_t.$$

The construction of the symmetric-power torsor commutes with this product. Indeed, after pullback to U^d, both sides are obtained by taking the contracted product of the d pullbacks along

multiplication in the structure group. Hence

$$\mathcal{P}^{[d]} \cong \mathcal{P}_1^{[d]} \times_{U^{(d)}} \cdots \times_{U^{(d)}} \mathcal{P}_s^{[d]} \times_{U^{(d)}} \mathcal{Q}_1^{[d]} \times_{U^{(d)}} \cdots \times_{U^{(d)}} \mathcal{Q}_t^{[d]}.$$

By the assumed cyclic cases, every factor on the right is tamely ramified at $\eta_{\mathfrak{n}}$. The fiber product is therefore tamely ramified there by Lemma 45. This proves the assertion for G. □

Thus it remains to understand cyclic groups. Their prime-to-p part is described by Kummer theory, while cyclic groups of order p and p^r are described by Artin–Schreier and Artin–Schreier–Witt theory, respectively.

2.3.2 Formal–Local Reduction from C to $\mathbb{P}^1$

We argue that it is enough to prove Theorem 33 for $C = \mathbb{P}^1_k$, $\mathfrak{n} = d \cdot 0$, and $\mathcal{P} \to \mathbb{G}_m = \mathbb{P}^1_k \setminus \{0, \infty\}$ a G-torsor given by an explicit global equation. Informally, this is because *the ramification of $\mathcal{P}^{[d]}$ at $\eta_{\mathfrak{n}}$ is a formal–local invariant of $\mathcal{P}$ at P*: it depends only on the completed local ring $\widehat{\mathcal{O}_{C,P}} \cong k[[x]]$ together with the restriction of $\mathcal{P}$ to the punctured formal disc $\mathrm{Spec}(k((x)))$, since all of the constructions involved — self products, quotients by finite groups, and blowups — commute with the flat base change to the completion; for blowups, this is Theorem 30. We emphasize that this is not a general principle that can simply be quoted: it is a statement about the particular constructions at hand, and it is proved by checking each construction separately. This is the content of Lemma 48 and Corollary 49 below. The second ingredient of the reduction is Lemma 54: every local torsor class appearing in Theorem 33 is realized by an explicit torsor over $\mathbb{G}_m \subset \mathbb{P}^1_k$.

Throughout this section we assume that k is algebraically closed; in particular every closed point of C is k-rational, and $\mathfrak{n} = dP$ with $P \in C(k)$ and $d = \deg \mathfrak{n}$.

Remark. The assumption that k is algebraically closed is harmless in our situation, for the following reasons.

1. **(Base change to $\bar{k}$.)** Assume k is perfect, so that $\bar{k} = k^{sep}$. The formation of $C^{(d)}$ (a quotient by a finite group), of the blowup $X_{\mathfrak{n}}$ along $Z_{\mathfrak{n}}$, and of $\mathcal{P}^{[d]}$ (Proposition 27) commutes with the flat base change $\mathrm{Spec}(\bar{k}) \to \mathrm{Spec}(k)$. Since $E_{\mathfrak{n}} \cong \mathbb{P}^{\deg \mathfrak{n}-1}_k$ is geometrically irreducible, $(E_{\mathfrak{n}})_{\bar{k}}$ is irreducible and its generic point $\bar{\eta}$ lies over $\eta_{\mathfrak{n}}$; moreover $\mathcal{O}_{X_{\mathfrak{n}},\eta_{\mathfrak{n}}} \to \mathcal{O}_{(X_{\mathfrak{n}})_{\bar{k}},\bar{\eta}}$ is a weakly unramified extension of discrete valuation rings with separable residue field extension. Therefore, by Lemma 44, $(\mathcal{P}_{\bar{k}})^{[d]} = (\mathcal{P}^{[d]})_{\bar{k}}$ is unramified (resp. tamely ramified) at $\bar{\eta}$ if and only if $\mathcal{P}^{[d]}$ is unramified (resp. tamely ramified) at $\eta_{\mathfrak{n}}$. Thus both properties may be checked after base change to $\bar{k}$.

2. **(Rationality of P — a caveat.)** If P is a closed point with $\kappa(P) \neq k$, then after base change to $\bar{k}$ the point $Z_{\mathfrak{n}}$ corresponds to the divisor $\sum_{\bar{P} \mapsto P} n\bar{P}$ on $C_{\bar{k}}$, whose support consists of $[\kappa(P) : k]$ distinct points, whereas the computation below treats a modulus supported at a single rational point. We therefore prove Theorem 33 only under the standing assumption that k is algebraically closed. This is the version used later: at the beginning of the proof of Theorem 63, all data are first base changed to $\bar{k}$. The modulus then decomposes into its rational-point-supported summands; the present theorem is applied to each summand, and Lemma 65 combines them before tameness is descended back to k.

We now formulate the formal–local invariance precisely. Fix $P \in C(k)$ lying in the support of $\mathfrak{m}$, set $\mathfrak{n} = dP$, and fix a uniformizer x of $\mathcal{O}_{C,P}$; it induces an isomorphism $\widehat{\mathcal{O}_{C,P}} \cong k[[x]]$. The scheme

$\mathrm{Spec}(k[[x]])$ has two points — the closed point, mapping to P, and the generic point, mapping to the generic point of C — so the punctured formal disc $\mathrm{Spec}(k((x)))$ maps to U, and we denote by

$$\mathcal{P}_x := \mathcal{P} \times_U \mathrm{Spec}(k((x)))$$

the corresponding restriction of $\mathcal{P}$; it is a G-torsor over $\mathrm{Spec}(k((x)))$. Let $e_1, \dots, e_d \in k[[x_1, \dots, x_d]]$ be the elementary symmetric polynomials in $x_1, \dots, x_d$, and set

$$V = \mathrm{Spec}\Big(k[[e_1, \dots, e_d]][e_d^{-1}]\Big), \qquad W = \mathrm{Spec}\Big(k[[x_1, \dots, x_d]][(x_1 \cdots x_d)^{-1}]\Big).$$

The inclusion $k[[e_1, \dots, e_d]] \subset k[[x_1, \dots, x_d]]$ induces a finite flat morphism $\hat{r} : W \to V$ (note that inverting $e_d = x_1 \cdots x_d$ inverts each x_i), and for each i we let

$$q_i : W \to \mathrm{Spec}(k((x))), \qquad x \mapsto x_i.$$

The group S_d acts on W over V by permuting $x_1, \dots, x_d$, and $q_{\sigma(i)} = q_i \circ \sigma$. Recall from Proposition 27 that

$$K_G := \ker\big(G^d \xrightarrow{m} G\big), \qquad \Xi_G := K_G \rtimes S_d,$$

where $m(g_1, \dots, g_d) = g_1 \cdots g_d$ and S_d acts on K_G by permuting coordinates. Thus quotienting by Ξ_G means first contracting the G^d-torsor along multiplication, by quotienting by K_G, and then descending from the ordered product to the symmetric product by quotienting by S_d.

Lemma 48 (Formal–local invariance)**.** *In the above notation:*

1. *The choice of x induces an isomorphism $\widehat{\mathcal{O}_{C^{(d)}, Z_{\mathfrak{n}}}} \cong k[[e_1, \dots, e_d]]$ carrying the maximal ideal to $(e_1, \dots, e_d)$, such that the induced morphism $c : \mathrm{Spec}(k[[e_1, \dots, e_d]]) \to C^{(d)}$ is flat and satisfies $c^{-1}(U^{(d)}) = V$; we write $c_V : V \to U^{(d)}$ for the restriction.*
2. *There is an induced isomorphism $\widehat{\mathcal{O}_{X_{\mathfrak{n}}, \eta_{\mathfrak{n}}}} \cong \kappa[[e_d]]$, where $\kappa = k(\frac{e_1}{e_d}, \dots, \frac{e_{d-1}}{e_d})$. In particular $\widehat{K}_{\eta_{\mathfrak{n}}} := \mathrm{Frac}(\widehat{\mathcal{O}_{X_{\mathfrak{n}}, \eta_{\mathfrak{n}}}}) \cong \kappa((e_d))$, and the canonical morphism $\mathrm{Spec}(\widehat{K}_{\eta_{\mathfrak{n}}}) \to U^{(d)}$ factors as $\mathrm{Spec}(\kappa((e_d))) \to V \xrightarrow{c_V} U^{(d)}$, where the first map does not depend on $(C, \mathcal{P})$.*
3. *There is a canonical isomorphism of G-torsors over V*

$$\mathcal{P}^{[d]} \times_{U^{(d)}} V \;\cong\; \Xi_G \backslash \Big(q_1^{-1}\mathcal{P}_x \times_W \cdots \times_W q_d^{-1}\mathcal{P}_x \Big).$$

Consequently, the isomorphism class of the G-torsor $\mathcal{P}^{[d]}|_{\mathrm{Spec}(\widehat{K}_{\eta_{\mathfrak{n}}})}$ depends only on d and on the isomorphism class of the G-torsor $\mathcal{P}_x$ over $k((x))$, and not otherwise on the pair $(C, \mathcal{P})$. In particular, whether $\mathcal{P}^{[d]}$ is tamely ramified or unramified at $\eta_{\mathfrak{n}}$ depends only on $(\mathcal{P}_x, d)$.

Proof. **(1)** The chosen parameter gives $\widehat{\mathcal{O}_{C^d, (P, \dots, P)}} \cong k[[x_1, \dots, x_d]]$. We use the standard facts that completion is compatible with finite morphisms and that the completed local ring of a finite quotient is the invariant ring of the completed local ring. Since $(P, \dots, P)$ is the unique point of C^d above dP, this gives

$$\widehat{\mathcal{O}_{C^{(d)}, dP}} \cong k[[x_1, \dots, x_d]]^{S_d} = k[[e_1, \dots, e_d]],$$

where the last equality is the formal-power-series version of the fundamental theorem of symmetric functions. The maximal ideal is thus $(e_1, \dots, e_d)$, and the resulting map c is flat by flatness of completion.

It remains only to identify the boundary. Near dP, the only component of $C^{(d)} \setminus U^{(d)}$ that occurs is the locus of divisors containing P. Its inverse image in the ordered formal neighbourhood is

$$\bigcup_i V(x_i) = V(x_1 \cdots x_d) = V(e_d).$$

Since the ordered-to-symmetric map is finite and surjective, the boundary in $\operatorname{Spec} k[[e_1, \ldots, e_d]]$ is also $V(e_d)$. Hence $c^{-1}(U^{(d)}) = V$ (and similarly the corresponding ordered inverse image is W).

(2) We use that blowups commute with flat base change. Thus the pullback of $X_{\mathfrak{n}}$ along c is the blowup of $\operatorname{Spec} k[[e_1, \ldots, e_d]]$ at its closed point. Completion preserves the exceptional divisor and its residue field, so it is enough to compute the completed local ring at the generic point of the exceptional divisor in this standard blowup.

On the chart where e_d is the exceptional parameter, put $t_i = e_i/e_d$ for $i < d$. The exceptional divisor is given by $e_d = 0$, and its generic point has residue field

$$\kappa = k(t_1, \ldots, t_{d-1}) = k\left(\frac{e_1}{e_d}, \ldots, \frac{e_{d-1}}{e_d}\right).$$

The usual local calculation for the blowup of a regular local scheme at its closed point therefore gives $\widehat{\mathcal{O}_{X_{\mathfrak{n}},\eta_{\mathfrak{n}}}} \cong \kappa[[e_d]]$ and $\widehat{K}_{\eta_{\mathfrak{n}}} \cong \kappa((e_d))$. Since e_d is invertible in the latter field, its map to the completed symmetric power factors through V, giving the asserted factorization through c_V.

(3) Let $h : \mathcal{P}^{\times_k d} \to U^{(d)}$ denote the canonical morphism; it is finite, being the composition of the finite étale morphism $\mathcal{P}^{\times_k d} \to U^d$ with the finite morphism $U^d \to U^{(d)}$. By Proposition 27, $\mathcal{P}^{[d]} = \Xi_G \backslash \mathcal{P}^{\times_k d} = \underline{\operatorname{Spec}}_{U^{(d)}}\left((h_* \mathcal{O}_{\mathcal{P}^{\times_k d}})^{\Xi_G}\right)$. Pushforward along the affine morphism h commutes with the flat base change $c_V : V \to U^{(d)}$, and the formation of Ξ_G-invariants is a kernel, hence also commutes with flat base change; therefore

$$\mathcal{P}^{[d]} \times_{U^{(d)}} V \cong \Xi_G \backslash \left(\mathcal{P}^{\times_k d} \times_{U^{(d)}} V\right).$$

Completing the finite map $C^d \to C^{(d)}$ at the unique point $(P, \ldots, P)$ above dP gives the Cartesian square defined by $k[[e_1, \ldots, e_d]] \subset k[[x_1, \ldots, x_d]]$. Restricting this square to $e_d = x_1 \cdots x_d \neq 0$ gives directly $U^d \times_{U^{(d)}} V \cong W$, and hence $\mathcal{P}^{\times_k d} \times_{U^{(d)}} V \cong \mathcal{P}^{\times_k d} \times_{U^d} W$.

The i-th component of the resulting map $W \to U^d$ is $W \xrightarrow{q_i} \operatorname{Spec}(k((x))) \to U$: indeed, it is induced in coordinates by $x \mapsto x_i$, and x_i is invertible on W. Since $\mathcal{P}^{\times_k d} = p_1^{-1}\mathcal{P} \times_{U^d} \cdots \times_{U^d} p_d^{-1}\mathcal{P}$, compatibility of fiber products with base change now gives

$$\mathcal{P}^{\times_k d} \times_{U^d} W \cong q_1^{-1}\mathcal{P}_x \times_W \cdots \times_W q_d^{-1}\mathcal{P}_x,$$

which is equivariant for the actions of G^d (in particular of K_G) and of S_d — the latter acting on W by permuting the x_i, compatibly with $q_{\sigma(i)} = q_i \circ \sigma$. Passing to Ξ_G-quotients yields the displayed isomorphism.

For the final statement: the schemes V and W, the maps q_i and $\hat{r}$, the Ξ_G-action, and the point $\operatorname{Spec}(\kappa((e_d))) \to V$ of (2) are all constructed from d alone; by (3), the torsor $\mathcal{P}^{[d]}|_{\operatorname{Spec}(\widehat{K}_{\eta_{\mathfrak{n}}})}$ is the pullback along $\operatorname{Spec}(\kappa((e_d))) \to V$ of a torsor constructed from $\mathcal{P}_x$ and these data. Since tame ramification and unramifiedness at $\eta_{\mathfrak{n}}$ are, by definition, determined by this restricted torsor together with the discrete valuation ring $\kappa[[e_d]]$ (Definition 38), both properties depend only on $(\mathcal{P}_x, d)$. □

Corollary 49 (Reduction to $\mathbb{P}^1$)**.** *Let C, C' be smooth projective geometrically connected curves over the algebraically closed field k, let $\mathcal{P} \to U = C \setminus \mathfrak{m}$ and $\mathcal{P}' \to U' = C' \setminus \mathfrak{m}'$ be G-torsors, let $P \in \mathrm{Supp}(\mathfrak{m})(k)$ and $P' \in \mathrm{Supp}(\mathfrak{m}')(k)$, and let x, x' be uniformizers at P, P' respectively. Assume that $\mathcal{P}_x \cong \mathcal{P}'_{x'}$ as G-torsors, compatibly with the isomorphism $k((x)) \cong k((x'))$, $x \mapsto x'$. Then, for every $d \geq 1$, $\mathcal{P}^{[d]}$ is unramified (resp. tamely ramified) at η_{dP} if and only if $\mathcal{P}'^{[d]}$ is unramified (resp. tamely ramified) at $\eta_{dP'}$.*

Proof. Immediate from Lemma 48: both $\mathcal{P}^{[d]}|_{\mathrm{Spec}(\widehat{K}_{\eta_{dP}})}$ and $\mathcal{P}'^{[d]}|_{\mathrm{Spec}(\widehat{K}_{\eta_{dP'}})}$ are identified, compatibly with the discrete valuation ring $\kappa[[e_d]]$, with the pullback along $\mathrm{Spec}(\kappa((e_d))) \to V$ of $\Xi_G \backslash (q_1^{-1}\mathcal{P}_x \times_W \cdots \times_W q_d^{-1}\mathcal{P}_x)$. □

The corollary reduces the geometry to $\mathbb{P}^1$ once the local cyclic torsor is realized by a global equation on $\mathbb{G}_m$. The next section introduces the three cyclic theories that provide precisely these equations and control their ramification.

2.4 Three Parallel Theories of Cyclic Torsors

The first reduction of the preceding section explains why cyclic theories suffice: it reduces the theorem from finite abelian groups to cyclic groups of order prime to p and cyclic groups of order p^r. The second reduction explains exactly what these theories must accomplish: they must realize a local cyclic torsor over $k((x))$ by an explicit equation on $\mathbb{G}_m \subset \mathbb{P}^1_k$, with the required control of its ramification. We therefore recall the three theories in the parallel form in which they will be used:

theory	H	$\varphi : H \to H$	$\ker(\varphi)$
Kummer	$\mathbb{G}_m$	$z \mapsto z^e$	$\mu_e \cong \mathbb{Z}/e\mathbb{Z}$
Artin–Schreier	$\mathbb{G}_a$	$z \mapsto z^p - z$	$\mathbb{F}_p \cong \mathbb{Z}/p\mathbb{Z}$
Artin–Schreier–Witt	$\mathbb{W}_r$	$F - \mathrm{id}$	$W_r(\mathbb{F}_p) \cong \mathbb{Z}/p^r\mathbb{Z}$.

For e invertible, and in characteristic p for the latter two rows, the corresponding exact sequences of étale sheaves are

$$1 \longrightarrow \mu_e \longrightarrow \mathbb{G}_m \xrightarrow{(-)^e} \mathbb{G}_m \longrightarrow 1,$$
$$0 \longrightarrow \mathbb{F}_p \longrightarrow \mathbb{G}_a \xrightarrow{z \mapsto z^p - z} \mathbb{G}_a \longrightarrow 0,$$
$$0 \longrightarrow W_r(\mathbb{F}_p) \longrightarrow \mathbb{W}_r \xrightarrow{F - \mathrm{id}} \mathbb{W}_r \longrightarrow 0.$$

Here the identification in the Kummer row is made after choosing a primitive e-th root of unity. In every row, pulling back φ along a section a produces the torsor given by the equation $\varphi(X) = a$, and changing a by an element of $\varphi(H)$ does not change its class. The kernel action, pullback construction, and resulting cohomology class are detailed in Section A.1.

Artin–Schreier–Witt theory contains Artin–Schreier theory as the case $r = 1$. We nevertheless treat the Artin–Schreier case separately: it displays, without the additional Witt-vector bookkeeping, the basic principle later used in the general case—choose a reduced representative of the local torsor, control its pole order by the ramification break, and use the same equation to realize the torsor on $\mathbb{G}_m$.

Only the conclusions used in Chapter 2 are stated here. The group-scheme constructions, classification arguments, normal-form results, and proofs are collected in Chapter A.

Throughout this section, let K be a complete discrete valuation field with perfect residue field κ of characteristic $p > 0$.

2.4.1 Kummer theory: order prime to p

Classes and equations. Assume that K contains μ_n and that $(n, p) = 1$. Write

$$H^1(K, \mu_n) := H^1_{\text{ét}}(\operatorname{Spec} K, \mu_n);$$

equivalently, this is the first continuous Galois cohomology group of K with coefficients in μ_n. Its elements classify μ_n-torsors over K. The Kummer sequence gives a natural isomorphism

$$H^1(K, \mu_n) \cong K^\times/(K^\times)^n.$$

Under this isomorphism, the coset of $a \in K^\times$ corresponds to the torsor

$$\operatorname{Spec} K[T]/(T^n - a),$$

on which μ_n acts by $T \mapsto \zeta T$. In particular, replacing a by ab^n, with $b \in K^\times$, does not change the isomorphism class of the torsor. If the class of a has order n, adjoining a solution of $T^n = a$ gives a cyclic field extension of degree n; conversely, every cyclic extension of degree n arises in this way.

When $K = k((x))$ with k algebraically closed, every unit in $k[[x]]$ is an nth power by Hensel's lemma. Thus valuation gives

$$K^\times/(K^\times)^n \cong \mathbb{Z}/n\mathbb{Z}, \qquad [x] \longmapsto 1. \tag{2.3}$$

In particular, a class has order n exactly when it is represented by x^a with $\gcd(a, n) = 1$.

Theorem 50 (Ramification in Kummer Extensions, [Koc97, Proposition 1.83])**.** *Assume K contains the n-th roots of unity μ_n and $(n, p) = 1$. Let L/K be the extension given by $X^n = a$, for $a \in K^\times$. Then:*

1. *If $v_K(a) \in n\mathbb{Z}$ and the image of $a\pi^{-v_K(a)}$ in κ is an n-th power, the extension is trivial.*
2. *If $v_K(a) \in n\mathbb{Z}$ and the image of $a\pi^{-v_K(a)}$ in κ is not an n-th power, the extension is unramified.*
3. *If $v_K(a) \notin n\mathbb{Z}$, the ramification index is $n/\gcd(n, v_K(a))$; in particular, if $\gcd(n, v_K(a)) = 1$, the extension is totally and tamely ramified of degree n.*

2.4.2 Artin–Schreier theory: order p

Classes and equations. Assume that K has characteristic p, and put $\wp(z) = z^p - z$. Write

$$H^1(K, \mathbb{Z}/p\mathbb{Z}) := H^1_{\text{ét}}(\operatorname{Spec} K, \mathbb{Z}/p\mathbb{Z}).$$

Its elements classify $\mathbb{Z}/p\mathbb{Z}$-torsors over K. The Artin–Schreier sequence gives a natural isomorphism

$$H^1(K, \mathbb{Z}/p\mathbb{Z}) \cong K/\wp(K).$$

Under this isomorphism, the coset $[a] \in K/\wp(K)$ corresponds to the $\mathbb{Z}/p\mathbb{Z}$-torsor

$$\operatorname{Spec} K[X]/(X^p - X - a),$$

on which $c \in \mathbb{F}_p \cong \mathbb{Z}/p\mathbb{Z}$ acts by $X \mapsto X + c$. In particular, replacing a by $a + (b^p - b)$, with $b \in K$, does not change the isomorphism class of the torsor. If $[a] \neq 0$, adjoining a solution of $X^p - X = a$ gives a cyclic field extension of degree p; conversely, every cyclic extension of degree p arises in this way. When $K = k((x))$ with k algebraically closed, every class has a representative in $x^{-1}k[x^{-1}]$ whose nonzero pole order is prime to p (Proposition 90, case $r = 1$).

Theorem 51 (Ramification in Artin–Schreier Extensions, [Tho05, Proposition 2.1])**.** *Let K be a complete discretely valued field of characteristic p with perfect residue field κ, and let L/K be given by $X^p - X = a$. Then:*

1. *If $v_K(a) > 0$, or if $v_K(a) = 0$ and $a \in \wp(K)$, the extension is trivial.*
2. *If $v_K(a) = 0$ and $a \notin \wp(K)$, the extension is cyclic of degree p and unramified.*
3. *If $v_K(a) = -m < 0$ with $p \nmid m$, the extension is cyclic, totally ramified of degree p, and has unique upper ramification break m.*

2.4.3 Artin–Schreier–Witt theory: order p^r

Classes and equations. For an $\mathbb{F}_p$-algebra A, write $W_r(A) = A^r$ for the ring of p-typical Witt vectors of length r, with Witt addition and subtraction denoted by $\oplus$ and $\ominus$, and put $\wp = F - \mathrm{id}$. For a field K of characteristic p, write

$$H^1(K, \mathbb{Z}/p^r\mathbb{Z}) := H^1_{\text{ét}}(\operatorname{Spec} K, \mathbb{Z}/p^r\mathbb{Z}).$$

Its elements classify $\mathbb{Z}/p^r\mathbb{Z}$-torsors over K. The Artin–Schreier–Witt sequence gives a natural isomorphism

$$H^1(K, \mathbb{Z}/p^r\mathbb{Z}) \cong W_r(K)/\wp W_r(K).$$

Under this isomorphism, the coset $[\vec{g}] \in W_r(K)/\wp W_r(K)$ corresponds to an affine solution scheme as follows. Let $\vec{X} = (X_0, \dots, X_{r-1})$ be the universal Witt vector on $\mathbb{W}_{r,K} = \operatorname{Spec} K[X_0, \dots, X_{r-1}]$. Then

$$T_{\vec{g}} := \operatorname{Spec}\Big(K[X_0, \dots, X_{r-1}]/\big(\wp\vec{X} \ominus \vec{g}\big)\Big),$$

where the parenthesized Witt vector denotes the ideal generated by its r coordinate polynomials. Equivalently, $T_{\vec{g}}$ is the fiber of $\wp : \mathbb{W}_{r,K} \to \mathbb{W}_{r,K}$ over the K-point $\vec{g}$. Its first defining equation is $X_0^p - X_0 - g_0 = 0$, and each subsequent equation has the triangular form

$$X_j^p - X_j - g_j + C_j(X_0, \dots, X_{j-1}; g_0, \dots, g_{j-1}) = 0.$$

Here C_j is the universal Witt correction polynomial determined by the addition law and depends only on the preceding coordinates. This is a finite étale K-scheme of degree p^r, on which $\vec{c} \in W_r(\mathbb{F}_p) \cong \mathbb{Z}/p^r\mathbb{Z}$ acts by Witt translation $\vec{X} \mapsto \vec{X} \oplus \vec{c}$. In particular, replacing $\vec{g}$ by $\vec{g} \oplus \wp\vec{h}$, with $\vec{h} \in W_r(K)$, gives an isomorphic torsor. If $[\vec{g}]$ has order p^s, its associated character cuts out a cyclic field extension of degree p^s; thus $T_{\vec{g}}$ is connected precisely when $[\vec{g}]$ has order p^r. Conversely, every cyclic extension of degree dividing p^r arises in this way. The Witt-vector group scheme is constructed in Section A.3. The Artin–Schreier–Witt operator, the proof that the displayed equation defines a $\mathbb{Z}/p^r\mathbb{Z}$-torsor, and the classification theorem—including its description of the degree and connected components and of induction from a torsor under the subgroup $\mathbb{Z}/p^s\mathbb{Z} \subseteq \mathbb{Z}/p^r\mathbb{Z}$—are given in Section A.4.

Definition 52. Let $K = k((x))$. For $g \in K$, put $m(g) := \max(0, -v_x(g))$. A vector $\vec{g} = (g_0, \dots, g_{r-1}) \in W_r(K)$ is **reduced** if, for every j, either $m(g_j) = 0$ or $p \nmid m(g_j)$.

Every class in $W_r(K)/\wp W_r(K)$ admits a reduced representative. The stronger reduced polynomial normal form for algebraically closed k is proved in the appendix (Proposition 90).

Theorem 53 (Ramification in Artin–Schreier–Witt Extensions (Schmid–Witt))**.** *Let $K = k((x))$, with k perfect of characteristic p, and let L/K be a cyclic totally ramified extension of degree p^r whose Artin–Schreier–Witt class is represented by a reduced vector $\vec{g} = (g_0, \dots, g_{r-1})$, with $m_j := m(g_j)$. Then the largest upper-numbering ramification break of L/K is*

$$u = \max_{0 \le j \le r-1} p^{r-1-j} m_j.$$

The largest upper break is the last index at which an upper-numbering ramification group is nontrivial. The formula reads this break from the pole orders of the coordinates of a reduced Witt vector. The jth coordinate has pole order m_j, and the formula assigns it the value $p^{r-1-j}m_j$. For example, when $r = 2$ it says $u = \max\{pm_0, m_1\}$. For any bound d, the inequality $u < d$ is therefore equivalent to $p^{r-1-j}m_j < d$ for every j, as required in (2.4) below.

The word "reduced" matters here. Already for $r = 1$, the elements x^{-p} and x^{-1} represent the same Artin–Schreier class, since $x^{-p} - x^{-1} = \wp(x^{-1})$. The first has pole order p, but the reduced representative x^{-1} shows that the break is 1. In this case the formula agrees with Theorem 51.

We use the Schmid–Witt formula as a known theorem. For finite residue fields, it is due to Kanesaka–Sekiguchi [KS79] and Thomas [Tho05], building on Schmid [Sch37] for $r = 1$; for arbitrary perfect residue fields, it is due to Elder–Keating [EK25].

2.4.4 The common realization conclusion

The three theories give the following single conclusion, which is the form needed after the formal–local reduction to $\mathbb{P}^1_k$.

Lemma 54 (Parallel realization over $\mathbb{G}_m$)**.** *Let k be algebraically closed of characteristic $p > 0$, and let $\mathcal{Q}$ be a G-torsor over* $\operatorname{Spec} k((x))$.

1. *If $G = \mathbb{Z}/e\mathbb{Z}$ with $(e, p) = 1$ and $\mathcal{Q}$ is connected, then $\mathcal{Q}$ is represented by*

$$T^e = x^a$$

for some a prime to e. The same equation defines a G-torsor on $\mathbb{G}_m$ that is tamely ramified at 0 and ∞.

2. *If $G = \mathbb{Z}/p\mathbb{Z}$ and the ramification of $\mathcal{Q}$ is bounded by d, then $\mathcal{Q}$ is represented by*

$$X^p - X = f(x), \qquad f \in x^{-1}k[x^{-1}],$$

where either $f = 0$, or $m := \deg_{x^{-1}} f < d$ and $p \nmid m$. The same equation defines a torsor on $\mathbb{G}_m$ extending étale across ∞.

3. *If $G = \mathbb{Z}/p^r\mathbb{Z}$, the torsor $\mathcal{Q}$ is connected, and its ramification is bounded by d, then $\mathcal{Q}$ is represented by*

$$\wp\vec{X} = \vec{f},$$

where $\vec{f} = (f_0, \dots, f_{r-1})$ *is reduced, each* $f_j \in x^{-1}k[x^{-1}]$*, and, writing* $m_j := \deg_{x^{-1}} f_j$ *(with* $m_j = 0$ *if* $f_j = 0$*),*

$$p^{r-1-j} m_j < d \qquad (0 \le j < r). \tag{2.4}$$

The same equation defines a torsor on $\mathbb{G}_m$ *extending étale across* ∞*.*

Proof. For (1), choose $\mu_e \cong G$. By Kummer theory, $\mathcal{Q}$ is represented by an equation $T^e = b$ for some $b \in k((x))^\times$. By (2.3), $[b] = [x^a]$ in $k((x))^\times/(k((x))^\times)^e$ for some integer a. Thus $b = x^a c^e$ for some $c \in k((x))^\times$, and replacing T by T/c gives the equation $T^e = x^a$. Connectedness means that the class $[x^a]$ has order e; under (2.3), this is equivalent to $\gcd(a, e) = 1$. The equation $T^e = x^a$ is étale over $k[x, x^{-1}]$ and defines the required torsor on $\mathbb{G}_m$. At 0 and ∞ the valuations of x^a are a and $-a$, respectively, so Theorem 50 shows that its ramification at both boundary points is tame.

For (2), Artin–Schreier theory represents $\mathcal{Q}$ as $\operatorname{Spec}\big(k((x))[X]/(X^p - X - b)\big)$ for some $b \in k((x))$, with $\mathbb{F}_p \cong G$ acting by $X \mapsto X + \lambda$. The length-one construction in the proof of Proposition 90 gives an $f \in x^{-1}k[x^{-1}]$ with $[f] = [b]$ in $k((x))/\wp(k((x)))$, where either $f = 0$, or $m := \deg_{x^{-1}} f$ is prime to p. Thus $b - f = c^p - c$ for some $c \in k((x))$, and replacing X by $X - c$ gives the equation $X^p - X = f$ for $\mathcal{Q}$. If $f = 0$, then $\mathcal{Q}$ is trivial; otherwise, Theorem 51 identifies m with its unique upper ramification break, so the assumed ramification bound is equivalent to $m < d$. The same equation defines a torsor over $\mathbb{G}_m$: it is finite étale over $k[x, x^{-1}]$ because its derivative with respect to X is -1, and since $f \in k[x^{-1}]$, it extends étale over $\operatorname{Spec} k[x^{-1}] = \mathbb{P}^1_k \setminus \{0\}$.

For (3), Artin–Schreier–Witt theory represents $\mathcal{Q}$ by an equation $\wp\vec{X} = \vec{g}$ for some $\vec{g} \in W_r(k((x)))$, with $W_r(\mathbb{F}_p) \cong G$ acting by Witt translation $\vec{X} \mapsto \vec{X} \oplus \vec{c}$. By Proposition 90, the class $[\vec{g}]$ has a reduced representative $\vec{f} = (f_0, \dots, f_{r-1})$ with each $f_j \in x^{-1}k[x^{-1}]$. Thus $\vec{g} = \vec{f} \oplus \wp\vec{h}$ for some $\vec{h} \in W_r(k((x)))$, and the change of variables $\vec{Y} = \vec{X} \ominus \vec{h}$ gives $\wp\vec{Y} = \vec{f}$; renaming $\vec{Y}$ as $\vec{X}$ gives the equation in the statement. Connectedness means that $[\vec{f}]$ has order p^r, so Corollary 87 gives $f_0 \neq 0$. Since k is algebraically closed, the corresponding degree-p^r extension is totally ramified. With the convention for m_j in the statement, Theorem 53 therefore gives

$$\max_j p^{r-1-j} m_j < d,$$

which is equivalent to (2.4). Finally, Lemma 85 applied over $k[x, x^{-1}]$ shows that the same equation defines a $\mathbb{Z}/p^r\mathbb{Z}$-torsor over $\mathbb{G}_m$ whose restriction is $\mathcal{Q}$. Since every f_j lies in $k[x^{-1}]$, applying the same lemma over $k[x^{-1}]$ extends this torsor étale across ∞. □

Remark. The weighted pole bound in part (3) has the following immediate consequence. If $p^{r-1-j} \ge d$, then m_j cannot be positive, so $f_j = 0$. Thus sufficiently early Witt coordinates must vanish. In particular, since connectedness requires $f_0 \neq 0$, a connected $\mathbb{Z}/p^r\mathbb{Z}$-torsor necessarily has $d > p^{r-1}$. This observation is only a sanity check explaining the meaning of the weights and will not be used below.

2.4.5 Three common inputs for the cyclic calculations

The explicit equations above enter the symmetric-power construction through one contracted-product calculation. We record it once for all three cyclic theories.

Let S be a scheme, let H be a commutative S-group scheme, and let $\varphi : H \to H$ be a finite étale surjective homomorphism with constant kernel G. For a section $a : X \to H$, write

$$\mathcal{T}_a := X \times_{a,H,\varphi} H.$$

Thus $\mathcal{T}_a$ is a G-torsor on X. Equivalently, for every X-scheme U,

$$\mathcal{T}_a(U) = \{h \in H(U) : \varphi(h) = a|_U\}.$$

In other words, a point of $\mathcal{T}_a$ is an element of H lying over a under φ. We write $\star$ for the group law of H: multiplication on $\mathbb{G}_m$, ordinary addition on $\mathbb{G}_a$, and Witt addition on $\mathbb{W}_r$.

Proposition 55 (Compatibility with contracted products)**.** *Let $a_i : X \to H$, $1 \le i \le d$, be sections. There is a canonical isomorphism of G-torsors*

$$\mathcal{T}_{a_1} \otimes^G \cdots \otimes^G \mathcal{T}_{a_d} \cong \mathcal{T}_{a_1 \star \cdots \star a_d}.$$

On local sections it is induced by

$$[(h_1, \dots, h_d)] \longmapsto h_1 \star \cdots \star h_d, \qquad h_i \in \mathcal{T}_{a_i}.$$

In other words, the construction $a \mapsto \mathcal{T}_a$ is compatible with the group law on $H(X)$ and the contracted-product group law on G-torsors. Moreover, if a group Γ acts on this construction by permuting the indexed factors, the isomorphism is Γ-equivariant and hence descends whenever the relevant quotients exist.

Proof. By the definition of the contracted product, the left-hand side is

$$K_G \backslash \prod_i \mathcal{T}_{a_i}, \qquad K_G = \ker(G^d \xrightarrow{m} G),$$

where K_G acts coordinatewise by

$$(g_1, \dots, g_d) \cdot (h_1, \dots, h_d) = (g_1 \star h_1, \dots, g_d \star h_d).$$

Consider the map

$$F : \prod_i \mathcal{T}_{a_i} \longrightarrow H, \qquad (h_1, \dots, h_d) \longmapsto h_1 \star \cdots \star h_d.$$

Since $\varphi(h_i) = a_i$ and φ is a homomorphism,

$$\varphi\big(F(h_1, \dots, h_d)\big) = a_1 \star \cdots \star a_d,$$

so F takes values in $\mathcal{T}_{a_1 \star \cdots \star a_d}$. Moreover, F is K_G-invariant because, for $(g_1, \dots, g_d) \in K_G$, the product $g_1 \star \cdots \star g_d$ is the identity. It therefore descends to the contracted product. The descended map is G-equivariant, and hence is an isomorphism of G-torsors by Proposition 5. Finally, commutativity of $\star$ gives the asserted Γ-equivariance. □

We shall also need to pass from a connected torsor under a subgroup $H \subseteq G$ to the G-torsor induced from it. The following compatibility ensures that this passage commutes with the symmetric-power construction.

Lemma 56 (Symmetric powers and induced torsors)**.** *Let $\iota : H \hookrightarrow G$ be an injection of finite abelian groups, and let $\mathcal{Q}$ be an H-torsor on U. There is a canonical isomorphism of G-torsors over $U^{(d)}$*

$$(\iota_* \mathcal{Q})^{[d]} \cong \iota_*(\mathcal{Q}^{[d]}).$$

Proof. The canonical morphism $j : \mathcal{Q} \to \iota_*\mathcal{Q} = (\mathcal{Q} \times G)/H$ is H-equivariant, where H acts on the target through ι. Its d-fold product commutes with the S_d-actions and is equivariant for $H^d \to G^d$. Writing $K_H := \ker(H^d \to H)$, it therefore descends to an H-equivariant morphism

$$\mathcal{Q}^{[d]} = (K_H \rtimes S_d)\backslash\mathcal{Q}^{\times d} \longrightarrow (K_G \rtimes S_d)\backslash(\iota_*\mathcal{Q})^{\times d} = (\iota_*\mathcal{Q})^{[d]}.$$

The universal property of induction extends this morphism uniquely to a G-equivariant morphism $\iota_*(\mathcal{Q}^{[d]}) \to (\iota_*\mathcal{Q})^{[d]}$. Both sides are G-torsors over $U^{(d)}$, so it is an isomorphism by Proposition 5. □

Finally, the two wild calculations use the same elementary fact about the exceptional valuation. Let $e_1, \dots, e_d$ be the elementary symmetric polynomials in $x_1, \dots, x_d$, put $y_i = x_i^{-1}$, and write

$$\kappa := k(e_1/e_d, \dots, e_{d-1}/e_d).$$

Lemma 57 (Symmetric regularity)**.** *If $\Phi \in k[y_1, \dots, y_d]^{S_d}$ and every monomial of Φ has total degree strictly less than d, then*

$$\Phi \in k(e_1/e_d, \dots, e_{d-1}/e_d) = \kappa.$$

In particular, Φ is regular at the generic point of the exceptional divisor.

Proof. The identity

$$e_j(y_1, \dots, y_d) = \frac{e_{d-j}(x_1, \dots, x_d)}{e_d(x_1, \dots, x_d)}$$

puts $e_1(y), \dots, e_{d-1}(y)$ in κ. By the fundamental theorem of symmetric polynomials, each homogeneous part of Φ is a polynomial in $e_1(y), \dots, e_d(y)$ with their natural degrees. Since every homogeneous part has degree $< d$, the generator $e_d(y)$ cannot occur. Hence $\Phi \in \kappa \subset \kappa[[e_d]]$. □

2.5 The Cyclic Exceptional–Divisor Calculation and Final Assembly

The reductions and cyclic theories of the preceding sections leave one explicit calculation. We state its three cyclic forms together and then prove them in parallel. This is the only intermediate statement needed for the proof of Theorem 33.

Theorem 58 (The three cyclic cases)**.** *Assume that k is algebraically closed. Let $\mathcal{P} \to U = C \setminus \mathfrak{m}$ be a G-torsor with G cyclic, and let $\mathfrak{n} = dP \subset \mathfrak{m}$ be a single-point submodulus.*

1. *If $G = \mathbb{Z}/e\mathbb{Z}$ with $(e, p) = 1$, then $\mathcal{P}^{[d]}$ is tamely ramified at $\eta_{\mathfrak{n}}$.*
2. *If $G = \mathbb{Z}/p\mathbb{Z}$ and the ramification of $\mathcal{P}$ at P is bounded by d, then $\mathcal{P}^{[d]}$ is unramified at $\eta_{\mathfrak{n}}$.*
3. *If $G = \mathbb{Z}/p^r\mathbb{Z}$, $r \geq 2$, and the ramification of $\mathcal{P}$ at P is bounded by d, then $\mathcal{P}^{[d]}$ is unramified at $\eta_{\mathfrak{n}}$.*

Proof. By Section 2.3, we may work over an algebraically closed field and regard P as rational. Choose a uniformizer x at P, let

$$\rho : \mathrm{Gal}(k((x))^{\mathrm{sep}}/k((x))) \longrightarrow G$$

classify the punctured-formal-disc restriction $\mathcal{P}_x$, and put $H := \mathrm{im}(\rho)$. If H is trivial, then $\mathcal{P}_x$ and its symmetric power are trivial, so the conclusion follows from Lemma 48.

Assume $H \neq 0$. The surjective character with image H defines a connected H-torsor $\mathcal{Q}$ over $\operatorname{Spec} k((x))$, and $\mathcal{P}_x \cong \iota_*\mathcal{Q}$ for the inclusion $\iota : H \hookrightarrow G$. Since $k[[x]]$ is complete with perfect residue field, whenever a ramification bound is assumed, it passes from $\mathcal{P}_x$ to $\mathcal{Q}$ by Lemma 46(3). By Lemma 54, the torsor $\mathcal{Q}$ is realized by an explicit connected H-torsor $\mathcal{Q}'$ on $\mathbb{G}_m$. Moreover,

$$(\iota_*\mathcal{Q}')^{[d]} \cong \iota_*(\mathcal{Q}'^{[d]})$$

by Lemma 56, and parts (1) and (2) of Lemma 46 show that the corresponding tame or unramified conclusion passes from $\mathcal{Q}'^{[d]}$ to $\iota_*(\mathcal{Q}'^{[d]})$. Therefore Corollary 49 reduces every case to the connected explicit model

$$C = \mathbb{P}^1_k, \qquad U = \mathbb{G}_m, \qquad \mathfrak{n} = d \cdot 0.$$

We rename H as G for the calculation.

Write $x_1, \dots, x_d$ for the coordinates on $\mathbb{G}_m^d$ and $e_1, \dots, e_d$ for their elementary symmetric polynomials. By Lemma 48, the completed local ring and field at the generic point $\eta = \eta_{d\cdot 0}$ of the exceptional divisor are

$$\widehat{\mathcal{O}}_\eta = \kappa[[e_d]], \qquad \widehat{K}_\eta = \kappa((e_d)), \qquad \kappa = k(e_1/e_d, \dots, e_{d-1}/e_d). \tag{2.5}$$

The three calculations have the common shape

	$\mathbb{Z}/e\mathbb{Z}$	$\mathbb{Z}/p\mathbb{Z}$	$\mathbb{Z}/p^r\mathbb{Z}$
normal form	$T^e = x^a$	$X^p - X = f$	$\wp\vec{X} = \vec{f}$
product coordinate	$\prod_i T_i$	$\sum_i X_i$	$\bigoplus_i \vec{X}_i$
new datum	e_d^a	$\sum_i f(x_i)$	$\bigoplus_i \vec{f}(x_i)$
exceptional behavior	$v(e_d^a) = a$	regular	componentwise regular
conclusion	tame	unramified	unramified.

Case (1): Kummer. The relevant group scheme is $\mathbb{G}_m$. More precisely, the connected torsor is the pullback of the finite étale homomorphism

$$[e] : \mathbb{G}_m \longrightarrow \mathbb{G}_m, \qquad T \longmapsto T^e,$$

whose kernel is $\mu_e \cong G$, along the section $x \mapsto x^a$. Thus it has equation

$$T^e = x^a, \qquad (a, e) = 1,$$

by Lemma 54(1). Put

$$R := k[x, x^{-1}], \qquad S := R[T]/(T^e - x^a).$$

Then $\operatorname{Spec} S \to \operatorname{Spec} R = \mathbb{G}_m$ is this G-torsor. After pulling it back along the d projections from $\mathbb{G}_m^d$, the product torsor has coordinate algebra $S^{\otimes_k d}$ over $R^{\otimes_k d}$. Contracting the product action G^d along $G^d \to G$ means taking invariants under $K_G = \ker(G^d \to G)$. By Proposition 55, multiplication in $\mathbb{G}_m$ gives

$$Y = \prod_{i=1}^d T_i, \qquad Y^e = \prod_{i=1}^d x_i^a = (x_1 \cdots x_d)^a = e_d^a,$$

and hence an isomorphism of G-torsor algebras

$$(S^{\otimes_k d})^{K_G} \cong R^{\otimes_k d}[Y]/(Y^e - e_d^a).$$

The group S_d permutes the tensor factors and fixes both Y and e_d^a. The algebra on the right is free over $R^{\otimes_k d}$ with the S_d-invariant basis $1, Y, \dots, Y^{e-1}$, so taking S_d-invariants acts only on its coefficients. Since

$$(R^{\otimes_k d})^{S_d} = k[x_1^{\pm 1}, \dots, x_d^{\pm 1}]^{S_d} = k[e_1, \dots, e_d, e_d^{-1}],$$

Proposition 27 identifies the descended torsor with

$$\mathcal{P}^{[d]} \cong \operatorname{Spec} \frac{k[e_1, \dots, e_d, e_d^{-1}][Y]}{(Y^e - e_d^a)}.$$

Its generic fiber over $\widehat{K}_\eta$ is therefore the restriction of $\mathcal{P}^{[d]}$ and is given by the same Kummer equation. The defining element e_d^a has valuation a; thus Theorem 50, together with $(e, p) = 1$, shows that every field factor is tamely ramified. This proves part (1).

Case (2): Artin–Schreier. Here the relevant group scheme is $\mathbb{G}_a$. The torsor is the pullback of the finite étale homomorphism

$$\wp : \mathbb{G}_a \longrightarrow \mathbb{G}_a, \qquad X \longmapsto X^p - X,$$

whose kernel is $\mathbb{F}_p \cong G$, along the section $x \mapsto f(x)$ on $\mathbb{G}_m$. By Lemma 54(2), its equation is

$$X^p - X = f(x), \qquad f \in x^{-1}k[x^{-1}],$$

where either $f = 0$, or $m := \deg_{x^{-1}} f < d$ and $p \nmid m$. Put

$$R := k[x, x^{-1}], \qquad S := R[X]/(X^p - X - f(x)).$$

Thus $\operatorname{Spec} S \to \operatorname{Spec} R = \mathbb{G}_m$ is the realized G-torsor. After pullback along the d projections, $S^{\otimes_k d}$ is the coordinate algebra of the product torsor over $R^{\otimes_k d}$. Contracting along $G^d \to G$ again amounts to taking K_G-invariants. This time the group law on $\mathbb{G}_a$ is addition, so Proposition 55 gives

$$Y = \sum_{i=1}^{d} X_i, \qquad Y^p - Y = \alpha, \qquad \alpha := \sum_{i=1}^{d} f(x_i),$$

and hence

$$(S^{\otimes_k d})^{K_G} \cong R^{\otimes_k d}[Y]/(Y^p - Y - \alpha).$$

The group S_d fixes Y and α. The displayed algebra is free over $R^{\otimes_k d}$ with the S_d-invariant basis $1, Y, \dots, Y^{p-1}$, so taking S_d-invariants acts only on the coefficients. Using $(R^{\otimes_k d})^{S_d} = k[e_1, \dots, e_d, e_d^{-1}]$ and Proposition 27, we obtain

$$\mathcal{P}^{[d]} \cong \operatorname{Spec} \frac{k[e_1, \dots, e_d, e_d^{-1}][Y]}{(Y^p - Y - \alpha)}.$$

Put $y_i = x_i^{-1}$. If $f \neq 0$, then α is a symmetric polynomial in $y_1, \dots, y_d$ whose monomials have total degree at most $m < d$; if $f = 0$, then $\alpha = 0$. Hence Lemma 57 gives

$$\alpha \in \kappa \subset \widehat{\mathcal{O}}_\eta.$$

Set $A := \widehat{\mathcal{O}}_\eta$ and

$$B := A[Y]/(Y^p - Y - \alpha).$$

Because the defining polynomial is monic of degree p, the algebra B is finite free of rank p over A. Its derivative with respect to Y is -1, a unit in B; hence B/A is finite étale. The preceding descended presentation identifies its generic fiber

$$B \otimes_A \widehat{K}_\eta = \widehat{K}_\eta[Y]/(Y^p - Y - \alpha)$$

with the restriction of $\mathcal{P}^{[d]}$ to $\widehat{K}_\eta$.

Since A is a complete discrete valuation ring, the finite étale A-algebra B is a product of finite étale local A-algebras. Each factor is a discrete valuation ring whose maximal ideal is generated by the uniformizer of A, and whose residue field is finite separable over κ. Consequently, every field factor of the generic fiber has ramification index 1 and separable residue extension, so it is unramified in the sense of Definition 35. This proves part (2).

Case (3): Artin–Schreier–Witt. A natural alternative to the direct calculation below would be to filter a cyclic p-power torsor by its degree-p layers and try to apply the Artin–Schreier calculation inductively. In Appendix B we examine this approach and explain why the two expected induction inputs do not by themselves control ramification before the intervening wild base change.

If the connected inertia group H produced by the common reduction has order p, the Artin–Schreier calculation in Case (2) already gives the required unramified conclusion. We may therefore assume that H has order p^r with $r \geq 2$, and continue to denote it by G. By Lemma 54(3), the connected torsor has an equation

$$\wp\vec{X} = \vec{f} = (f_0, \ldots, f_{r-1}), \qquad f_j \in x^{-1}k[x^{-1}],$$

where, putting $m_j = \deg_{x^{-1}} f_j$ for $f_j \neq 0$ and $m_j = 0$ for $f_j = 0$, the pole bounds are

$$p^{r-1-j} m_j < d \qquad (0 \leq j < r). \tag{2.6}$$

Put $R = k[x, x^{-1}]$ and

$$S = R[\vec{X}]/(\wp\vec{X} \ominus \vec{f}).$$

After pulling back to $\mathbb{G}_m^d$, define the single Witt vectors

$$\vec{Y} := \bigoplus_{i=1}^{d} \vec{X}_i, \qquad \vec{\alpha} := \bigoplus_{i=1}^{d} \vec{f}(x_i).$$

Here and below $\bigoplus$ denotes iterated Witt addition, not a direct sum.

Proposition 59 (Identification of the Witt product torsor)**.** *Every component of $\vec{Y}$ is K_G-invariant, every component of $\vec{\alpha}$ is a symmetric Laurent polynomial, and*

$$(S^{\otimes_k d})^{K_G} \cong R^{\otimes_k d}[\vec{Y}]/(\wp\vec{Y} \ominus \vec{\alpha})$$

as G-torsors over $R^{\otimes_k d}$. Taking S_d-invariants gives

$$\left((S^{\otimes_k d})^{K_G}\right)^{S_d} \cong k[e_1, \ldots, e_d, e_d^{-1}][\vec{Y}]/(\wp\vec{Y} \ominus \vec{\alpha}).$$

Consequently,

$$\mathcal{P}^{[d]}\big|_{\operatorname{Spec} \widehat{K}_\eta} \cong \operatorname{Spec} \widehat{K}_\eta[\vec{Y}]/(\wp\vec{Y} \ominus \vec{\alpha}).$$

Proof. Write an element of $K_G \cong G^{d-1}$ as $(g_1, \dots, g_{d-1})$. Under the usual embedding in G^d, it acts in Witt coordinates by

$$\begin{aligned}
\vec{X}_1 &\longmapsto \vec{X}_1 \oplus \underline{g}_1, \\
\vec{X}_i &\longmapsto \vec{X}_i \oplus \underline{g}_i \ominus \underline{g}_{i-1} \quad (1 < i < d), \\
\vec{X}_d &\longmapsto \vec{X}_d \ominus \underline{g}_{d-1},
\end{aligned}$$

where $\underline{g}$ is the image of g in $W_r(\mathbb{F}_p)$. Functoriality of W_r means that this action is componentwise and commutes with $\oplus$ and $\ominus$. The translations telescope in $\vec{Y}$, so each component of $\vec{Y}$ is invariant. Additivity of $\wp$ gives

$$\wp\vec{Y} = \bigoplus_{i=1}^{d} \wp\vec{X}_i = \bigoplus_{i=1}^{d} \vec{f}(x_i) = \vec{\alpha}.$$

The resulting coordinate map is equivariant for the residual G-action. Both the displayed Artin–Schreier–Witt cover and the contracted product are G-torsors over $R^{\otimes_k d}$, by Lemma 85 and proposition 55; hence the map is an isomorphism by Proposition 5.

The group S_d permutes the tensor factors. Commutativity and associativity of Witt addition show that it fixes $\vec{Y}$ and $\vec{\alpha}$ componentwise. Thus every component of $\vec{\alpha}$ lies in

$$(R^{\otimes_k d})^{S_d} = k[e_1, \dots, e_d, e_d^{-1}].$$

By Lemma 85, the ASW algebra is free over $R^{\otimes_k d}$ with basis $\prod_n Y_n^{a_n}$, $0 \le a_n < p$. Every basis element is S_d-invariant, so an element is invariant exactly when all its coefficients are invariant. This proves the second displayed isomorphism. Finally, Proposition 27 identifies the quotient by $K_G \rtimes S_d$ with $\mathcal{P}^{[d]}$, and base change to $\widehat{K}_\eta$ gives the last assertion. □

It remains to show that every component of $\vec{\alpha}$ is regular at the exceptional valuation. The Witt components are not ordinary componentwise sums, so this is the weighted analogue of the Artin–Schreier degree count. We claim that every component α_n, $0 \le n < r$, belongs to κ.

Give $f_j(x_i)$ weight p^j. By Lemma 91, every nonzero monomial

$$\prod_{i,j} f_j(x_i)^{a_{ij}}$$

appearing in α_n satisfies

$$\sum_{i,j} a_{ij} p^j = p^n.$$

Put $A_j := \sum_i a_{ij}$. Since $f_j(x_i)$ is a polynomial in $y_i = x_i^{-1}$ of degree m_j, every monomial obtained after expansion in the y_i has total degree at most

$$\sum_j A_j m_j < d \sum_j A_j p^{j-(r-1)} = d\, p^{n-(r-1)} \le d.$$

The inequality is strict: a nonzero monomial has a positive exponent on at least one nonzero f_j, and every pole bound in (2.6) is strict. By Proposition 59, the whole component α_n is symmetric. Therefore Lemma 57 gives $\alpha_n \in \kappa$. Thus

$$\vec{\alpha} \in W_r(\kappa) \subset W_r(\widehat{\mathcal{O}}_\eta).$$

It follows that the equation

$$\wp\vec{Y} = \vec{\alpha}$$

is defined over $\widehat{\mathcal{O}}_\eta$. By Lemma 85, it defines a finite étale $\widehat{\mathcal{O}}_\eta$-algebra. Its generic fiber is the restriction of $\mathcal{P}^{[d]}$ to $\widehat{K}_\eta$ by Proposition 59. Hence every field factor is unramified. This proves part (3). □

Proof of Theorem 33. Fix a single-point submodulus $\mathfrak{n} = dP$. By Proposition 47, it is enough to consider cyclic groups of order prime to p and cyclic p-groups. The prime-to-p case is Theorem 58(1). For $G = \mathbb{Z}/p^r\mathbb{Z}$, the ramification bound on $\mathcal{P}$ gives the unramified conclusion by part (2) when $r = 1$ and by part (3) when $r \geq 2$. The reduction proposition reassembles the cyclic factors and proves the theorem for every finite abelian group. □

Chapter 3

Geometric Class Field Theory

The purpose of this chapter is to deduce the reduced form of geometric class field theory, Theorem 2, from the ramification calculation established in Chapter 2.

Let C be a smooth, projective, geometrically connected curve over the perfect field k, let $\mathfrak{m}$ be a modulus on C, and put $U = C \setminus \mathfrak{m}$. Given a rank-one local system $\mathcal{F}$ on U whose ramification is bounded by $\mathfrak{m}$, our goal is to descend its symmetric power $\mathcal{F}^{[d]}$ along the Abel–Jacobi morphism

$$\Phi_d \colon U^{(d)} \longrightarrow \mathrm{Pic}^d_{C,\mathfrak{m}}$$

for all sufficiently large d.

When $\mathfrak{m} = 0$, the fibers of the Abel–Jacobi morphism are projective spaces. Their trivial étale fundamental groups allow the desired descent directly. When $\mathfrak{m} > 0$, however, the fibers of Φ_d are affine spaces, so this argument cannot be applied to Φ_d itself. Following Takeuchi, we resolve this difficulty by extending Φ_d to a projective space bundle

$$\widetilde{\Phi}_d \colon \widetilde{C}^{(d)}_{\mathfrak{m}} \longrightarrow \mathrm{Pic}^d_{C,\mathfrak{m}},$$

where $\widetilde{C}^{(d)}_{\mathfrak{m}}$ is an open subscheme of the blowup of $C^{(d)}$ along the locus obtained by adding the modulus.

The proof then has three remaining steps. First, we use the single-point ramification calculation of Chapter 2, together with compatibility under products and addition of divisors, to prove that $\mathcal{F}^{[d]}$ is tamely ramified along the boundary

$$H = \widetilde{C}^{(d)}_{\mathfrak{m}} \setminus U^{(d)}.$$

Second, on the geometric generic Abel–Jacobi fiber, the boundary is the hyperplane at infinity. A Tendler-style prime-to-p power argument, together with the structure of the abelian fundamental group of affine space, forces the corresponding boundary inertia representation to be trivial. Purity then extends $\mathcal{F}^{[d]}$ uniquely across the boundary. Finally, since $\widetilde{\Phi}_d$ is a projective space bundle, it induces an isomorphism on étale fundamental groups. The extended local system therefore descends uniquely to $\mathrm{Pic}^d_{C,\mathfrak{m}}$, proving Theorem 2.

We begin by recalling the generalized Picard scheme and the geometry of the Abel–Jacobi morphism. We then introduce Takeuchi's compactification, prove the required ramification and extension results, and conclude with the descent argument.

3.1 Generalized Picard Schemes and the Abel–Jacobi Map

3.1.1 The generalized Picard scheme

We recall the notion of generalized Jacobian varieties and study their fundamental properties. The material presented here is primarily adapted from [Gui19] and [Tak19]. For further background on the general theory of abelian varieties and Jacobians, the reader may also consult [Mil08]. Although the main theorem is over the fixed perfect field k, we recall the generalized Picard construction temporarily over an arbitrary base scheme S. Let S be a scheme and let $p : C \to S$ be a projective smooth morphism whose geometric fibers are connected and of dimension 1. Let $\mathfrak{m}$ be a modulus on C, defined as an effective Cartier divisor of C/S (i.e., a closed subscheme of C which is finite flat of finite presentation over S). We denote the projection $C \times_S T \to T$ by pr for any S-scheme T.

The Functor of Points

Let d be an integer. For an S-scheme T, we consider the set of data $(\mathcal{L}, \psi)$ where:

- $\mathcal{L}$ is an invertible sheaf of degree d on C_T.
- $\psi : \mathcal{O}_{\mathfrak{m}_T} \xrightarrow{\sim} \mathcal{L}|_{\mathfrak{m}_T}$ is a trivialization of $\mathcal{L}$ along the modulus.

Two such pairs $(\mathcal{L}, \psi)$ and $(\mathcal{L}', \psi')$ are said to be isomorphic if there exists an isomorphism of invertible sheaves $f : \mathcal{L} \to \mathcal{L}'$ such that the following diagram commutes:

$$\begin{array}{ccc} & \mathcal{O}_{\mathfrak{m}_T} & \\ {\scriptstyle \psi}\swarrow & & \searrow{\scriptstyle \psi'} \\ \mathcal{L}|_{\mathfrak{m}_T} & \xrightarrow[f|_{\mathfrak{m}_T}]{} & \mathcal{L}'|_{\mathfrak{m}_T} \end{array}$$

We define the presheaf $\mathrm{Pic}^{d,\mathrm{pre}}_{C,\mathfrak{m}}$ on Sch/S by assigning to T the set of isomorphism classes of such pairs. Let $\mathrm{Pic}^{d}_{C,\mathfrak{m}}$ denote the étale sheafification of this presheaf.

Representability and Structure

The fundamental properties of this functor are as follows:

1. $\mathrm{Pic}^{d}_{C,\mathfrak{m}}$ is represented by an S-scheme. (Note: If $\mathfrak{m}$ is faithfully flat over S, the presheaf is already a étale sheaf).
2. $\mathrm{Pic}^{0}_{C,\mathfrak{m}}$ is a smooth commutative group S-scheme with geometrically connected fibers, referred to as the *generalized Jacobian variety* of C with modulus $\mathfrak{m}$.
3. For any d, $\mathrm{Pic}^{d}_{C,\mathfrak{m}}$ is a $\mathrm{Pic}^{0}_{C,\mathfrak{m}}$-torsor.

In the case where $\mathfrak{m} = 0$, we recover the standard Jacobian variety, denoted simply as Pic^{d}_{C}.

Relation to the Standard Jacobian

We now examine the behavior of the generalized Picard scheme under the variation of the modulus. By viewing the structure along the modulus as an additional rigidification, we obtain natural transition maps corresponding to the inclusion of moduli.

Let $\mathfrak{m}_1$ and $\mathfrak{m}_2$ be moduli such that $\mathfrak{m}_1 \subset \mathfrak{m}_2$. There exists a natural map

$$\mathrm{Pic}^d_{C,\mathfrak{m}_2} \to \mathrm{Pic}^d_{C,\mathfrak{m}_1}$$

obtained by restricting the rigidification from $\mathfrak{m}_2$ to $\mathfrak{m}_1$. More precisely, it is an epimorphism for the étale topology: if $(\mathcal{L}, \psi_1)$ is an object over an S-scheme T, then, after an étale cover $T' \to T$, the rigidification ψ_1 of $\mathcal{L}|_{(\mathfrak{m}_1)_{T'}}$ extends to a rigidification of $\mathcal{L}|_{(\mathfrak{m}_2)_{T'}}$. Thus every local section of $\mathrm{Pic}^d_{C,\mathfrak{m}_1}$ lifts étale-locally to a section of $\mathrm{Pic}^d_{C,\mathfrak{m}_2}$. In particular, for any modulus $\mathfrak{m}$, there is a natural surjective morphism of étale sheaves:

$$\mathrm{Pic}^d_{C,\mathfrak{m}} \to \mathrm{Pic}^d_C.$$

Local Freeness and Base Change

Let $\mathfrak{m}$ be a modulus which is everywhere strictly positive. Let g denote the genus of C, which is a locally constant function on S. We restrict our attention to degrees d satisfying the condition:

$$d \geq \max\{2g - 1 + \deg \mathfrak{m}, \deg \mathfrak{m}\}. \tag{3.1}$$

Assuming S is quasi-compact, such a d always exists.

Fix an integer d satisfying the condition above. Let T be an S-scheme and let $\mathcal{L}$ be an invertible sheaf of degree d on C_T. One can show that the pushforwards $\mathrm{pr}_*\mathcal{L}$ and $\mathrm{pr}_*\mathcal{L}(-\mathfrak{m})$ are locally free sheaves and their formations commute with any base change. Explicitly, for any morphism of S-schemes $u : T' \to T$, the base change morphisms are isomorphisms:

$$u^*\mathrm{pr}_*\mathcal{L} \xrightarrow{\sim} \mathrm{pr}_* u^*\mathcal{L}$$

and

$$u^*\mathrm{pr}_*(\mathcal{L}(-\mathfrak{m})) \xrightarrow{\sim} \mathrm{pr}_* u^*(\mathcal{L}(-\mathfrak{m})).$$

In particular, by [Gui19, Proposition 4.3], if $\mathcal{L}$ is an invertible $\mathcal{O}_C$-module of degree d on each fiber of p, where d satisfies (3.1), then $p_*\mathcal{L}$ is a locally free $\mathcal{O}_S$-module of rank $d - g + 1$.

For further background and verification of these constructions, we refer the reader to Milne's notes on abelian Varieties ([Mil08]).

3.1.2 The Abel–Jacobi Morphism and Its Fibers

Let $U = C \backslash \mathfrak{m}$ be the complement of the modulus in C. The effective cartier divisors of degree d which are prime to $\mathfrak{m}$ are parameterized by the symmetric power $\mathrm{Sym}^d_S(U) = U^{(d)}$ over S (See [Gui19] Proposition 4.12, [Mil08] Theorem 3.13). For any such divisor $D \in U^{(d)}$, the associated line bundle $\mathcal{O}_C(D)$ admits a canonical trivialization along $\mathfrak{m}$. Specifically, the canonical section 1_D is regular and non-vanishing on $\mathfrak{m}$ because $\mathrm{supp}(D) \cap \mathrm{supp}(\mathfrak{m}) = \emptyset$. This section restricts to a nowhere-vanishing section on the subscheme $\mathfrak{m}$, thereby determining a trivialization $\psi_D^{-1} : \mathcal{O}_C(D)|_{\mathfrak{m}} \xrightarrow{\sim} \mathcal{O}_{\mathfrak{m}}$. This

is done functorially in families, yielding a morphism from the symmetric power to the generalized Picard scheme (over S):

$$\Phi_d : U^{(d)} \to \mathrm{Pic}^d_{C,\mathfrak{m}}, \quad D \mapsto [(\mathcal{O}_C(D), \psi_D)], \tag{3.2}$$

If $\mathfrak{m} = 0$, $d \geq 2g-1$, and $C \to S$ admits a section, then the Abel–Jacobi morphism $C^{(d)} \to \mathrm{Pic}^d_C$ is a projective space bundle. In particular, it is proper and smooth with geometrically connected fibers.

Guignard ([Gui19] Theorem 4.14) proves that for $\mathfrak{m} > 0$ and d satisfying (3.1), the Abel-Jacobi morphism Φ_d is surjective smooth of relative dimension $d - \deg \mathfrak{m} - g + 1$, with geometrically connected fibers.

We now return to the standing situation $S = \mathrm{Spec}(k)$, which remains in force for the rest of the chapter. The geometric fibers of Φ_d are well understood:

Theorem 60. *Assume that $S = \mathrm{Spec}(k)$ and $d \geq \max\{2g-1+\deg \mathfrak{m}, \deg \mathfrak{m}\}$. Let $\bar{s} = \mathrm{Spec}\,\Omega \to \mathrm{Pic}^d_{C,\mathfrak{m}}$ be a geometric point, where Ω is separably closed. Then the geometric fiber of the Abel–Jacobi morphism*

$$\Phi_d : U^{(d)} \longrightarrow \mathrm{Pic}^d_{C,\mathfrak{m}}$$

at $\bar{s}$ is isomorphic to

$$\begin{cases} \mathbb{A}_\Omega^{d-\deg \mathfrak{m}-g+1} & \text{if } \mathfrak{m} > 0, \\ \mathbb{P}_\Omega^{d-g} & \text{if } \mathfrak{m} = 0. \end{cases}$$

Thus every geometric fiber is an affine space when $\mathfrak{m} > 0$ and a projective space when $\mathfrak{m} = 0$. This statement does not, by itself, assert that Φ_d is Zariski-locally trivial.

Proof. See [Ten15] Propositions 3.13-3.14, or [Tót11] Proposition 2.1.4.

□

3.2 Takeuchi's Compactification of the Abel–Jacobi Morphism

We recall that our objective is to descend the local system $\mathcal{F}^{[d]}$ from $U^{(d)}$ to $\mathrm{Pic}^d_{C,\mathfrak{m}}$ along the Abel-Jacobi map Φ_d:

$$\begin{array}{ccc} \mathcal{F}^{[d]} & & \\ \downarrow & & \\ U^{(d)} & \xrightarrow{\Phi_d} & \mathrm{Pic}^d_{C,\mathfrak{m}} \end{array}$$

(Here, the purple arrow emphasizes that the morphism is of sheaves on the étale site).

However, we encounter an obstruction: in the case we are considering ($\mathfrak{m} > 0$), the fibers of Φ_d are affine spaces (of the same degree) rather than the better-behaved projective spaces. This suggests that a solution to this problem is to compactify the morphism to yield projective fibers.

This section describes the result of the compactification constructed by [Tak19] by blowing up the symmetric power.

Let $\mathfrak{m} = \sum_{i=1}^{n} k_i P_i$, where the P_i are distinct closed points of C, be a modulus on C, and let d satisfy (3.1). Takeuchi ([Tak19]) defines $Z_0 = Z_0(\mathfrak{m}, d)$ as the closed subscheme of $C^{(d)}$ defined by the map $C^{(d-\deg \mathfrak{m})} \to C^{(d)}$ adding $\mathfrak{m}$. He also defines $X_{\mathfrak{m},d}$ as the blowup of $C^{(d)}$ along Z_0. Let $E_0 = E_{\mathfrak{m},d} = Z_0(\mathfrak{m}, d) \times_{C^{(d)}} X_{\mathfrak{m},d}$ be the exceptional divisor of the blowup. It is irreducible of codimension 1, and we let $\eta_0 = \eta_{\mathfrak{m},d}$ be its generic point.

Diagrammatically:

$$\begin{array}{ccc} \overline{\{\eta_0\}} = E_0 & \longrightarrow & X_{\mathfrak{m},d} \\ \downarrow & & \downarrow{\scriptstyle \pi} \\ Z_0 & \xrightarrow{c.i} & C^{(d)} \end{array}$$

Incorporating $U^{(d)}$, the local system $\mathcal{F}^{[d]}$ and the Abel-Jacobi map, we have:

$$\begin{array}{ccccccc} & & & & & & \mathcal{F}^{[d]} \\ & & & & & & \downarrow \\ \overline{\{\eta_0\}} = E_0 & \longrightarrow & X_{\mathfrak{m},d} & \hookleftarrow & U^{(d)} & \xrightarrow{\Phi_d} & \mathrm{Pic}^d_{C,\mathfrak{m}} \\ \downarrow & & \downarrow{\scriptstyle \pi} & \swarrow & & & \\ Z_0 & \xrightarrow{c.i} & C^{(d)} & & & & \end{array}$$

In Section 3 of [Tak19] Takeuchi constructs, for d satisfying (3.1) a compactification denoted by $\tilde{C}^{(d)}_{\mathfrak{m}}$ and proves the following:

Theorem 61 (Takeuchi)**.** *The scheme $\tilde{C}^{(d)}_{\mathfrak{m}}$ is an open subscheme of $X_{\mathfrak{m},d}$ containing $U^{(d)}$. The morphism $\Phi_d : U^{(d)} \to \mathrm{Pic}^d_{C,\mathfrak{m}}$ extends to a morphism $\tilde{\Phi}_d : \tilde{C}^{(d)}_{\mathfrak{m}} \to \mathrm{Pic}^d_{C,\mathfrak{m}}$ which makes $\tilde{C}^{(d)}_{\mathfrak{m}}$ a projective space bundle over $\mathrm{Pic}^d_{C,\mathfrak{m}}$. Furthermore, the complement of $U^{(d)}$ in $\tilde{C}^{(d)}_{\mathfrak{m}}$ is the scheme-theoretic intersection*

$$E_0 \times_{X_{\mathfrak{m},d}} \tilde{C}^{(d)}_{\mathfrak{m}} \cong Z_0 \times_{C^{(d)}} \tilde{C}^{(d)}_{\mathfrak{m}}.$$

Diagrammatically we have:

$$\begin{array}{ccccc} & & & & \mathcal{F}^{[d]} \\ & & & & \downarrow \\ E_0 \times_{X_{\mathfrak{m},d}} \tilde{C}^{(d)}_{\mathfrak{m}} & \longrightarrow & \tilde{C}^{(d)}_{\mathfrak{m}} & \hookleftarrow & U^{(d)} \\ \downarrow & & \downarrow & \searrow^{\tilde{\Phi}_d} & \downarrow{\scriptstyle \Phi_d} \\ \overline{\{\eta_0\}} = E_0 & \longrightarrow & X_{\mathfrak{m},d} & & \mathrm{Pic}^d_{C,\mathfrak{m}} \\ \downarrow & & \downarrow{\scriptstyle \pi} & & \\ Z_0 & \xrightarrow{c.i} & C^{(d)} & & \end{array}$$

Here the boundary may equivalently be written as $Z_0 \times_{C^{(d)}} \tilde{C}^{(d)}_{\mathfrak{m}}$, since $E_0 = Z_0 \times_{C^{(d)}} X_{\mathfrak{m},d}$.

The construction gives more than the description of the open complement in Theorem 61: it identifies the boundary as a relative hyperplane.

Corollary 62 (The boundary is a relative hyperplane)**.** *The boundary*

$$H = \tilde{C}_{\mathfrak{m}}^{(d)} \setminus U^{(d)}$$

is a smooth relative hyperplane divisor in the projective-space bundle $\widetilde{\Phi}_d : \tilde{C}_{\mathfrak{m}}^{(d)} \to \mathrm{Pic}^d_{C,\mathfrak{m}}$*. Consequently, if* $\bar{s} = \operatorname{Spec}\Omega \to \mathrm{Pic}^d_{C,\mathfrak{m}}$ *is any geometric point and* $r = d - \deg \mathfrak{m} - g + 1$*, then there is an isomorphism of pairs*

$$\Big((\tilde{C}_{\mathfrak{m}}^{(d)})_{\bar{s}}, H_{\bar{s}}\Big) \cong \Big(\mathbb{P}^r_{\Omega}, \mathbb{P}^{r-1}_{\Omega}\Big),$$

and hence $U^{(d)}_{\bar{s}} \cong \mathbb{A}^r_{\Omega}$*.*

Proof. In Takeuchi's construction, the projective-space bundle is obtained from a locally free sheaf $E_{\mathfrak{m}}$ fitting into an exact sequence whose kernel is $\mathrm{pr}_*\,\mathcal{L}(-\mathfrak{m})$ and whose quotient is invertible. The resulting closed immersion

$$\mathbb{P}\big(\mathrm{pr}_*\,\mathcal{L}(-\mathfrak{m})\big) \hookrightarrow \mathbb{P}(E_{\mathfrak{m}})$$

is therefore a relative hyperplane. Takeuchi identifies this hyperplane scheme-theoretically with the inverse image of Z_0, and identifies $\mathbb{P}(E_{\mathfrak{m}})$ with the blowup [Tak19, Lemmas 3.2 and 3.3(1), Theorem 3.4]. Pulling these identifications back along the open immersion defining $\tilde{C}_{\mathfrak{m}}^{(d)}$ gives the asserted description of H. The statement on geometric fibers follows by base change. □

3.3 Tame Ramification Along the Full Boundary

The ramification calculation of Chapter 2 concerns a modulus supported at a single point and a symmetric power of the corresponding degree. We now combine those local calculations over the different points of $\mathfrak{m}$ and then pass from degree $\deg \mathfrak{m}$ to every sufficiently large degree d.

We retain the blowup notation of Chapter 2. For every nonzero effective submodulus $\mathfrak{n} \leq \mathfrak{m}$, let $z_{\mathfrak{n}} \in C^{(\deg \mathfrak{n})}$ be the point representing the divisor $\mathfrak{n}$, let $Z_{\mathfrak{n}}$ be the corresponding closed point, and put

$$X_{\mathfrak{n}} = \mathrm{Bl}_{Z_{\mathfrak{n}}} C^{(\deg \mathfrak{n})}, \qquad E_{\mathfrak{n}} = X_{\mathfrak{n}} \times_{C^{(\deg \mathfrak{n})}} Z_{\mathfrak{n}}.$$

We write $\eta_{\mathfrak{n}}$ for the generic point of $E_{\mathfrak{n}}$. For the full modulus in degree d, the corresponding center, blowup, exceptional divisor, and generic point are denoted by $Z_0 = Z_0(\mathfrak{m}, d)$, $X_{\mathfrak{m},d}$, $E_0 = E_{\mathfrak{m},d}$, and $\eta_0 = \eta_{\mathfrak{m},d}$, as in Section 3.2.

Theorem 63. *Let* Λ *be a finite commutative ring of cardinality invertible in* k*, and let* $\mathcal{F}$ *be an étale sheaf of* Λ*-modules, locally free of rank* 1 *on* U*, with ramification bounded by* $\mathfrak{m}$*. For every sufficiently large integer* d*, the local system* $\mathcal{F}^{[d]}$ *is tamely ramified along*

$$H = \widetilde{C}_{\mathfrak{m}}^{(d)} \setminus U^{(d)} = E_0 \times_{X_{\mathfrak{m},d}} \widetilde{C}_{\mathfrak{m}}^{(d)} \cong Z_0 \times_{C^{(d)}} \widetilde{C}_{\mathfrak{m}}^{(d)}.$$

3.3.1 From Point-Supported Moduli to the Full Modulus

The first step is to combine the single-point calculations. The following product-compatibility result is placed here because it is used only in the coprime-modulus reduction that immediately follows it.

Compatibility under products

Let X and Y be smooth integral schemes over a field k, and let $x \in X(k)$ and $y \in Y(k)$. We denote the blowups of these schemes at the given points by $\pi_X : \mathrm{Bl}_x(X) \to X$ and $\pi_Y : \mathrm{Bl}_y(Y) \to Y$. Furthermore, let $\pi_{X\times Y} : \mathrm{Bl}_{(x,y)}(X \times_k Y) \to X \times_k Y$ be the blowup of the product scheme at the point (x, y). We denote by E_X, E_Y, and $E_{X\times Y}$ the respective exceptional divisors, and let η_X, η_Y, and $\eta_{X\times Y}$ be their generic points.

In this section, we establish the following result concerning the stability of tame ramification under the external product of torsors.

Proposition 64. *Let G be a finite abelian group. Suppose $\mathcal{G}_X$ and $\mathcal{G}_Y$ are G-torsors defined on open subsets $U_X \subset \mathrm{Bl}_x(X)$ and $U_Y \subset \mathrm{Bl}_y(Y)$ that are disjoint from the exceptional divisors. If $\mathcal{G}_X$ is tamely ramified at η_X and $\mathcal{G}_Y$ is tamely ramified at η_Y, then the external product torsor*

$$\mathcal{G}_{X\times Y} := pr_1^{-1}\mathcal{G}_X \otimes pr_2^{-1}\mathcal{G}_Y$$

is tamely ramified at the generic point $\eta_{X\times Y}$ of the exceptional divisor in the product blowup.

Proof. The proposition is purely local in nature, so it suffices to consider the case where X and Y are affine. More precisely, by the smoothness of X and Y, we may restrict our attention to open neighborhoods of x and y that are etale over affine spaces. The rest of this section treats that case.

The Affine Case

Let $X = \mathbb{A}^n_k$ be the affine n-space over a field k, and let $0 \in X$ be the origin. Let $\tilde{X} = \mathrm{Bl}_0(X)$ be the blowup of X at the origin. Recall that $\tilde{X} \subset X \times_k \mathbb{P}^{n-1}_k$ is defined by the equations $x_i u_j = x_j u_i$, where $[u_1 : \cdots : u_n]$ are the homogeneous coordinates of $\mathbb{P}^{n-1}_k$. The exceptional divisor $E \subset \tilde{X}$ is the fiber over the origin, $E = \{(0, [u_1 : \cdots : u_n])\}$, which is of codimension 1 in $\tilde{X}$. Let $\eta \in E$ be the generic point of E, and let $R = \mathcal{O}_{\tilde{X},\eta}$ be the associated local ring. This ring R is a discrete valuation ring (DVR) with fraction field $K = K(X) = k(x_1, \ldots, x_n)$.

On the affine chart U_1 where $u_1 \neq 0$, we have $x_i = \frac{u_i}{u_1} x_1$. The coordinate ring is:

$$\mathcal{O}_{\tilde{X}}(U_1) = k\left[x_1, \frac{u_2}{u_1}, \ldots, \frac{u_n}{u_1}\right]$$

In this chart, the generic point η corresponds to the prime ideal $\mathfrak{p}_1 = (x_1)$. Thus, the local ring is $R = k[x_1, \frac{u_2}{u_1}, \ldots, \frac{u_n}{u_1}]_{(x_1)}$. The residue field is $\kappa(\eta) = k(\frac{u_2}{u_1}, \ldots, \frac{u_n}{u_1})$, and the completion of R with respect to its maximal ideal is:

$$\hat{R} = \kappa(\eta)[[x_1]]$$

In this local ring, x_1 is a uniformizer. Note that any x_i (for $i > 1$) can also serve as a uniformizer, as $x_i = (\frac{u_i}{u_1})x_1$ and $\frac{u_i}{u_1}$ is a unit in R.

For any monomial $M = x_1^{a_1} \ldots x_n^{a_n} \in K$, we can write:

$$M = x_1^{\sum a_i} \left(\frac{u_2}{u_1}\right)^{a_2} \ldots \left(\frac{u_n}{u_1}\right)^{a_n}$$

Since the term in parentheses is a unit in R, the valuation ν_E associated with E satisfies:

$$\nu_E(M) = \sum a_i = \deg M$$

Consequently, for any polynomial $f = f_d + f_{d+1} + \cdots + f_l$, where f_i is the homogeneous part of degree i, we have $\nu_E(f) = d$ (the order of vanishing at the origin).

The Product Case

Now, let $X = \mathbb{A}^n$ and $Y = \mathbb{A}^m$ with origins $x = 0$ and $y = 0$. As before, $\mathrm{Bl}_0(X) \subset X \times \mathbb{P}^{n-1}$ and $\mathrm{Bl}_0(Y) \subset Y \times \mathbb{P}^{m-1}$ have exceptional divisors E_X and E_Y respectively. Consider the product $X \times_k Y \cong \mathbb{A}_k^{n+m}$. The blowup of the product at the origin $(0,0)$, denoted $\mathrm{Bl}_{(0,0)}(X \times_k Y)$, is a subscheme of $(X \times Y) \times \mathbb{P}^{n+m-1}$ defined by:

$$\begin{cases} x_i w_j = x_j w_i & 1 \le i, j \le n \\ y_k w_{n+l} = y_l w_{n+k} & 1 \le k, l \le m \\ x_i w_{n+j} = y_j w_i & 1 \le i \le n,\ 1 \le j \le m \end{cases}$$

where $[w_1 : \cdots : w_{n+m}]$ are the homogeneous coordinates of $\mathbb{P}^{n+m-1}$. The exceptional divisor $E_{X\times Y}$ is isomorphic to $\mathbb{P}^{n+m-1}$.

Comparison of Blowups

Both $\mathrm{Bl}_0(X) \times_k \mathrm{Bl}_0(Y)$ and $\mathrm{Bl}_{(0,0)}(X \times_k Y)$ are birational to $X \times_k Y$; their common dense open is

$$(X \setminus \{0\}) \times_k (Y \setminus \{0\}).$$

Put $B = \mathrm{Bl}_{(0,0)}(X \times_k Y)$, and let

$$q \colon B \longrightarrow \mathbb{P}_k^{n+m-1}$$

be the second projection in the above realization of B. Define

$$\Omega_X = \bigcup_{i=1}^{n} D_+(w_i), \qquad \Omega_Y = \bigcup_{j=1}^{m} D_+(w_{n+j}),$$

and

$$\widetilde{U} := q^{-1}(\Omega_X \cap \Omega_Y) = \bigcup_{\substack{1\le i\le n \\ 1\le j\le m}} q^{-1}\big(D_+(w_i w_{n+j})\big).$$

There is a morphism

$$\begin{aligned} f \colon \widetilde{U} &\longrightarrow \mathrm{Bl}_0(X) \times_k \mathrm{Bl}_0(Y), \\ ((x,y),[w_1 : \cdots : w_{n+m}]) &\longmapsto \big((x,[w_1 : \cdots : w_n]),(y,[w_{n+1} : \cdots : w_{n+m}])\big). \end{aligned}$$

Thus we have a diagram

$$\begin{array}{ccccc} & & \widetilde{U} & & \\ & \swarrow^{f} & & \searrow^{j} & \\ \mathrm{Bl}_0(X) \times_k \mathrm{Bl}_0(Y) & & & & B, \end{array}$$

where j is the open immersion. The restriction of q induces an isomorphism $E_{X\times Y} \xrightarrow{\sim} \mathbb{P}_k^{n+m-1}$. Since neither $V_+(w_1, \ldots, w_n)$ nor $V_+(w_{n+1}, \ldots, w_{n+m})$ contains the generic point of $\mathbb{P}_k^{n+m-1}$, we have $\eta_{X\times Y} \in \widetilde{U}$.

Extensions of DVRs

Let

$$S = \mathcal{O}_{B,\eta_{X\times Y}}, \qquad R_X = \mathcal{O}_{\mathrm{Bl}_0(X),\eta_X}, \qquad R_Y = \mathcal{O}_{\mathrm{Bl}_0(Y),\eta_Y}.$$

The morphisms $\mathrm{pr}_X \circ f$ and $\mathrm{pr}_Y \circ f$ send $\eta_{X\times Y}$ to η_X and η_Y, respectively, and therefore induce local homomorphisms of DVRs

$$R_X \longrightarrow S, \qquad R_Y \longrightarrow S.$$

The point $\eta_{X\times Y}$ belongs to $q^{-1}(D_+(w_1 w_{n+1}))$. On this chart,

$$\begin{aligned} S &= k\left[x_1, \frac{x_2}{x_1}, \dots, \frac{x_n}{x_1}, \frac{y_1}{x_1}, \dots, \frac{y_m}{x_1}\right]_{(x_1)} \\ &= k\left[y_1, \frac{x_1}{y_1}, \dots, \frac{x_n}{y_1}, \frac{y_2}{y_1}, \dots, \frac{y_m}{y_1}\right]_{(y_1)}. \end{aligned}$$

Consequently, x_1 is a uniformizer in both R_X and S, while y_1 is a uniformizer in both R_Y and S. Both extensions have ramification index one. Their residue-field extensions are

$$k\left(\frac{x_2}{x_1}, \dots, \frac{x_n}{x_1}\right) \subset k\left(\frac{x_2}{x_1}, \dots, \frac{x_n}{x_1}, \frac{y_1}{x_1}, \dots, \frac{y_m}{x_1}\right)$$

and

$$k\left(\frac{y_2}{y_1}, \dots, \frac{y_m}{y_1}\right) \subset k\left(\frac{x_1}{y_1}, \dots, \frac{x_n}{y_1}, \frac{y_2}{y_1}, \dots, \frac{y_m}{y_1}\right),$$

respectively. In particular, both are purely transcendental and hence separable.

Ramification of G-Torsors

The extensions $R_X \to S$ and $R_Y \to S$ are weakly unramified and have separable residue-field extensions. Hence Lemma 44 shows that the pullbacks of $\mathcal{G}_X$ and $\mathcal{G}_Y$ to $\mathrm{Frac}(S)$ are tamely ramified. Their contracted product is the restriction of $\mathcal{G}_{X\times Y}$ to $\mathrm{Frac}(S)$. Therefore Lemma 43 shows that $\mathcal{G}_{X\times Y}$ is tamely ramified at $\eta_{X\times Y}$. □

Lemma 65. *Let $\mathfrak{n}_1, \mathfrak{n}_2 \subset \mathfrak{m}$ be two coprime nonzero submoduli. Assume that $\mathcal{P}^{[\deg \mathfrak{n}_1]}$ and $\mathcal{P}^{[\deg \mathfrak{n}_2]}$ are tamely ramified at $\eta_{\mathfrak{n}_1}$ and $\eta_{\mathfrak{n}_2}$, respectively. Then $\mathcal{P}^{[\deg \mathfrak{n}_1 + \deg \mathfrak{n}_2]}$ is tamely ramified at $\eta_{\mathfrak{n}_1+\mathfrak{n}_2}$.*

Proof. Apply Proposition 64 to the two torsors and then use the compatibility of symmetric powers with addition of divisors from Proposition 28. □

3.3.2 From the Minimal Degree to Large Degrees

We now propagate the result in degree $\deg \mathfrak{m}$ to degree d by adding an arbitrary effective divisor of degree $d - \deg \mathfrak{m}$.

The following addition lemma is adapted from [Tak19, Lemma 4.1].

Lemma 66. *Let C be a projective, smooth, and geometrically connected curve over a perfect field k. Let $\mathfrak{m} = \sum_{i=1}^{r} k_i P_i$ be an effective divisor where $P_1, \dots, P_r$ are distinct closed points. Let $U = C \setminus \mathfrak{m}$ and let $d \geq \deg \mathfrak{m}$.*

Suppose $\mathfrak{n}_1, \ldots, \mathfrak{n}_l$ are pairwise coprime submoduli of $\mathfrak{m}$ such that $\mathfrak{m} = \sum_{j=1}^{l} \mathfrak{n}_j$. Consider the summation morphism:

$$\pi : C^{(\deg \mathfrak{n}_1)} \times_k \cdots \times_k C^{(\deg \mathfrak{n}_l)} \times_k C^{(d-\deg \mathfrak{m})} \longrightarrow C^{(d)}$$

defined by $(D_1, \ldots, D_l, D_{extra}) \mapsto \sum_{j=1}^{l} D_j + D_{extra}$.

Then π is étale at the generic point of the closed subvariety

$$V = \{\mathfrak{n}_1\} \times_k \cdots \times_k \{\mathfrak{n}_l\} \times_k C^{(d-\deg \mathfrak{m})}$$

inside the domain $C^{(\deg \mathfrak{n}_1)} \times_k \cdots \times_k C^{(\deg \mathfrak{n}_l)} \times_k C^{(d-\deg \mathfrak{m})}$.

Proof. We may assume that k is algebraically closed (hence $\deg P_i = 1$ for all i). The morphism π is finite: it is induced by the finite quotient morphism

$$C^d/\left(\mathfrak{S}_{\deg \mathfrak{n}_1} \times \cdots \times \mathfrak{S}_{\deg \mathfrak{n}_l} \times \mathfrak{S}_{d-\deg \mathfrak{m}}\right) \longrightarrow C^d/\mathfrak{S}_d.$$

Its source and target are smooth k-schemes of pure dimension d. Hence π is flat by miracle flatness, and therefore finite flat. It is enough to show that there exists a closed point Q of $\mathfrak{n}_1 + \ldots \mathfrak{n}_l + C^{(d-\deg \mathfrak{m})} \subset C^{(d)}$ over which there are $\deg \pi$ points on $C^{(\deg \mathfrak{n}_1)} \times_k \cdots \times_k C^{(\deg \mathfrak{n}_l)} \times_k C^{(d-\deg \mathfrak{m})}$. (Because it will be unramified at this point and thus also at the generic point of V.) Choose Q as a point corresponding to a divisor $\mathfrak{n}_1 + \cdots + \mathfrak{n}_l + P_{r+1} + \cdots + P_{r+d-\deg \mathfrak{m}}$, where $P_1, \ldots, P_{r+d-\deg \mathfrak{m}}$ are distinct points of $U(k)$. □

Corollary 67. *The morphism $C^{(\deg \mathfrak{m})} \times_k C^{(d-\deg \mathfrak{m})} \xrightarrow{\pi} C^{(d)}$ is finite flat everywhere, and étale at the generic point of the closed subvariety $Z_{\mathfrak{m}} \times_k C^{(d-\deg \mathfrak{m})} \subset C^{(\deg \mathfrak{m})} \times_k C^{(d-\deg \mathfrak{m})}$.*

Proof. This is the case $l = 1$ and $\mathfrak{n}_1 = \mathfrak{m}$ of Lemma 66. □

Following from this, we look at the following diagram, coming from the flat base change $C^{(d-\deg \mathfrak{m})} \to \mathrm{Spec}(k)$ (Proposition 20) of (2.1):

$$\begin{array}{ccccc}
E_{\mathfrak{m}} \times_k C^{(d-\deg \mathfrak{m})} & \longrightarrow & X_{\mathfrak{m}} \times_k C^{(d-\deg \mathfrak{m})} & & \mathcal{F}^{[\deg \mathfrak{m}]} \times_k C^{(d-\deg \mathfrak{m})} \\
\downarrow & & \downarrow & \nearrow & \downarrow \\
Z_{\mathfrak{m}} \times_k C^{(d-\deg \mathfrak{m})} & \longrightarrow & C^{(\deg \mathfrak{m})} \times_k C^{(d-\deg \mathfrak{m})} & & U^{(\deg \mathfrak{m})} \times_k C^{(d-\deg \mathfrak{m})}
\end{array}$$

Note that $U^{(\deg \mathfrak{m})} \times_k C^{(d-\deg \mathfrak{m})}$ is dense open subscheme of $X_{\mathfrak{m}} \times_k C^{(d-\deg \mathfrak{m})}$, And $E_{\mathfrak{m}} \times_k C^{(d-\deg \mathfrak{m})}$ is a prime divisor of $X_{\mathfrak{m}} \times_k C^{(d-\deg \mathfrak{m})}$. Hence it is a well-defined question according to Definition 38 to ask whether $\mathcal{F}^{[\deg \mathfrak{m}]} \times_k C^{(d-\deg \mathfrak{m})}$ is tamely ramified at the generic point θ of $E_{\mathfrak{m}} \times_k C^{(d-\deg \mathfrak{m})}$.

Lemma 68. *If $\mathcal{F}^{[\deg \mathfrak{m}]}$ is tamely ramified at $\eta_{\mathfrak{m}}$, then $\mathcal{F}^{[\deg \mathfrak{m}]} \times_k C^{(d-\deg \mathfrak{m})}$ is tamely ramified at the generic point θ of $E_{\mathfrak{m}} \times_k C^{(d-\deg \mathfrak{m})}$.*

Proof. This is the stability of tame ramification in codimension one under base change ([Stacks, Tag 0EYD]), applied to the projection

$$X_{\mathfrak{m}} \times_k C^{(d-\deg \mathfrak{m})} \longrightarrow X_{\mathfrak{m}}.$$

□

By Lemma 68, replacing $C^{(d-\deg \mathfrak{m})}$ with the dense open subscheme $U^{(d-\deg \mathfrak{m})} \subset C^{(d-\deg \mathfrak{m})}$, we get that the G-torsor $p_1^{-1}\mathcal{P}^{[\deg \mathfrak{m}]}$, where $\mathcal{P} = \mathrm{Fr}(\mathcal{F})$, is tamely ramified at θ the generic point of $E_{\mathfrak{m}} \times_k U^{(d-\deg \mathfrak{m})}$, viewed as a prime divisor of $X_{\mathfrak{m}} \times_k U^{(d-\deg \mathfrak{m})}$; the torsor itself is defined on the dense open $U^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})}$, where $p_1 : U^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})} \to U^{(\deg \mathfrak{m})}$ is the projection to the first factor.

Looking at the second projection $p_2 : X_{\mathfrak{m}} \times_k U^{(d-\deg \mathfrak{m})} \to U^{(d-\deg \mathfrak{m})}$, and the fact that $\mathcal{P}^{[d-\deg \mathfrak{m}]}$ is étale on $U^{(d-\deg \mathfrak{m})}$ we get that $p_2^{-1}\mathcal{P}^{[d-\deg \mathfrak{m}]} = X_{\mathfrak{m}} \times_k \mathcal{P}^{[d-\deg \mathfrak{m}]}$ is étale on $X_{\mathfrak{m}} \times_k U^{(d-\deg \mathfrak{m})}$. Hence, its restriction to $U^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})}$ is unramified at θ.

Thus, by Lemma 69, we conclude that $\mathcal{P}^{[\deg \mathfrak{m}]} \boxtimes \mathcal{P}^{[d-\deg \mathfrak{m}]} = p_1^{-1}\mathcal{P}^{[\deg \mathfrak{m}]} \otimes^G p_2^{-1}\mathcal{P}^{[d-\deg \mathfrak{m}]}$ is tamely ramified at θ the generic point of $E_{\mathfrak{m}} \times_k U^{(d-\deg \mathfrak{m})}$.

Lemma 69. *Let $X \to \operatorname{Spec} k$ be a scheme over a field k, and let $\mathcal{P}_1, \mathcal{P}_2$ be two G-torsors on U_{et}. Let ξ be the generic point of a prime divisor $D \subset X$. If $\mathcal{P}_1$ is tamely ramified at ξ, and $\mathcal{P}_2$ is unramified at ξ, then the product $\mathcal{P}_1 \otimes^G \mathcal{P}_2$ is tamely ramified at ξ.*

Proof. This follows from Lemma 43. □

Combining this with Corollary 67 we get

Corollary 70. *Let $\mathfrak{m}$ be a modulus as above, and let $\eta_{\mathfrak{m}}$ be the generic point of $E_{\mathfrak{m}}$. Let $\mathcal{P}$ be a G-torsor on U_{et} with ramification bounded by $\mathfrak{m}$. Assume $\mathcal{P}^{[\deg \mathfrak{m}]}$ is tamely ramified at $\eta_{\mathfrak{m}}$. Then $\mathcal{P}^{[\deg \mathfrak{m}]} \boxtimes \mathcal{P}^{[d-\deg \mathfrak{m}]}$ is tamely ramified at the generic point θ of $E_{\mathfrak{m}} \times_k C^{(d-\deg \mathfrak{m})} \subset C^{(\deg \mathfrak{m})} \times_k C^{(d-\deg \mathfrak{m})}$, and $\mathcal{P}^{[d]}$ is tamely ramified at the generic point $\eta_0 = \eta_{\mathfrak{m},d}$ of $E_0 = E_{\mathfrak{m},d}$*

Proof. The first assertion, that $\mathcal{P}^{[\deg \mathfrak{m}]} \boxtimes \mathcal{P}^{[d-\deg \mathfrak{m}]}$ is tamely ramified at θ, follows from the preceding discussion. Thus, it remains to show that $\mathcal{P}^{[d]}$ is tamely ramified at η_0.

Consider the blowup diagram defining $X_{\mathfrak{m},d}$:

$$\begin{array}{ccc} \overline{\{\eta_0\}} = E_0 & \longrightarrow & X_{\mathfrak{m},d} \\ \downarrow & & \downarrow \pi \\ Z_0 & \xrightarrow{c.i} & C^{(d)} \end{array}$$

By performing a base change along the flat addition map $+ : C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})} \to C^{(d)}$, we obtain the following commutative diagram:

$$\begin{array}{ccccc} & & \left(C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})}\right) \times_{C^{(d)}} E_0 & \longrightarrow & \overline{\{\eta_0\}} = E_0 \\ & & \downarrow & & \downarrow \\ & & \left(C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})}\right) \times_{C^{(d)}} X_{\mathfrak{m},d} & \longrightarrow & X_{\mathfrak{m},d} \\ & & \downarrow & & \downarrow \pi \\ \mathfrak{m} \times_k U^{(d-\deg \mathfrak{m})} & \xrightarrow{c.i} & C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})} & \xrightarrow{+} & C^{(d)} \end{array}$$

By Theorem 30, blowups commute with flat base change. Since the inverse image of the center Z_0 under the map $+$ is $\mathfrak{m} \times_k U^{(d-\deg \mathfrak{m})}$, the scheme $\left(C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})}\right) \times_{C^{(d)}} X_{\mathfrak{m},d}$ is the blowup

of $C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})}$ along $\mathfrak{m} \times_k U^{(d-\deg \mathfrak{m})}$. This, in turn, is isomorphic to the base change of the blowup $X_{\mathfrak{m}}$ (of $C^{(\deg \mathfrak{m})}$ along $\mathfrak{m}$) via the (flat) projection $C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})} \to C^{(\deg \mathfrak{m})}$.

Assembling these facts, we obtain the following Cartesian square:

$$\begin{array}{ccccc}
 & & E_{\mathfrak{m}} \times U^{(d-\deg \mathfrak{m})} & \xrightarrow{\tilde{+}} & \overline{\{\eta_0\}} = E_0 \\
 & & \downarrow & & \downarrow \\
 & & X_{\mathfrak{m}} \times_k U^{(d-\deg \mathfrak{m})} & \xrightarrow{\tilde{+}} & X_{\mathfrak{m},d} \\
 & & \downarrow & & \downarrow \pi \\
\mathfrak{m} \times_k U^{(d-\deg \mathfrak{m})} & \xrightarrow{c.i} & C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})} & \xrightarrow{+} & C^{(d)}
\end{array}$$

The point θ defined in the Corollary is the generic point of $E_{\mathfrak{m}} \times_k U^{(d-\deg \mathfrak{m})}$. Let η be the generic point of $\mathfrak{m} \times_k U^{(d-\deg \mathfrak{m})}$. By Corollary 67, the map $+$ is étale at η. Consequently, the lifted map $\tilde{+}$ is étale at θ. Given the isomorphism $(+^{-1})(\mathcal{P}^{[d]}) \cong \mathcal{P}^{[\deg \mathfrak{m}]} \boxtimes \mathcal{P}^{[d-\deg \mathfrak{m}]}$ from Proposition 28, the tame ramification of the box product at θ descends to the tame ramification of $\mathcal{P}^{[d]}$ at η_0 by applying Lemma 44.

□

3.3.3 Conclusion of the Tameness Argument

Proof of Theorem 63. We first reduce to the case where k is algebraically closed. The formations of symmetric powers, Takeuchi's blowup and its boundary, and the frame torsor commute with the flat base change $\mathrm{Spec}(\bar{k}) \to \mathrm{Spec}(k)$. At the generic points of the boundary, this base change induces a weakly unramified extension of discrete valuation rings with separable residue-field extension. Hence Lemma 44 shows that tameness over k is equivalent to tameness after base change to $\bar{k}$. We may therefore replace k by $\bar{k}$.

Because the original field is perfect, each closed point P_i splits into distinct $\bar{k}$-points, and the base-changed modulus is

$$\mathfrak{m}_{\bar{k}} = \sum_i \sum_{\bar{P} \mapsto P_i} n_i \bar{P}.$$

In the remainder of the proof we write k, C, $\mathfrak{m}$, and $\mathcal{F}$ for these base-changed objects.

Let $\mathcal{P} = \mathrm{Fr}(\mathcal{F})$ be the $\Lambda^\times$-torsor of frames. For each single-point submodulus $\mathfrak{n} = k_P P$ of $\mathfrak{m}$, Theorem 3 shows that $\mathcal{P}^{[\deg \mathfrak{n}]}$ is tamely ramified at $\eta_{\mathfrak{n}}$. Repeated application of Lemma 65 shows that $\mathcal{P}^{[\deg \mathfrak{m}]}$ is tamely ramified at $\eta_{\mathfrak{m}}$. Therefore Corollary 70 implies that $\mathcal{P}^{[d]}$ is tamely ramified at the generic point η_0 of E_0 for every sufficiently large d. The same inertia character governs $\mathcal{F}^{[d]}$, so the same conclusion holds for $\mathcal{F}^{[d]}$. Since H is the nonempty open part of E_0 described in Corollary 62, its generic point is η_0. Thus $\mathcal{F}^{[d]}$ is tamely ramified along H. □

3.4 Extension Across the Boundary

We now use the geometry of the Abel–Jacobi fibers to upgrade the tameness proved above to extension across the boundary. On the geometric generic fiber, a Tendler-style prime-to-p power argument forces the boundary inertia representation to be trivial.

Lemma 71. *Let Λ be a finite commutative ring whose cardinality is invertible in k, and let $\mathcal{F}^{[d]}$ be a locally constant sheaf of free rank-one Λ-modules on $U^{(d)}$. If $\mathcal{F}^{[d]}$ is tamely ramified at the generic point of*

$$H = \widetilde{C}^{(d)}_{\mathfrak{m}} \setminus U^{(d)},$$

then it extends uniquely to a locally constant sheaf $\widetilde{\mathcal{F}}^{[d]}$ on $\widetilde{C}^{(d)}_{\mathfrak{m}}$.

Proof. Put $r = d - \deg \mathfrak{m} - g + 1$, let ξ be the generic point of $\mathrm{Pic}^d_{C,\mathfrak{m}}$, let $\Omega = \kappa(\xi)^{\mathrm{sep}}$, and write $\bar{\xi} = \operatorname{Spec} \Omega$. By Corollary 62, the compactified geometric generic Abel–Jacobi fiber and its boundary are

$$\left(\widetilde{C}^{(d)}_{\mathfrak{m},\bar{\xi}}, H_{\bar{\xi}}\right) \cong \left(\mathbb{P}^r_\Omega, \mathbb{P}^{r-1}_\Omega\right),$$

and hence the geometric generic open fiber is

$$U^{(d)}_{\bar{\xi}} \cong \mathbb{A}^r_\Omega.$$

Let

$$\chi \colon \pi_1\left(\mathbb{A}^r_\Omega\right) \longrightarrow \Lambda^\times$$

be the character corresponding to the restriction of $\mathcal{F}^{[d]}$ to this fiber.

Let η be the generic point of H, and let $\bar{\eta}$ be the generic point of $H_{\bar{\xi}}$. Because H is a relative hyperplane divisor, the passage from η to $\bar{\eta}$ preserves a local equation of the boundary and induces a separable extension of residue fields. The corresponding extension of henselian discrete valuation rings is therefore unramified, and it identifies the inertia group at η with the inertia group

$$I_\infty := I_{\bar{\eta}}$$

at the hyperplane at infinity. In particular, tameness and triviality of the finite inertia representation may both be checked after this base change.

If $\mathrm{char}(k) = 0$, then

$$\pi_1\left(\mathbb{A}^r_\Omega\right) = 1,$$

so χ is trivial [Ten15, Example 4.14].

Suppose now that $\mathrm{char}(k) = p > 0$. Since $\mathcal{F}^{[d]}$ is tame along H, the finite group $\chi(I_\infty)$ has order prime to p. Let N be its exponent. Then $(N, p) = 1$, and the character χ^N is trivial on I_∞.

Consequently, the local system corresponding to χ^N is unramified at the generic point of $\mathbb{P}^{r-1}_\Omega$. By purity, it extends to $\mathbb{P}^r_\Omega$. Since

$$\pi_1\left(\mathbb{P}^r_\Omega\right) = 1,$$

this extension is constant, and therefore

$$\chi^N = 1.$$

On the other hand, χ factors through

$$\pi_1\left(\mathbb{A}^r_\Omega\right)^{\mathrm{ab}},$$

which is a pro-p group [Ten15, Example 4.15]. Hence the finite group $\mathrm{im}(\chi)$ is a p-group. Since its exponent divides N, and N is prime to p, it follows that

$$\mathrm{im}(\chi) = 1.$$

Thus χ is trivial.

In either characteristic, the restriction of χ to I_∞ is trivial. By the inertia comparison above, the inertia group at η acts trivially on $\mathcal{F}^{[d]}$.

Therefore $\mathcal{F}^{[d]}$ is unramified at the generic point of H. Since $\widetilde{C}_{\mathfrak{m}}^{(d)}$ is regular and its complement of $U^{(d)}$ is the single smooth relative hyperplane divisor H, purity of the branch locus ([Stacks, Tag 0BMB]) gives an extension of $\mathcal{F}^{[d]}$ to $\widetilde{C}_{\mathfrak{m}}^{(d)}$. The extension is unique because

$$\pi_1\big(U^{(d)}\big) \longrightarrow \pi_1\big(\widetilde{C}_{\mathfrak{m}}^{(d)}\big)$$

is surjective. □

Remark. The argument does not assert that tame ramification is generally unramified. Tameness is used only to choose N prime to p. The conclusion that the inertia character is trivial then follows from the global geometry of the geometric generic fiber and from the fact that the abelian fundamental group of affine space is pro-p.

3.5 Descent and Reduced Geometric Class Field Theory

We finish by recording the fundamental-group input and applying it to the ordinary and generalized Abel–Jacobi morphisms.

3.5.1 The Fundamental-Group Input

Proposition 72 ([Stacks, Tag 0C0J])**.** *Let $f\colon X \to S$ be a flat proper morphism of finite presentation whose geometric fibers are connected and reduced. Assume that S is connected, and let $\bar{s}$ be a geometric point of S. Then there is an exact sequence*

$$\pi_1(X_{\bar{s}}) \longrightarrow \pi_1(X) \longrightarrow \pi_1(S) \longrightarrow 1.$$

Corollary 73. *Let $f\colon X \to S$ be a proper smooth morphism of finite presentation whose geometric fibers are connected. Then the sequence of Proposition 72 is exact.*

Proof. A smooth geometric fiber is reduced, so this is an immediate application of Proposition 72. □

Corollary 74. *Let $f\colon X \to S$ be a projective space bundle over a connected scheme. Then pullback induces an isomorphism*

$$\pi_1^{\mathrm{et}}(X) \xrightarrow{\sim} \pi_1^{\mathrm{et}}(S).$$

Proof. The geometric fibers are projective spaces and therefore have trivial étale fundamental group. The assertion follows from Corollary 73. □

3.5.2 The Descent Argument

We work over $S = \operatorname{Spec} k$, where k is perfect. It is convenient first to prove the following torsor formulation.

Theorem 75 (Torsor formulation). *Let $G = \Lambda^\times$, and let $\mathcal{P}$ be a G-torsor on U whose ramification is bounded by $\mathfrak{m}$. For every sufficiently large integer d, there exists a G-torsor $\mathcal{Q}_d$ on $\mathrm{Pic}^d_{C,\mathfrak{m}}$, unique up to isomorphism, such that*

$$\Phi_d^* \mathcal{Q}_d \cong \mathcal{P}^{[d]}.$$

Proof. We divide the proof into two cases.

Case 1: $\mathfrak{m} = 0$. For d sufficiently large, the usual Abel–Jacobi morphism

$$\Phi_d \colon C^{(d)} \longrightarrow \mathrm{Pic}^d_C$$

is proper and smooth with geometrically connected fibers, and Theorem 60 identifies every geometric fiber with $\mathbb{P}^{d-g}$. Applying Proposition 72, and using $\pi_1^{\mathrm{et}}(\mathbb{P}^{d-g}) = 1$, gives

$$\pi_1^{\mathrm{et}}\big(C^{(d)}\big) \xrightarrow{\sim} \pi_1^{\mathrm{et}}\big(\mathrm{Pic}^d_C\big).$$

The G-torsor $\mathcal{P}^{[d]}$ therefore descends uniquely to a G-torsor on Pic^d_C.

Case 2: $\mathfrak{m} > 0$. Let $\mathcal{F} = \mathcal{P} \times^G \Lambda$ be the rank-one local system associated with $\mathcal{P}$; its symmetric power has frame torsor $\mathcal{P}^{[d]}$. For d sufficiently large, Theorem 61 extends Φ_d to a projective space bundle

$$\widetilde{\Phi}_d \colon \widetilde{C}^{(d)}_{\mathfrak{m}} \longrightarrow \mathrm{Pic}^d_{C,\mathfrak{m}}\,.$$

By Theorem 63, $\mathcal{F}^{[d]}$ is tamely ramified along $H = \widetilde{C}^{(d)}_{\mathfrak{m}} \setminus U^{(d)}$. Hence Lemma 71 gives a unique extension $\widetilde{\mathcal{F}}^{[d]}$ to $\widetilde{C}^{(d)}_{\mathfrak{m}}$.

By Corollary 74,

$$\pi_1^{\mathrm{et}}\big(\widetilde{C}^{(d)}_{\mathfrak{m}}\big) \xrightarrow{\sim} \pi_1^{\mathrm{et}}\big(\mathrm{Pic}^d_{C,\mathfrak{m}}\big).$$

Thus $\widetilde{\mathcal{F}}^{[d]}$ descends uniquely to a rank-one local system $\mathcal{G}_d$ on $\mathrm{Pic}^d_{C,\mathfrak{m}}$. Restricting to $U^{(d)}$ gives $\Phi_d^* \mathcal{G}_d \cong \mathcal{F}^{[d]}$. Taking frame torsors yields

$$\Phi_d^* \,\mathrm{Fr}(\mathcal{G}_d) \cong \mathcal{P}^{[d]},$$

so $\mathcal{Q}_d = \mathrm{Fr}(\mathcal{G}_d)$ has the required property. Uniqueness follows from the fundamental-group isomorphism and from the surjection

$$\pi_1^{\mathrm{et}}(U^{(d)}) \twoheadrightarrow \pi_1^{\mathrm{et}}(\widetilde{C}^{(d)}_{\mathfrak{m}})$$

induced by the dense open immersion into the normal scheme $\widetilde{C}^{(d)}_{\mathfrak{m}}$. □

Proof of Theorem 2. Let $\mathcal{F}$ be as in Theorem 2, and set $\mathcal{P} = \mathrm{Fr}(\mathcal{F})$. Apply Theorem 75 to obtain $\mathcal{Q}_d$ on $\mathrm{Pic}^d_{C,\mathfrak{m}}$, and let

$$\mathcal{G}_d = \mathcal{Q}_d \times^{\Lambda^\times} \Lambda.$$

The associated-module construction commutes with pullback and carries $\mathcal{P}^{[d]}$ to $\mathcal{F}^{[d]}$, so $\Phi_d^* \mathcal{G}_d \cong \mathcal{F}^{[d]}$. If $\mathcal{G}'_d$ is another such local system, then $\mathrm{Fr}(\mathcal{G}'_d)$ has the defining property of $\mathcal{Q}_d$. The uniqueness of $\mathcal{Q}_d$, followed by the associated-module construction, gives $\mathcal{G}'_d \cong \mathcal{G}_d$. □

Appendix A

Cyclic Torsor Theories and Witt Vectors

This appendix contains the supporting theory and proofs for the three cyclic torsor descriptions stated in Chapter 2. We begin with the common group-scheme construction and the classical Kummer and Artin–Schreier cases. We then develop the finite-length p-typical Witt-vector algebra needed for Artin–Schreier–Witt theory, followed by its torsor classification and normal-form arguments. The final section isolates the weighted-homogeneity property used in the cyclic symmetric-power calculation. The Schmid–Witt ramification-break formula is stated, referenced, and applied in Chapter 2; we do not reproduce its proof.

A.1 The common group-scheme construction

The three cyclic theories used in Chapter 2 are instances of the following construction.

A.1.1 The kernel torsor construction

Let H be a commutative group scheme over a scheme S, written additively, and let $\varphi : H \to H$ be a homomorphism. Its scheme-theoretic kernel $K := \ker(\varphi)$ is defined by the Cartesian square

$$\begin{array}{ccc} K & \longrightarrow & H \\ \downarrow & & \downarrow{\scriptstyle \varphi} \\ S & \xrightarrow[0_H]{} & H. \end{array} \tag{A.1}$$

Thus, for every S-scheme U,

$$K(U) = \ker(\varphi_U) = \{k \in H(U) : \varphi_U(k) = 0_{H(U)}\}, \tag{A.2}$$

where $H(U) = \mathrm{Hom}_S(U, H)$ and φ_U is induced by φ.

For an S-scheme X and a section $a \in H(X)$, define the fiber

$$T_a := X \times_{a,H,\varphi} H, \tag{A.3}$$

equivalently by the Cartesian square

$$\begin{array}{ccc} T_a & \longrightarrow & H \\ \downarrow & & \downarrow{\scriptstyle \varphi} \\ X & \xrightarrow[a]{} & H. \end{array}$$

For every X-scheme $U \to X$, with a_U the pullback of a,

$$T_a(U) = \{y \in H(U) : \varphi_U(y) = a_U\}.$$

Thus "the torsor $\varphi(Y) = a$" means the scheme of solutions after every base change to X; it has no preferred identity element in general.

Write $H_X := H \times_S X$, $K_X := K \times_S X$, let $\varphi_X : H_X \to H_X$ be the base change of φ, and let $\iota_K : K_X \to H_X$ and $\iota_a : T_a \to H_X$ be the canonical maps. Addition on H_X restricts to the action

$$K_X \times_X T_a \longrightarrow T_a, \qquad (k, y) \longmapsto y + k. \tag{A.4}$$

It is the top arrow in the commutative diagram

$$\begin{array}{ccc} K_X \times_X T_a & \xrightarrow{(k,y)\mapsto y+k} & T_a \\ {\scriptstyle \iota_K \times_X \iota_a}\downarrow & & \downarrow{\scriptstyle \iota_a} \\ H_X \times_X H_X & \xrightarrow[+_X]{} & H_X. \end{array}$$

Indeed,

$$\varphi_U(y + k) = \varphi_U(y) + \varphi_U(k) = a_U + 0_{H(U)} = a_U,$$

so translation by K preserves the fiber T_a. The action is simply transitive: the morphism

$$K_X \times_X T_a \longrightarrow T_a \times_X T_a, \qquad (k, y) \longmapsto (y, y + k), \tag{A.5}$$

is an isomorphism with inverse

$$(y_1, y_2) \longmapsto (y_2 - y_1, y_1).$$

This inverse is well-defined because

$$\varphi_U(y_2 - y_1) = \varphi_U(y_2) - \varphi_U(y_1) = a_U - a_U = 0_{H(U)}.$$

If φ is finite étale and surjective, then its base change $T_a \to X$ is finite étale and surjective. Hence T_a is a K_X-torsor, namely

$$T_a = a^*(H \xrightarrow{\varphi} H)$$

as a pullback of the K-torsor $\varphi : H \to H$.

Suppose that $a' = a + \varphi_X(b)$ for some $b \in H(X)$. Translation by b defines

$$\tau_b : T_a \longrightarrow T_{a'}, \qquad y \longmapsto y + b,$$

because

$$\varphi_U(y + b_U) = a_U + \varphi_U(b_U) = a'_U,$$

and it is K_X-equivariant since $\tau_b(y + k) = y + k + b = \tau_b(y) + k$. Its inverse is τ_{-b}, so T_a depends only on the class of a modulo $\varphi_X(H(X))$.

When φ is finite étale and surjective, regard the group schemes K_X and H_X as the sheaves that they represent on the étale site of X. Then

$$0 \longrightarrow K_X \longrightarrow H_X \xrightarrow{\varphi_X} H_X \longrightarrow 0$$

is a short exact sequence of sheaves. Applying étale cohomology gives the beginning of the associated long exact sequence

$$0 \longrightarrow H^0_{\text{ét}}(X, K_X) \longrightarrow H^0_{\text{ét}}(X, H_X) \xrightarrow{\varphi_X} H^0_{\text{ét}}(X, H_X) \xrightarrow{\delta} H^1_{\text{ét}}(X, K_X).$$

Because H_X is representable, its global sections are

$$H^0_{\text{ét}}(X, H_X) = \operatorname{Hom}_X(X, H_X) = \operatorname{Hom}_S(X, H) = H(X).$$

For $a \in H(X)$, the connecting class $\delta(a)$ is represented by the sheaf of local lifts of a through φ_X. This sheaf is precisely the pullback $T_a = X \times_{a,H,\varphi} H$, with its K_X-action, and hence $\delta(a) = [T_a]$. Exactness gives $\ker(\delta) = \varphi_X(H(X))$, so δ descends to the injection

$$H(X)/\varphi_X(H(X)) \hookrightarrow H^1_{\text{ét}}(X, K_X), \qquad [a] \longmapsto [T_a]. \tag{A.6}$$

The class $[T_a]$ is trivial exactly when $a \in \varphi_X(H(X))$, equivalently when $\varphi(Y) = a$ has a global solution.

For a multiplicative group scheme the same construction is written multiplicatively: $K(U) = \{k \in H(U) : \varphi_U(k) = 1_{H(U)}\}$, the action is $(k, y) \mapsto ky$, and replacing a by $a\varphi_X(b)$ gives the isomorphic torsor $y \mapsto by$.

A.1.2 The three cyclic theories

For an arbitrary S-scheme X, the fiber over a section $a \in H(X)$ is

$$T_a = \varphi_X^{-1}(a) = X \times_{a,H,\varphi} H.$$

For nonaffine X, explicit equations are interpreted using the structure sheaf $\mathcal{O}_X$. In all three applications in Chapter 2, however, the base is affine: first $\operatorname{Spec} k((x))$ and then $\mathbb{G}_m = \operatorname{Spec} k[x, x^{-1}]$. We therefore work affinely from now on and write

$$X = \operatorname{Spec} A,$$

where the capital letter A is the coordinate ring of X, while the lowercase letter a is the section specifying the fiber. More generally, if $S = \operatorname{Spec} R$ and $H = \operatorname{Spec} C$, then

$$T_a = \operatorname{Spec}\big(A \otimes_C C\big),$$

where C acts on A through $a^{\#} : C \to A$ and on the second copy of C through $\varphi^{\#} : C \to C$.

Kummer. Take $H = \mathbb{G}_m$ and $\varphi(z) = z^e$. A section is an element $a \in \mathbb{G}_m(X) = A^\times$, and its fiber is

$$T_a = \operatorname{Spec}\big(A[Y]/(Y^e - a)\big),$$

where Y is automatically a unit because a is a unit. The kernel is

$$\mu_{e,X} = \operatorname{Spec}\big(A[t,t^{-1}]/(t^e-1)\big),$$

and it acts on T_a through a morphism

$$\alpha : \mu_{e,X} \times_X T_a \longrightarrow T_a.$$

Writing

$$B = A[Y]/(Y^e - a) \quad \text{and} \quad D = A[t,t^{-1}]/(t^e-1),$$

the corresponding homomorphism of coordinate rings is

$$\alpha^{\#} : B \longrightarrow D \otimes_A B, \qquad \overline{Y} \longmapsto t \otimes \overline{Y},$$

where $\overline{Y}$ denotes the class of Y in B. This is well-defined because

$$(t \otimes \overline{Y})^e = t^e \otimes \overline{Y}^e = 1 \otimes a.$$

Equivalently, for every X-scheme U,

$$T_a(U) = \{y \in \Gamma(U,\mathcal{O}_U)^\times : y^e = a_U\}, \qquad \mu_e(U) = \{\xi \in \Gamma(U,\mathcal{O}_U)^\times : \xi^e = 1\},$$

and the action sends (ξ, y) to ξy; indeed, $(\xi y)^e = \xi^e y^e = a_U$. If e is invertible on S and a primitive root $\zeta_e \in \mu_e(S)$ is chosen, then $\bar{j} \mapsto \zeta_e^j$ identifies $(\mathbb{Z}/e\mathbb{Z})_S$ with μ_e. Only this identification, not the μ_e-torsor or its action, depends on the choice of ζ_e.

Artin–Schreier. Let A be an $\mathbb{F}_p$-algebra, take $H = \mathbb{G}_a$, and put $\varphi(Y) = Y^p - Y$. A section is an element $a \in \mathbb{G}_a(X) = A$, and its fiber is

$$T_a = \operatorname{Spec}\big(A[Y]/(Y^p - Y - a)\big).$$

The kernel is the constant group scheme $(\mathbb{F}_p)_X$, acting by $(c, Y) \mapsto Y + c$.

Artin–Schreier–Witt. Let A be an $\mathbb{F}_p$-algebra. The third theory takes $H = \mathbb{W}_r$ and

$$\varphi = \wp := F - \mathrm{id}, \qquad \wp\vec{Y} = F\vec{Y} \ominus \vec{Y}.$$

For a section $\vec{a} \in W_r(A)$, the fiber is explicitly

$$T_{\vec{a}} = \operatorname{Spec}\Big(A[Y_0,\dots,Y_{r-1}]/\big(\wp\vec{Y} \ominus \vec{a}\big)\Big), \qquad \wp\vec{Y} = \vec{a}.$$

Here the parenthesized Witt vector denotes the ideal generated by its r coordinate polynomials. Its kernel is $W_r(\mathbb{F}_p)_X \cong (\mathbb{Z}/p^r\mathbb{Z})_X$, acting by Witt translation $(\vec{c}, \vec{Y}) \mapsto \vec{Y} \oplus \vec{c}$.

This paragraph states the affine model in advance for comparison with the two preceding theories. The operations F, $\oplus$, and $\ominus$, and the coordinate polynomials occurring in the displayed quotient, are defined in Section A.3. The fact that this fiber is finite étale and that the displayed kernel action makes it a torsor is proved in Lemma 85.

A.2 Kummer and Artin–Schreier theory

Let K be a complete discretely valued field with residue characteristic $p > 0$.

A.2.1 Kummer theory

If $(e, p) = 1$ and K contains μ_e, the Kummer sequence

$$1 \longrightarrow \mu_e \longrightarrow \mathbb{G}_m \xrightarrow{(-)^e} \mathbb{G}_m \longrightarrow 1$$

and Hilbert 90 give

$$H^1_{\text{ét}}(\operatorname{Spec} K, \mu_e) \cong K^\times/(K^\times)^e.$$

Equivalently, since the base is a field, this étale cohomology group is the continuous Galois cohomology group

$$H^1_{\text{ét}}(\operatorname{Spec} K, \mu_e) \cong H^1_{\text{cont}}\big(\operatorname{Gal}(K^{\text{sep}}/K), \mu_e(K^{\text{sep}})\big).$$

Under this identification, the coset $[a] \in K^\times/(K^\times)^e$ corresponds to the μ_e-torsor

$$\operatorname{Spec} K[T]/(T^e - a),$$

on which $\zeta \in \mu_e$ acts by $T \mapsto \zeta T$. Thus replacing a by ab^e, for $b \in K^\times$, gives an isomorphic torsor. If $[a]$ has order e, adjoining a solution of $T^e = a$ gives a cyclic field extension of degree e.

A.2.2 Artin–Schreier theory

In characteristic p, the Artin–Schreier sequence

$$0 \longrightarrow \mathbb{F}_p \longrightarrow \mathbb{G}_a \xrightarrow{\wp} \mathbb{G}_a \longrightarrow 0, \qquad \wp(z) = z^p - z,$$

and additive Hilbert 90 give

$$H^1(K, \mathbb{Z}/p\mathbb{Z}) := H^1_{\text{ét}}(\operatorname{Spec} K, \mathbb{Z}/p\mathbb{Z}) \cong K/\wp(K).$$

Under this identification, the coset $[a] \in K/\wp(K)$ corresponds to the $\mathbb{Z}/p\mathbb{Z}$-torsor

$$\operatorname{Spec} K[X]/(X^p - X - a),$$

on which $c \in \mathbb{F}_p \cong \mathbb{Z}/p\mathbb{Z}$ acts by $X \mapsto X + c$. Thus replacing a by $a + (b^p - b)$, for $b \in K$, gives an isomorphic torsor. If $[a] \neq 0$, adjoining a solution of $X^p - X = a$ gives a cyclic field extension of degree p; conversely, every cyclic extension of degree p arises in this way.

Lemma 76 (Classical cyclic equations are étale torsors)**.** *Let A be a ring.*

(i) If n is invertible in A and $a \in A^\times$, then

$$A[T]/(T^n - a)$$

is finite free over A, with basis $1, T, \ldots, T^{n-1}$, and is étale over A. Multiplication on the T-coordinate makes its spectrum a representable $\mu_{n,\operatorname{Spec} A}$-torsor over $\operatorname{Spec} A$ for the étale topology.

(ii) If A is an $\mathbb{F}_p$-algebra and $a \in A$, then

$$A[X]/(X^p - X - a)$$

is finite free over A, with basis $1, X, \dots, X^{p-1}$, and is étale over A. Translation on the X-coordinate makes its spectrum a representable $(\mathbb{Z}/p\mathbb{Z})_{\operatorname{Spec} A}$-torsor over $\operatorname{Spec} A$ for the étale topology.

Proof. Both defining polynomials are monic, giving the stated bases and finite freeness. In the Kummer algebra, T is a unit because $T^n = a \in A^\times$; therefore the derivative nT^{n-1} is a unit. The derivative of $X^p - X - a$ is -1. Thus both algebras are étale over A. They are the pullbacks from Appendix A.1 of $(-)^n : \mathbb{G}_m \to \mathbb{G}_m$ and $\wp : \mathbb{G}_a \to \mathbb{G}_a$, respectively, so translation by the corresponding kernels μ_n and $\mathbb{F}_p \cong \mathbb{Z}/p\mathbb{Z}$ gives the asserted torsor structures. □

A.3 Witt-vector algebra

A.3.1 Ring structure and ghost coordinates

For $n \geq 0$, the n-th **Witt polynomial** in the variables $X_0, \dots, X_n$ is

$$w_n(X_0, \dots, X_n) \;=\; \sum_{j=0}^{n} p^j X_j^{p^{n-j}} \;=\; X_0^{p^n} + pX_1^{p^{n-1}} + \cdots + p^n X_n.$$

Theorem 77 (Witt; [Ser79, II, §6, Theorem 6], [Rab14, §1])**.** *There exists a unique family of polynomials $S_n, P_n \in \mathbb{Z}[X_0, \dots, X_n; Y_0, \dots, Y_n]$, $n \geq 0$, such that*

$$w_n(S_0, \dots, S_n) = w_n(X_0, \dots, X_n) + w_n(Y_0, \dots, Y_n),$$
$$w_n(P_0, \dots, P_n) = w_n(X_0, \dots, X_n) \cdot w_n(Y_0, \dots, Y_n).$$

for all n. For any commutative ring A, the operations

$$\vec{a} \oplus \vec{b} := \big(S_n(\vec{a}; \vec{b})\big)_{0 \leq n \leq r-1}, \qquad \vec{a} \odot \vec{b} := \big(P_n(\vec{a}; \vec{b})\big)_{0 \leq n \leq r-1}$$

*make the set A^r into a commutative ring $W_r(A)$, the ring of **p-typical Witt vectors of length r over A**, with zero $(0, \dots, 0)$ and unit $(1, 0, \dots, 0)$. Additive inverses are likewise given by universal integral polynomials; we write $\ominus$ for Witt subtraction. The **ghost map***

$$w \colon W_r(A) \longrightarrow A^r, \qquad \vec{a} \longmapsto \big(w_0(\vec{a}), w_1(\vec{a}), \dots, w_{r-1}(\vec{a})\big)$$

is a ring homomorphism to the product ring A^r; it is injective when p is a non-zero-divisor in A, and bijective when $p \in A^\times$.

(Equivalently, the universal Witt addition and inverse polynomials make $\mathbb{A}^r_{\mathbb{Z}}$ an affine commutative group scheme. After base change to $\mathbb{F}_p$, this is the additive Witt-vector group scheme $\mathbb{W}_r$ used in Section A.4.)

We index the components of a Witt vector $\vec{a} = (a_0, \dots, a_{r-1})$ by $0, \dots, r-1$. For $r = 1$ the polynomials reduce to $S_0 = X_0 + Y_0$ and $P_0 = X_0 Y_0$, so $W_1(A) = A$ as a ring.

Since the ring operations are given by universal polynomials with integer coefficients, W_r is a functor: a ring homomorphism $\sigma\colon A \to B$ induces the ring homomorphism

$$W_r(\sigma)\colon W_r(A) \to W_r(B), \qquad W_r(\sigma)(a_0, \dots, a_{r-1}) = (\sigma a_0, \dots, \sigma a_{r-1}),$$

acting *componentwise*. Two consequences of this trivial-looking observation are used repeatedly and deserve emphasis: a ring automorphism σ of A commutes with $\oplus$ and $\ominus$ on $W_r(A)$, and a Witt vector $\vec{a} \in W_r(A)$ satisfies $W_r(\sigma)(\vec{a}) = \vec{a}$ if and only if *each component* a_n is σ-invariant. In particular, for a group Γ acting on A by ring automorphisms,

$$W_r(A)^\Gamma \;=\; W_r\big(A^\Gamma\big) \qquad \text{inside } W_r(A). \tag{A.7}$$

The uniqueness statement in Theorem 77 is an instance of a general principle that we invoke several times, so we record it explicitly.

Lemma 78 (ghost principle)**.** *Let Φ and Ψ be two operations on Witt vectors given by universal polynomials with integer coefficients in the components of their arguments. If $w \circ \Phi = w \circ \Psi$ holds over the polynomial ring $\mathbb{Z}[X_0, X_1, \dots]$ on the components of the arguments, then $\Phi = \Psi$ over every commutative ring.*

Proof. Over $\mathbb{Z}[X_0, X_1, \dots]$ the prime p is a non-zero-divisor, so the ghost map is injective by Theorem 77 and $\Phi = \Psi$ there; this is an identity of integral polynomials, which persists under every base change $\mathbb{Z}[X_0, X_1, \dots] \to A$. □

The only structural feature of the addition polynomials S_n that we shall need is their "triangular" shape:

Proposition 79 (shape of Witt addition)**.** *For every $n \geq 0$ there is a polynomial*

$$C_n \in \mathbb{Z}[X_0, \dots, X_{n-1}; Y_0, \dots, Y_{n-1}]$$

involving only the variables of index strictly less *than n, such that*

$$(\vec{a} \oplus \vec{b})_n \;=\; a_n + b_n + C_n(a_0, \dots, a_{n-1}; b_0, \dots, b_{n-1}). \tag{A.8}$$

We refer to C_n as the ***carry polynomial****. The analogous statement holds for $\ominus$.*

Proof. By induction on n, using the recursion defining S_n:

$$p^n S_n \;=\; \sum_{j \leq n} p^j \big(X_j^{p^{n-j}} + Y_j^{p^{n-j}}\big) \;-\; \sum_{j<n} p^j S_j^{p^{n-j}}.$$

The terms with $j = n$ contribute $p^n(X_n + Y_n)$, and all remaining terms involve only variables and polynomials S_j of index $j < n$, which by induction lie in $\mathbb{Z}[X_{<n}; Y_{<n}]$. Since S_n has integer coefficients (Theorem 77), $C_n := S_n - X_n - Y_n$ is an integral polynomial in the variables of index $< n$. The statement for $\ominus$ follows by the same argument applied to the inversion polynomials. □

A.3.2 Frobenius, Verschiebung, and truncation

From now on, A denotes an $\mathbb{F}_p$-algebra; this is the only case we use. Three operators act on the rings $W_r(A)$:

Definition 80. Let A be an $\mathbb{F}_p$-algebra.

1. The **Frobenius** $F\colon W_r(A) \to W_r(A)$ is $F := W_r(\varphi_A)$, where $\varphi_A\colon a \mapsto a^p$ is the absolute Frobenius of A. Explicitly, F acts componentwise: $F(a_0, \dots, a_{r-1}) = (a_0^p, \dots, a_{r-1}^p)$. By functoriality, F is a ring endomorphism of $W_r(A)$.

2. The **Verschiebung** is the shift

$$V\colon W_{r-1}(A) \to W_r(A), \qquad V(a_0, \dots, a_{r-2}) = (0, a_0, \dots, a_{r-2}).$$

   Iterating, $V^j\colon W_{r-j}(A) \to W_r(A)$; in particular $V^{r-1}\colon A = W_1(A) \to W_r(A)$ sends $c \mapsto (0, \dots, 0, c)$.

3. The **truncation** is

$$t\colon W_r(A) \to W_{r-1}(A), \qquad t(a_0, \dots, a_{r-1}) = (a_0, \dots, a_{r-2}).$$

Remark. For a general (not necessarily $\mathbb{F}_p$-) algebra A, the Frobenius is defined by universal polynomials characterized by $w_n \circ F = w_{n+1}$, and the componentwise formula above holds precisely in characteristic p; see [Ser79, II, §6] or [Rab14, §2]. We will not need the general case.

We record from the general theory ([Ser79, II, §6], [Rab14, §2]) how multiplication by p interacts with these operators.

Proposition 81. *Let A be an $\mathbb{F}_p$-algebra. Then:*

1. *t is a natural ring homomorphism commuting with F, and its kernel is*

$$\ker\big(t\colon W_r(A) \to W_{r-1}(A)\big) \;=\; V^{r-1}A \;=\; \{(0, \dots, 0, c) : c \in A\}.$$

2. *V is additive, and $FV = VF$.*

3. *Multiplication by p on $W_r(A)$ equals $V \circ F = F \circ V$ (composed with the appropriate truncation); explicitly,*

$$p \cdot (a_0, \dots, a_{r-1}) \;=\; (0,\, a_0^p,\, \dots,\, a_{r-2}^p).$$

   Iterating, $p^j \cdot \vec{a} = (0, \dots, 0, a_0^{p^j}, \dots, a_{r-1-j}^{p^j})$; in particular $p^r = 0$ on $W_r(A)$, and

$$p^{r-1} \cdot (a_0, \dots, a_{r-1}) \;=\; \big(0, \dots, 0,\, a_0^{p^{r-1}}\big). \tag{A.9}$$

Proof. (1) That t is a ring homomorphism follows from the recursive definition of the S_n, P_n: the first $r-1$ components of a sum or product depend only on the first $r-1$ components of the arguments (Proposition 79). Naturality and the commutation with the (componentwise) F are clear, and the kernel is computed componentwise. (2) Additivity of V follows from Lemma 78 and the ghost identity $w_n(V\vec{a}) = p\, w_{n-1}(\vec{a})$ (with $w_{-1} := 0$); the commutation $FV = VF$ is immediate

since F is componentwise. (3) Let $M_n \in \mathbb{Z}[X_0, \dots, X_n]$ be the n-th universal coordinate polynomial for multiplication by p. Since the ghost map is a ring homomorphism, these polynomials satisfy

$$p^n M_n = p\, w_n(\vec{X}) - \sum_{i=0}^{n-1} p^i M_i^{p^{n-i}}. \tag{A.10}$$

We claim that

$$M_0 = pX_0, \qquad M_n \equiv X_{n-1}^p \pmod{p} \quad (n \geq 1).$$

The first identity is clear. For the induction step, use $(u + pv)^{p^m} \equiv u^{p^m} \pmod{p^{m+1}}$. If the claim holds below index n, then, modulo p^{n+1}, the sum in equation (A.10) is

$$\sum_{i=1}^{n-1} p^i X_{i-1}^{p^{n-i+1}};$$

the $i = 0$ term is also 0 modulo p^{n+1}. Cancelling this sum against the corresponding terms of $p\, w_n(\vec{X})$ gives

$$p^n M_n \equiv p^n X_{n-1}^p \pmod{p^{n+1}},$$

which proves the claim. After evaluating in the $\mathbb{F}_p$-algebra A, we therefore obtain

$$p \cdot (a_0, \dots, a_{r-1}) = (0, a_0^p, \dots, a_{r-2}^p) = VF(\vec{a}),$$

with the appropriate truncation. Since $FV = VF$ by (2), this also gives $FV = p$. See [Ser79, II, §6, Proposition 8 ff.] for details. □

Example 82. $W_r(\mathbb{F}_p) \cong \mathbb{Z}/p^r\mathbb{Z}$ canonically. Indeed, by (A.9) the unit $\underline{1} = (1, 0, \dots, 0)$ satisfies $p^j \cdot \underline{1} = (0, \dots, 0, 1, 0, \dots, 0)$ (1 in position j), so $\underline{1}$ has additive order p^r; since $|W_r(\mathbb{F}_p)| = p^r$, the additive group is cyclic of order p^r generated by the unit, and $m \mapsto m \cdot \underline{1}$ is a ring isomorphism $\mathbb{Z}/p^r\mathbb{Z} \xrightarrow{\cong} W_r(\mathbb{F}_p)$. We write $\underline{m} \in W_r(\mathbb{F}_p)$ for the image of $m \in \mathbb{Z}/p^r\mathbb{Z}$. (Similarly, the full ring of Witt vectors satisfies $W(\mathbb{F}_p) \cong \mathbb{Z}_p$; we shall not need this.)

A.4 Artin–Schreier–Witt theory

We retain the notation for Witt vectors from the beginning of this appendix. For an $\mathbb{F}_p$-algebra A, Frobenius acts componentwise on $W_r(A)$.

A.4.1 The ASW operator and bookkeeping

Definition 83. The **Artin–Schreier–Witt operator** is

$$\wp := F - \mathrm{id}, \qquad \wp\vec{u} = F\vec{u} \ominus \vec{u}.$$

It is an endomorphism of the additive group of $W_r(A)$ and commutes with truncation and Verschiebung.

Lemma 84 (Witt bookkeeping)**.** *Let A be an $\mathbb{F}_p$-algebra.*

(i) The truncation $t : W_r(A) \to W_{r-1}(A)$ is a ring homomorphism commuting with F, hence with $\wp$, and $\ker(t) = V^{r-1}A$.

(ii) The map $V^{r-1} : A \to W_r(A)$ is additive and $\wp \circ V^{r-1} = V^{r-1} \circ \wp$.

(iii) For every $\vec{a} \in W_r(A)$ and $c \in A$,

$$\vec{a} \oplus V^{r-1}(c) = (a_0, \dots, a_{r-2}, a_{r-1} + c).$$

Proof. The first two assertions follow from Proposition 81. For (iii), consider the two operations

$$\Phi(\vec{X}, C) := \vec{X} \oplus V^{r-1}(C) = \big(S_n(\vec{X}; 0, \dots, 0, C)\big)_{0 \le n \le r-1},$$
$$\Psi(\vec{X}, C) := (X_0, \dots, X_{r-2}, X_{r-1} + C).$$

Both are given by universal polynomials with integer coefficients, as required in Lemma 78. Over $\mathbb{Z}[X_0, \dots, X_{r-1}, C]$, for $n < r-1$ we have

$$w_n\big(\Phi(\vec{X}, C)\big) = w_n(\vec{X}) + w_n\big(V^{r-1}(C)\big) = w_n(\vec{X}) = w_n\big(\Psi(\vec{X}, C)\big).$$

For the final ghost component,

$$\begin{aligned} w_{r-1}\big(\Phi(\vec{X}, C)\big) &= w_{r-1}(\vec{X}) + p^{r-1}C \\ &= \sum_{i=0}^{r-2} p^i X_i^{p^{r-1-i}} + p^{r-1}(X_{r-1} + C) \\ &= w_{r-1}\big(\Psi(\vec{X}, C)\big). \end{aligned}$$

Thus $w \circ \Phi = w \circ \Psi$, so Lemma 78 gives $\Phi = \Psi$ over every commutative ring, and in particular over A. □

The last assertion is special to the top Verschiebung slice. Witt addition is not componentwise in lower positions, and the carry polynomials in (A.8) cannot in general be discarded.

A.4.2 Torsors and classification

The Witt addition polynomials make $\mathbb{A}^r_{\mathbb{F}_p}$ into the additive Witt vector group scheme $\mathbb{W}_r$ over $\mathbb{F}_p$. We distinguish this group scheme from its groups of points: for an $\mathbb{F}_p$-algebra A, its group of A-valued points is $\mathbb{W}_r(A) = W_r(A)$. If X is an $\mathbb{F}_p$-scheme, we write $W_r(\mathbb{F}_p)_X$ for the constant group scheme over X associated with the abstract group $W_r(\mathbb{F}_p)$. The operator $\wp = F - \mathrm{id}$ is an endomorphism of $\mathbb{W}_r$ and fits into an exact sequence of group schemes for the étale topology

$$0 \longrightarrow W_r(\mathbb{F}_p)_{\mathbb{F}_p} \cong (\mathbb{Z}/p^r\mathbb{Z})_{\mathbb{F}_p} \longrightarrow \mathbb{W}_r \xrightarrow{\wp} \mathbb{W}_r \longrightarrow 0. \tag{A.11}$$

The following is the Witt-vector analogue of Lemma 76. Its coordinate proof requires the triangular form of Witt addition because, unlike the two classical equations, Witt subtraction is not componentwise.

Lemma 85 (ASW covers are étale torsors)**.** *Let A be an $\mathbb{F}_p$-algebra, let $r \ge 1$, and let $\vec{g} \in W_r(A)$. Let $\vec{X} = (X_0, \dots, X_{r-1})$ denote the universal Witt vector of length r over A, so that $A[\vec{X}] = A[X_0, \dots, X_{r-1}]$. Define*

$$T_{\vec{g}} := \operatorname{Spec} B, \qquad B := A[\vec{X}]/\big(\wp\vec{X} \ominus \vec{g}\big),$$

where the parenthesized Witt vector denotes the ideal generated by its r coordinate polynomials. Here $\vec{g} \in W_r(A) = \mathbb{W}_r(A)$ is viewed as a morphism $\operatorname{Spec} A \to \mathbb{W}_r$. Thus $T_{\vec{g}}$ is the fiber of $\wp$ over $\vec{g}$, given by the Cartesian square

$$\begin{array}{ccc} T_{\vec{g}} & \longrightarrow & \mathbb{W}_r \\ \downarrow & & \downarrow{\scriptstyle \wp} \\ \operatorname{Spec} A & \xrightarrow[\vec{g}]{} & \mathbb{W}_r \\ & \searrow & \downarrow \\ & & \operatorname{Spec} \mathbb{F}_p, \end{array}$$

where the maps to $\operatorname{Spec} \mathbb{F}_p$ are the structure maps. Then the following hold:

(i) The algebra B is finite étale and free of rank p^r over A, with basis $\{\prod_{n=0}^{r-1} X_n^{a_n} : 0 \le a_n \le p-1\}$.

(ii) Witt translation $\vec{X} \mapsto \vec{X} \oplus \vec{c}$, for $\vec{c} \in W_r(\mathbb{F}_p) \cong \mathbb{Z}/p^r\mathbb{Z}$, makes $T_{\vec{g}}$ a $\mathbb{Z}/p^r\mathbb{Z}$-torsor over $\operatorname{Spec} A$.

Proof. For (i), the triangular shape of Witt addition ((A.8)) shows that the n-th equation in $\wp\vec{X} \ominus \vec{g} = 0$ has the form

$$X_n^p - X_n - g_n + C'_n(X_0, \dots, X_{n-1}; g_0, \dots, g_{n-1}) = 0.$$

Starting with A, first adjoin X_0 subject to the equation of index 0. Then, for each $1 \le n < r$, adjoin X_n subject to the displayed equation of index n; its coefficients lie in the algebra obtained at the preceding step. Because this equation is monic of degree p in X_n, the n-th step is free of rank p, with basis $1, X_n, \dots, X_n^{p-1}$. After all r steps we obtain B, finite free over A with the stated basis. The Jacobian matrix is lower triangular with diagonal entries -1, so B/A is étale. For (ii), the cover is surjective because B is free of positive rank. By construction, it is the pullback of $\wp : \mathbb{W}_r \to \mathbb{W}_r$ along $\vec{g}$. Recall that the kernel of the group-scheme morphism $\wp$ is the constant group scheme $W_r(\mathbb{F}_p)_{\mathbb{F}_p}$, characterized by $F\vec{u} = \vec{u}$. Translation by the base change of this kernel to $\operatorname{Spec} A$ preserves the fiber because, for $\vec{c} \in \ker(\wp)$,

$$\wp(\vec{X} \oplus \vec{c}) = \wp(\vec{X}) \oplus \wp(\vec{c}) = \wp(\vec{X}).$$

Any two points in the same fiber differ by a unique such $\vec{c}$, so this translation action makes the pullback a torsor. □

Theorem 86 (Artin–Schreier–Witt theory; [Tho05, §3–4])**.** *Let M be a field of characteristic p, and let $r \ge 1$. For $\vec{g} \in W_r(M)$, define*

$$T_{\vec{g}} := \operatorname{Spec}\Big(M[\vec{X}]/\big(\wp\vec{X} \ominus \vec{g}\big)\Big).$$

By Lemma 85, Witt translation makes $T_{\vec{g}}$ a $\mathbb{Z}/p^r\mathbb{Z}$-torsor over $\operatorname{Spec} M$. Then the following hold:

(i) The assignment $[\vec{g}] \mapsto [T_{\vec{g}}]$ is a canonical isomorphism of abelian groups

$$W_r(M)/\wp W_r(M) \xrightarrow{\cong} H^1_{\text{ét}}\big(\operatorname{Spec} M, (\mathbb{Z}/p^r\mathbb{Z})_{\operatorname{Spec} M}\big).$$

Thus this quotient classifies $\mathbb{Z}/p^r\mathbb{Z}$-torsors over $\operatorname{Spec} M$.

(ii) *If $[\vec{g}]$ has order p^s, where $0 \le s \le r$, then $T_{\vec{g}}$ has p^{r-s} connected components, each noncanonically isomorphic to* $\operatorname{Spec} L$ *for the same cyclic extension L/M of degree p^s. Let $H \subseteq \mathbb{Z}/p^r\mathbb{Z}$ be the unique subgroup of order p^s. Each component is an H-torsor, and $T_{\vec{g}}$ is induced from any one of them. In particular, $T_{\vec{g}}$ is connected if and only if $[\vec{g}]$ has order p^r.*

(iii) *Every cyclic extension L/M of degree dividing p^r occurs in this way: for some $\vec{g} \in W_r(M)$, the connected components of $T_{\vec{g}}$ are isomorphic to* $\operatorname{Spec} L$.

Proof. For (i), put $X = \operatorname{Spec} M$, and write $\mathbb{W}_{j,M}$ for the base change of $\mathbb{W}_j$ to X. Base-changing (A.11) to X gives the short exact sequence of abelian sheaves on $X_{\text{ét}}$

$$0 \longrightarrow (\mathbb{Z}/p^r\mathbb{Z})_X \longrightarrow \mathbb{W}_{r,M} \xrightarrow{\wp} \mathbb{W}_{r,M} \longrightarrow 0.$$

Its associated long exact cohomology sequence contains

$$W_r(M) \xrightarrow{\delta} H^1_{\text{ét}}\big(X, (\mathbb{Z}/p^r\mathbb{Z})_X\big) \longrightarrow H^1_{\text{ét}}(X, \mathbb{W}_{r,M}).$$

By (A.6), $\delta(\vec{g}) = [T_{\vec{g}}]$, and δ induces the well-defined injective homomorphism $[\vec{g}] \mapsto [T_{\vec{g}}]$ on $W_r(M)/\wp W_r(M)$. Thus it remains to show that $H^1_{\text{ét}}(X, \mathbb{W}_{r,M}) = 0$. For $r = 1$, we have $\mathbb{W}_{1,M} = \mathbb{G}_{a,M}$, and this is additive Hilbert 90. For $r > 1$, Lemma 84(i)–(ii) gives the short exact sequence of abelian sheaves on $X_{\text{ét}}$

$$0 \longrightarrow \mathbb{G}_{a,M} \xrightarrow{V^{r-1}} \mathbb{W}_{r,M} \xrightarrow{t} \mathbb{W}_{r-1,M} \longrightarrow 0.$$

The associated long exact sequence contains

$$H^1_{\text{ét}}(X, \mathbb{G}_{a,M}) \longrightarrow H^1_{\text{ét}}(X, \mathbb{W}_{r,M}) \longrightarrow H^1_{\text{ét}}(X, \mathbb{W}_{r-1,M}).$$

The outer groups vanish by additive Hilbert 90 and induction, so the middle group vanishes. Hence δ is surjective, proving the claimed isomorphism.

For (ii), suppose $[\vec{g}]$ has order p^s. By (i) and Corollary 14, the character associated with $T_{\vec{g}}$ has the same order. Let H be its image. The order of the character is the smallest positive integer annihilating every element of H. Since H is cyclic, this integer is $|H|$, so $|H| = p^s$. By Theorem 13, $T_{\vec{g}}$ is induced from a connected H-torsor $\operatorname{Spec} L$, where L/M is cyclic of degree $|H|$. This induced torsor is a disjoint union of $[\mathbb{Z}/p^r\mathbb{Z} : H] = p^{r-s}$ copies of $\operatorname{Spec} L$, each an H-torsor. Translation by $\mathbb{Z}/p^r\mathbb{Z}$ permutes these components, so $T_{\vec{g}}$ is induced from any one of them. It is connected precisely when $s = r$.

For (iii), let L/M be cyclic of degree dividing p^r. Choose an injection $\operatorname{Gal}(L/M) \hookrightarrow \mathbb{Z}/p^r\mathbb{Z}$ and induce the torsor $\operatorname{Spec} L$ along it. The resulting $\mathbb{Z}/p^r\mathbb{Z}$-torsor is isomorphic to some $T_{\vec{g}}$ by the classification just proved, and its connected components are copies of $\operatorname{Spec} L$. □

Corollary 87. *Let $M \supseteq \mathbb{F}_p$ and let $\vec{g} \in W_r(M)$ have $g_0 = 0$. Then*

$$p^{r-1}[\vec{g}] = 0 \qquad \text{in } W_r(M)/\wp W_r(M).$$

Consequently, a class of order p^r has nonzero zeroth component in every representative.

Proof. By (A.9), $p^{r-1}\vec{g} = (0, \dots, 0, g_0^{p^{r-1}}) = 0$ already in $W_r(M)$. □

A.4.3 Reduced polynomial representatives

Definition 88. A vector $\vec{g} = (g_0, \dots, g_{r-1}) \in W_r(k((x)))$ is a **reduced polynomial vector** if it is reduced in the sense of Definition 52 and every $g_j \in x^{-1}k[x^{-1}]$.

Lemma 89 (Surjectivity on integral Witt vectors)**.** *If k is algebraically closed, then*

$$\wp : W_r(k[[x]]) \longrightarrow W_r(k[[x]])$$

is surjective.

Proof. Proceed by induction on r. For $r = 1$, given $h = \sum_{i\geq 0} b_i x^i$, choose u_0 with $u_0^p - u_0 = b_0$. For $i \geq 1$, define

$$u_i = \begin{cases} u_{i/p}^p - b_i, & p \mid i, \\ -b_i, & p \nmid i. \end{cases}$$

This is recursive because $i/p < i$. Put $u = \sum_{i\geq 0} u_i x^i \in k[[x]]$. In characteristic p,

$$u^p = \sum_{i\geq 0} u_i^p x^{pi}.$$

Thus the constant coefficient of $u^p - u$ is $u_0^p - u_0 = b_0$. For $i \geq 1$, its coefficient of x^i is $u_{i/p}^p - u_i = b_i$ if $p \mid i$, and $-u_i = b_i$ otherwise. Hence every coefficient of $u^p - u$ equals the corresponding coefficient of h, so $u^p - u = h$.

For $r > 1$, choose by induction $\vec{u}' \in W_{r-1}(k[[x]])$ with $\wp\vec{u}' = t(\vec{h})$ and lift it to $\widetilde{u} = (\vec{u}', 0)$. By Lemma 84(i), $\wp\widetilde{u} \ominus \vec{h} = V^{r-1}(g)$ for some $g \in k[[x]]$. Choose $w \in k[[x]]$ with $\wp w = -g$. Then Lemma 84(ii) gives $\wp(\widetilde{u} \oplus V^{r-1}w) = \vec{h}$. □

Proposition 90 (Reduced polynomial representatives)**.** *Let k be an algebraically closed field of characteristic p, and let $r \geq 1$. Every class in $W_r(k((x)))/\wp W_r(k((x)))$ has a reduced polynomial representative.*

Proof. Proceed by induction on r. For $r = 1$, write a Laurent series as the sum of its principal part and an element of $k[[x]]$. The latter can be removed by Lemma 89. If the remaining principal part has leading term cx^{-m} with $p \mid m$, subtract

$$\wp(c^{1/p}x^{-m/p}) = cx^{-m} - c^{1/p}x^{-m/p}.$$

This strictly decreases the pole order; iteration produces zero or a pole order prime to p.

For the inductive step, let $\vec{g} \in W_r(k((x)))$. By induction, there is $\vec{v} \in W_{r-1}(k((x)))$ such that

$$\vec{f}' := t(\vec{g}) \ominus \wp\vec{v}$$

is a reduced polynomial vector. Lift $\vec{v}$ to $\widetilde{\vec{v}} := (v_0, \dots, v_{r-2}, 0)$ and put

$$\vec{h} := \vec{g} \ominus \wp\widetilde{\vec{v}}.$$

Since truncation commutes with $\wp$, we have $t(\vec{h}) = \vec{f}'$. Thus $\vec{h} = (f_0', \dots, f_{r-2}', h_{r-1})$, and only its last component remains to be reduced.

By the length-one case, choose $z \in k((x))$ such that $f_{r-1} := h_{r-1} - \wp z$ is a reduced polynomial. Now set

$$\begin{aligned}\vec{f} &:= \vec{h} \ominus \wp(V^{r-1}z) \\ &= \vec{h} \ominus V^{r-1}(\wp z) = (f_0', \dots, f_{r-2}', f_{r-1}),\end{aligned}$$

where the last two equalities follow from Lemma 84(ii)–(iii). Every component of $\vec{f}$ is now a reduced polynomial, and

$$\vec{f} = \vec{g} \ominus \wp\big(\widetilde{\vec{v}} \oplus V^{r-1}z\big).$$

Hence $\vec{f}$ represents the same class as $\vec{g}$, as required. □

A.5 Weighted Witt addition in the cyclic calculation

The following homogeneity property of the universal addition polynomials is what makes degree estimates on Witt sums possible: even though the carry polynomials C_n of (A.8) are complicated, they are *isobaric* for the weighting in which the j-th component has weight p^j.

Lemma 91 (isobaricity of Witt addition)**.** *Assign to the variables X_j and Y_j the weight p^j. Then the n-th addition polynomial S_n is isobaric of weight p^n: every monomial of S_n has total weight exactly p^n. The same holds for each component of a d-fold iterated Witt sum $\bigoplus_{i=1}^{d} \vec{a}_i$, viewed as a universal polynomial in the components $a_{i,j}$ of weight p^j.*

Proof. We first prove the assertion for two summands by induction on n. For $n = 0$, we have $S_0 = X_0 + Y_0$, which is isobaric of weight 1. Suppose now that S_j is isobaric of weight p^j for every $j < n$. The ghost identities give

$$p^n S_n = \sum_{j \le n} p^j \big(X_j^{p^{n-j}} + Y_j^{p^{n-j}}\big) - \sum_{j<n} p^j S_j^{p^{n-j}}$$

Each $X_j^{p^{n-j}}$ and $Y_j^{p^{n-j}}$ has weight $p^j p^{n-j} = p^n$. Also, every monomial in the expansion of $S_j^{p^{n-j}}$ is a product of p^{n-j} monomials of weight p^j, so it has weight p^n. Thus the right-hand side is isobaric of weight p^n. This identity holds in the universal polynomial ring over $\mathbb{Z}$, where multiplication by the nonzero integer p^n does not change which monomials occur. Hence S_n is isobaric of weight p^n as well.

We next induct on the number d of summands. Write

$$\vec{A}^{(d)} := \bigoplus_{i=1}^{d} \vec{a}_i = \vec{a}_1 \oplus \cdots \oplus \vec{a}_d,$$

where each variable $a_{i,j}$ has weight p^j. For $d = 1$, the component $A_n^{(1)} = a_{1,n}$ has weight p^n. Suppose the assertion holds for $d-1$ summands. Then

$$A_n^{(d)} = S_n\big(A_0^{(d-1)}, \dots, A_n^{(d-1)}; a_{d,0}, \dots, a_{d,n}\big).$$

By induction, $A_j^{(d-1)}$ is isobaric of weight p^j. Thus each substituted polynomial has the same weight as the variable it replaces, so substitution preserves weights. Since S_n is isobaric of weight p^n, so is $A_n^{(d)}$, completing the induction. □

Appendix B

A Failure of an Artin–Schreier Induction Strategy

B.1 Motivation: can the Artin–Schreier calculation be iterated?

Part (2) of Theorem 58 proves the order-p case by an explicit Artin–Schreier calculation. Part (3) treats $G = \mathbb{Z}/p^r\mathbb{Z}$ by working directly with an Artin–Schreier–Witt equation. It is natural to ask whether the second calculation can instead be obtained by iterating the first.

Indeed, let

$$G = \mathbb{Z}/p^r\mathbb{Z}, \qquad H = p^{r-1}G \cong \mathbb{Z}/p\mathbb{Z}, \qquad G/H \cong \mathbb{Z}/p^{r-1}\mathbb{Z},$$

and factor a G-torsor $\mathcal{P} \to U$ as

$$\mathcal{P} \longrightarrow \mathcal{P}/H \longrightarrow U.$$

The tempting induction has three steps:

1. apply the induction hypothesis to the G/H-torsor $\mathcal{P}/H \to U$;
2. regard $\mathcal{P} \to \mathcal{P}/H$ as an $H \cong \mathbb{Z}/p\mathbb{Z}$-torsor and apply the Artin–Schreier calculation after compactifying its base;
3. combine the two unramifiedness statements to recover the result for the original G-torsor.

The first two steps do produce natural statements, but they live over different exceptional valuations. Passing from one to the other involves a base change induced by the comparison between the symmetric power of a torsor as a scheme and its symmetric power as a torsor. That base change can be wildly ramified. The purpose of this appendix is to formulate the proposed induction precisely, follow both inputs through the resulting valuation tower, and explain why they do not by themselves imply the required conclusion.

B.2 Preliminaries for the attempted induction

B.2.1 The symmetric quotient tower

Let G be a finite abelian group, let $H \subseteq G$ be a subgroup, and let $\mathcal{P} \to X$ be a G-torsor over an S-scheme X that is Zariski-locally quasi-projective over S. Recall the canonical factorization

$$\mathcal{P} \longrightarrow \mathcal{P}/H \longrightarrow X,$$

where $\mathcal{P}/H = (G/H) \times^G \mathcal{P}$ is a G/H-torsor over X and $\mathcal{P} \to \mathcal{P}/H$ is an H-torsor.

In addition to the kernel $K_G = \ker(G^d \to G)$ of the summation morphism, put

$$K_H := \ker(H^d \to H), \qquad K_{G/H} := \ker((G/H)^d \to G/H).$$

The snake lemma applied to the two summation maps gives an exact sequence

$$0 \longrightarrow K_H \longrightarrow K_G \longrightarrow K_{G/H} \longrightarrow 0. \tag{B.1}$$

Construction 92 (The symmetric-power comparison morphism)**.** For the G-torsor $\mathcal{P} \to X$ fixed above, put

$$\Xi_G := K_G \rtimes S_d.$$

The inclusion $S_d \hookrightarrow \Xi_G$, given by $\sigma \mapsto ((1, \dots, 1), \sigma)$, induces a canonical comparison morphism

$$q_{\mathcal{P}} : \mathcal{P}^{(d)} = S_d \backslash \mathcal{P}^{\times_S d} \longrightarrow \Xi_G \backslash \mathcal{P}^{\times_S d} = \mathcal{P}^{[d]}.$$

Indeed, the quotient map from $\mathcal{P}^{\times_S d}$ to its Ξ_G-quotient is S_d-invariant and therefore factors uniquely through the S_d-quotient.

We use the following four quotient schemes:

$$\begin{aligned} \mathcal{P}^{[d]_G} &:= (K_G \rtimes S_d) \backslash \mathcal{P}^{\times_S d}, \\ \mathcal{P}^{[d]_H} &:= (K_H \rtimes S_d) \backslash \mathcal{P}^{\times_S d}, \\ (\mathcal{P}/H)^{(d)} &:= (H^d \rtimes S_d) \backslash \mathcal{P}^{\times_S d}, \\ (\mathcal{P}/H)^{[d]} &:= ((H^d + K_G) \rtimes S_d) \backslash \mathcal{P}^{\times_S d}. \end{aligned}$$

The first is the symmetric G-power of $\mathcal{P}$. The second is the symmetric H-power relative to $\mathcal{P} \to \mathcal{P}/H$, the third is the symmetric power of the scheme $\mathcal{P}/H$, and the fourth is the symmetric G/H-power of the G/H-torsor $\mathcal{P}/H \to X$. For the last two identifications, use $(\mathcal{P}/H)^{\times_S d} \cong H^d \backslash \mathcal{P}^{\times_S d}$ and note that the preimage of $K_{G/H}$ under $G^d \twoheadrightarrow (G/H)^d$ is $H^d + K_G$.

Proposition 93. *The four quotient schemes fit into a canonical commutative diagram*

$$\begin{array}{ccc} \mathcal{P}^{[d]_H} & \xrightarrow{\ H\ } & (\mathcal{P}/H)^{(d)} \\ \downarrow & & \downarrow{\scriptstyle q} \\ \mathcal{P}^{[d]_G} & \xrightarrow[\ H\ ]{} & (\mathcal{P}/H)^{[d]} \\ & {\scriptstyle G}\searrow & \downarrow{\scriptstyle G/H} \\ & & X^{(d)}, \end{array}$$

in which every labelled arrow is a torsor under the indicated constant group. The upper square is Cartesian:

$$\mathcal{P}^{[d]_H} \cong \mathcal{P}^{[d]_G} \times_{(\mathcal{P}/H)^{[d]}} (\mathcal{P}/H)^{(d)}.$$

The vertical map $q = q_{\mathcal{P}/H}$ is the comparison morphism of Construction 92, applied to the G/H-torsor $\mathcal{P}/H \to X$.

Proof. Apply Proposition 25 to the G-torsor $\mathcal{P} \to X$, the H-torsor $\mathcal{P} \to \mathcal{P}/H$, and the G/H-torsor $\mathcal{P}/H \to X$. This gives the three labelled torsors; the comparison map q is supplied by Construction 92, applied to $\mathcal{P}/H \to X$. The factorization

$$\mathcal{P}^{[d]_G} \longrightarrow (\mathcal{P}/H)^{[d]} \longrightarrow X^{(d)}$$

is the canonical quotient decomposition of the G-torsor $\mathcal{P}^{[d]_G} \to X^{(d)}$ by H, because both its intermediate quotient and $(\mathcal{P}/H)^{[d]}$ are realized as

$$((H^d + K_G) \rtimes S_d)\backslash \mathcal{P}^{\times_S d}.$$

Let

$$T := \mathcal{P}^{[d]_G} \times_{(\mathcal{P}/H)^{[d]}} (\mathcal{P}/H)^{(d)}.$$

Base change makes T an H-torsor over $(\mathcal{P}/H)^{(d)}$. The quotient maps give an H-equivariant morphism $\mathcal{P}^{[d]_H} \to T$. A morphism of H-torsors is an isomorphism, proving the Cartesian assertion. □

Remark. The two vertical comparison maps in the diagram need not be torsors under a constant group. For example, $K_H \rtimes S_d$ need not be normal in $K_G \rtimes S_d$, because coordinate permutations act nontrivially on K_G/K_H. Thus the map q is not another torsor layer to which the cyclic induction hypotheses automatically apply. Its ramification at the boundary is precisely the extra datum that will matter below.

B.2.2 The geometric boundary and its valuations

Let C be a smooth projective curve, let $U = C \setminus \mathfrak{m}$, and let $\mathcal{P} \to U$ be a G-torsor. Fix $p_\infty \in \mathfrak{m}$ and an integer $d \geq 1$, and assume that the ramification of $\mathcal{P}$ at p_∞ is bounded by d in the sense of Definition 39.

The G/H-torsor $\mathcal{P}/H \to U$ compactifies to a finite morphism of smooth projective curves

$$f : C' \longrightarrow C.$$

Choose $p'_\infty \in f^{-1}(p_\infty)$. The induced morphism on symmetric powers sends $d \cdot p'_\infty$ to $d \cdot p_\infty$:

$$f^{(d)} : C'^{(d)} \longrightarrow C^{(d)}.$$

Let

$$\pi_C : X_C \longrightarrow C^{(d)}, \qquad \pi_{C'} : X_{C'} \longrightarrow C'^{(d)}$$

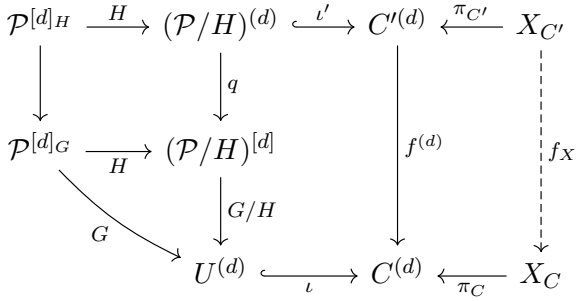
be the blowups at these two diagonal points. Write $E_C, E_{C'}$ for their exceptional divisors, $\eta_C, \eta_{C'}$ for the respective generic points, and

$$V_C := \mathcal{O}_{X_C,\eta_C} \subset K(C^{(d)}), \qquad V_{C'} := \mathcal{O}_{X_{C'},\eta_{C'}} \subset K(C'^{(d)})$$

for the corresponding DVRs.

B.2.3 The master diagram

The quotient tower, compactifications, and blowups fit into the diagram



The left block is the torsor tower of Proposition 93. The middle block records the open immersions and compactifications. The right block records the blowups. The rational lift f_X is defined at the generic point of the exceptional divisor and sends $\eta_{C'}$ to η_C, producing the extension of DVRs

$$V_C \hookrightarrow V_{C'}.$$

B.2.4 The proposed induction step

In the cyclic p-power setting of Section B.1, the two intended inputs are:

(IH$_{r-1}$) $(\mathcal{P}/H)^{[d]} \to U^{(d)}$ is unramified at η_C;

(AS) $\mathcal{P}^{[d]_H} \to (\mathcal{P}/H)^{(d)}$ is unramified at $\eta_{C'}$.

The first is the desired contribution of induction applied to the quotient G/H. The second is the desired contribution of the degree-p Artin–Schreier calculation applied to the H-torsor over the compactified curve C'. The proposed induction step asks whether these imply that

$$\mathcal{P}^{[d]_G} \longrightarrow U^{(d)}$$

is unramified at η_C. We will see that the two assertions do not, by themselves, control the ramification needed for this conclusion.

B.2.5 A base-change lemma

We use the following standard comparison between a Galois extension and its base change.

Lemma 94. *Let E/F be a finite Galois extension with group H, let K/F be a field extension, and form all fields inside a common overfield. Put $M := E \cap K$. Then restriction gives*

$$\mathrm{Gal}(EK/K) \cong \mathrm{Gal}(E/M),$$

and, as K-algebras,

$$E \otimes_F K \cong \prod_{i=1}^{[M:F]} EK.$$

In particular, $E \otimes_F K$ is a field if and only if E and K are linearly disjoint over F.

Proof. Write $E = F[\alpha]$ and let $g \in F[x]$ be the minimal polynomial of α. The translation theorem identifies $\mathrm{Gal}(EK/K)$ with the subgroup of $\mathrm{Gal}(E/F)$ fixing $E \cap K = M$, hence with $\mathrm{Gal}(E/M)$.

Factor $g = g_1 \cdots g_m$ into monic irreducible polynomials over K. Adjoining any root of g to K produces EK, so every g_i has degree $[E : M]$. Consequently $m = [M : F]$, and

$$E \otimes_F K \cong K[x]/(g) \cong \prod_{i=1}^{[M:F]} K[x]/(g_i) \cong \prod_{i=1}^{[M:F]} EK.$$

The product is a field exactly when $M = F$. □

B.3 Following the hypotheses through the valuation tower

We first make two harmless reductions. We may assume that $\mathcal{P} \to U$ is connected: replace G by the image G_0 of its monodromy and H by $H \cap G_0$. The four quotient schemes then decompose componentwise, and the two unramifiedness inputs pass to the connected component. We may also assume $H \neq 0$, since for $H = 0$ the statement (IH_{r-1}) is already the desired conclusion.

Under these reductions the four schemes in the Cartesian square of Proposition 93 are integral, so we may pass to function fields. Put

$$K_C := \mathrm{Frac}\, V_C = K(C^{(d)}), \qquad K_{C'} := \mathrm{Frac}\, V_{C'} = K(C'^{(d)}),$$

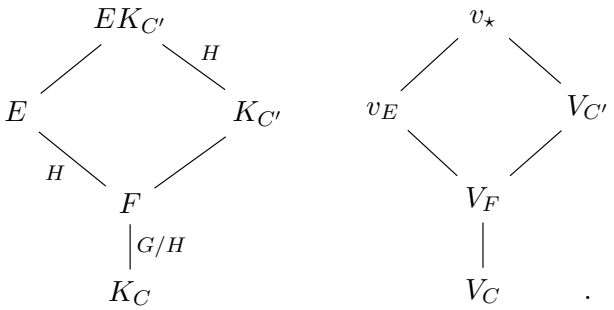
and

$$F := K((\mathcal{P}/H)^{[d]}), \qquad E := K(\mathcal{P}^{[d]_G}).$$

Then F/K_C is Galois with group G/H, E/F is Galois with group H, and E/K_C is Galois with group G. The Cartesian square identifies the function field of $\mathcal{P}^{[d]_H}$ with $E \cdot K_{C'}$. Its generic fibre over $(\mathcal{P}/H)^{(d)}$ is the finite domain $E \otimes_F K_{C'}$, hence a field. By Lemma 94,

$$E \cap K_{C'} = F, \qquad \mathrm{Gal}(EK_{C'}/K_{C'}) \cong \mathrm{Gal}(E/F) = H.$$

Let $V_F := V_{C'}|_F$. Choose a place v_E of E above V_F and a place $v_\star$ of $EK_{C'}$ above both v_E and $V_{C'}$. The relevant field and valuation towers are



We can now see exactly what each part of the proposed induction contributes:

(IH_{r-1}): F/K_C is unramified at V_C. In particular, $e(V_F \mid V_C) = 1$.

(AS): $EK_{C'}/K_{C'}$ is unramified at $V_{C'}$.

Missing step: one must prove that E/F is unramified at V_F.

Goal: once the missing step holds, the tower $K_C \subset F \subset E$ shows that E/K_C is unramified at V_C.

The gap lies between the second and third items. It is measured by the following inertia comparison.

Lemma 95. *Let E/F be finite Galois and linearly disjoint from K/F. Let w be a place of EK and write w_K, w_E, w_F for its restrictions. Under the restriction isomorphism $\mathrm{Gal}(EK/K) \cong \mathrm{Gal}(E/F)$ there is an injection*

$$I(w \mid w_K) \hookrightarrow I(w_E \mid w_F).$$

After completion, the source is the inertia group of the local compositum $\widehat{E}\,\widehat{K}/\widehat{K}$. In particular, if $\widehat{E} \subseteq \widehat{K}$, then $I(w \mid w_K) = 1$, regardless of the ramification of E/F at w_F.

Proof. Let $\sigma \in I(w \mid w_K)$. It fixes K, hence F, and restriction gives $\sigma|_E \in \mathrm{Gal}(E/F)$. Because σ preserves w, its restriction preserves w_E. Moreover, σ acts trivially on the residue field at w, and therefore on the subfield given by the residue of E. Thus $\sigma|_E \in I(w_E \mid w_F)$. Injectivity follows from the restriction isomorphism. The completed statement is the usual identification of the decomposition and inertia groups with those of the corresponding extension of completed fields. □

Applied to the tower above, the lemma gives

$$I(v_\star \mid V_{C'}) \hookrightarrow I(v_E \mid V_F).$$

Statement (AS) makes the group on the left trivial, whereas the missing step requires the group on the right to be trivial. An injection with trivial source gives no such conclusion. Surjectivity would repair the argument, but it can fail when $K_{C'}/F$ is wildly ramified: after completion, the base change may contain the ramified degree-p layer and thereby make that layer invisible.

B.4 Failure under wild base change

The smallest possible local model already exhibits the problem. It models the top $H \cong \mathbb{Z}/p\mathbb{Z}$ layer in the first nontrivial induction step $G = \mathbb{Z}/p^2\mathbb{Z}$.

Example 96 (Absorption of an Artin–Schreier layer)**.** Let $k = \bar{\mathbb{F}}_p$ and put $F = k((t))$. Define $K = k((\pi))$ and embed $F \hookrightarrow K$ by

$$t = \frac{\pi^p}{1 - \pi^{p-1}}.$$

Equivalently,

$$t^{-1} = \pi^{-p} - \pi^{-1} = \wp(\pi^{-1}), \qquad \wp(z) = z^p - z.$$

The element t^{-1} has reduced pole order one over F. Hence the Artin–Schreier extension

$$E = F(y), \qquad y^p - y = t^{-1},$$

is totally wildly ramified of degree p and has break one by Theorem 51. In particular,

$$I(E/F) = \mathbb{Z}/p\mathbb{Z}.$$

Over K, however, π^{-1} is already a solution of the same equation. After identifying y with π^{-1} in a common overfield, one has $E = K$. Consequently

$$EK/K = K/K$$

is trivial and therefore unramified. The inertia comparison is exactly

$$\underbrace{I(EK/K)}_{1} \hookrightarrow \underbrace{I(E/F)}_{\mathbb{Z}/p\mathbb{Z}}.$$

In the notation of the valuation tower, this models the possible equality of completed fields $\widehat{E} = \widehat{K_{C'}}$ over $\widehat{F}$. The global fields E and $K_{C'}$ may still be linearly disjoint even when their chosen completions coincide. Thus the example disproves the general local inference "unramified after base change implies unramified before base change." It does not assert that this particular extension K/F is produced by the map q for a specified global symmetric-power torsor.

The mechanism is visible in the valuations. Over F, the pole order of t^{-1} is one and cannot be reduced by an element of F. Under the wildly ramified extension K/F, that pole order becomes p, and K supplies the element π^{-1} of intermediate valuation needed for the Artin–Schreier change of variable. The base change does not show that the original ramification was absent; it absorbs it.

B.5 What the example proves

The example does not contradict part (3) of Theorem 58. That theorem is proved in Chapter 2 by a direct Artin–Schreier–Witt calculation. Rather, the example isolates the missing implication in the proposed alternative proof:

1. the induction hypothesis for G/H controls the lower extension F/K_C;
2. the degree-p Artin–Schreier calculation controls the top layer only after base change to $K_{C'}$;
3. a wild base change can absorb the ramification of E/F, so unramifiedness of $EK_{C'}/K_{C'}$ does not recover unramifiedness of E/F;
4. repairing the induction would require additional geometric control of the comparison extension $K_{C'}/F$, strong enough to prevent this absorption;
5. the direct Witt-vector calculation avoids this descent problem by computing the full $\mathbb{Z}/p^r\mathbb{Z}$-torsor at the exceptional valuation in one step.

Thus the obstruction is not the decomposition of the torsor itself. The obstruction is that the two induction inputs measure the top layer on opposite sides of an uncontrolled wild base change.

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